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(54) **LOX3 GENE MODULATION AND
ARMYWORM TOLERANCE**

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A01H 6/46 (2018.01)
C12N 9/22 (2006.01)

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(2018.05); **A01H 6/4684** (2018.05); **C12N**
9/22 (2013.01); **C12N 15/8218** (2013.01);
C12N 2310/20 (2017.05)

(58) **Field of Classification Search**

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USPC **800/279**
See application file for complete search history.

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(57) **ABSTRACT**

The present application provides a new technology to confer
or enhance insect resistance and, optionally also resistance
to fungal pathogens in plants. In particular, the present
invention provides a method for conferring or increasing
resistance or tolerance to insect and optionally also to fungal
pathogens in maize and oil seed rape (OSR) by targeting the
endogenous Lox3 gene. By introducing either a gene silenc-
ing construct, a genome editing system or a genome modi-
fication, which leads to a targeted knock-down or knock-out
of the Lox3 gene endogenous to the plant, a new or increased
resistance to insect and, optionally fungal pathogens can be
created.

16 Claims, 8 Drawing Sheets

Specification includes a Sequence Listing.

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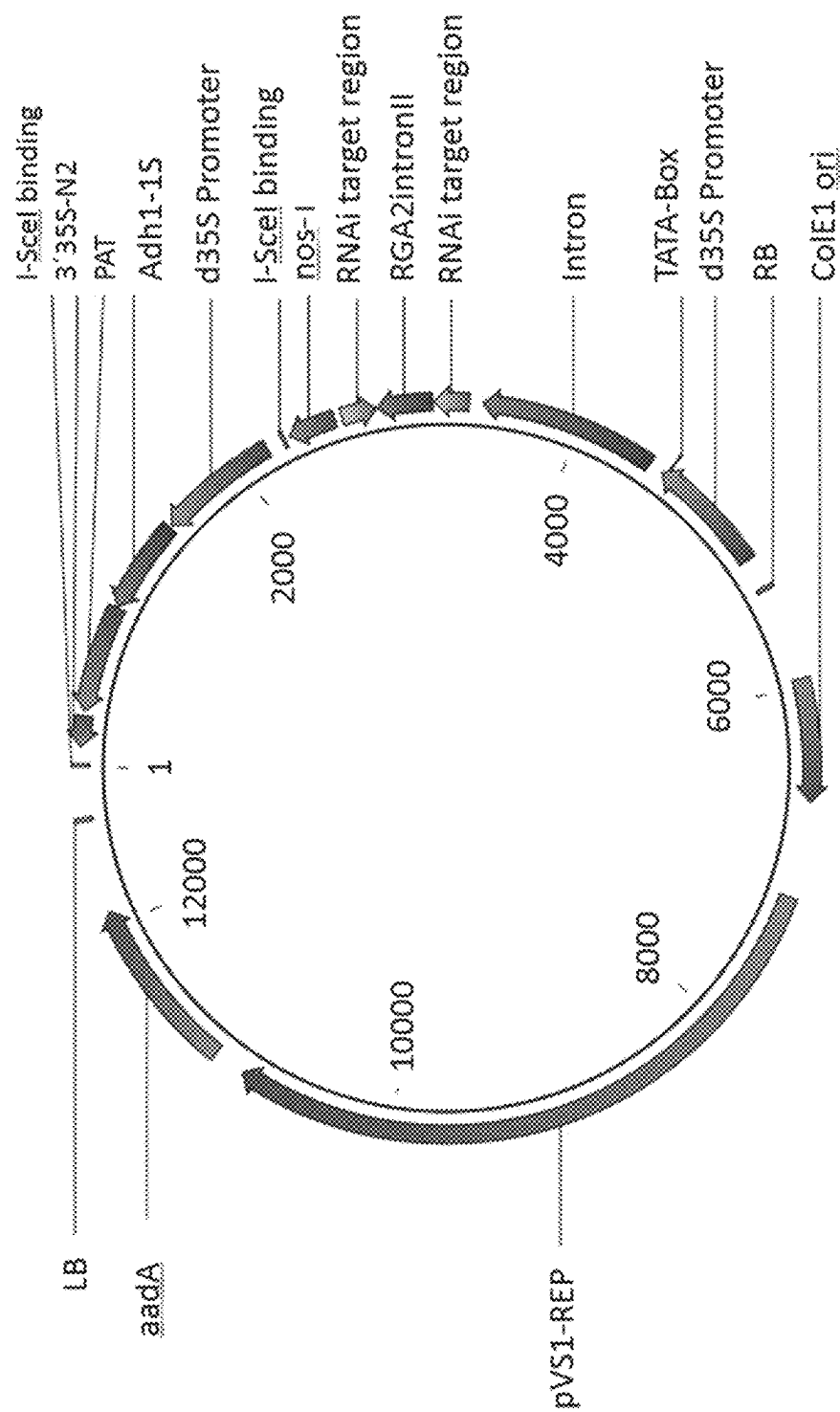


Figure 1

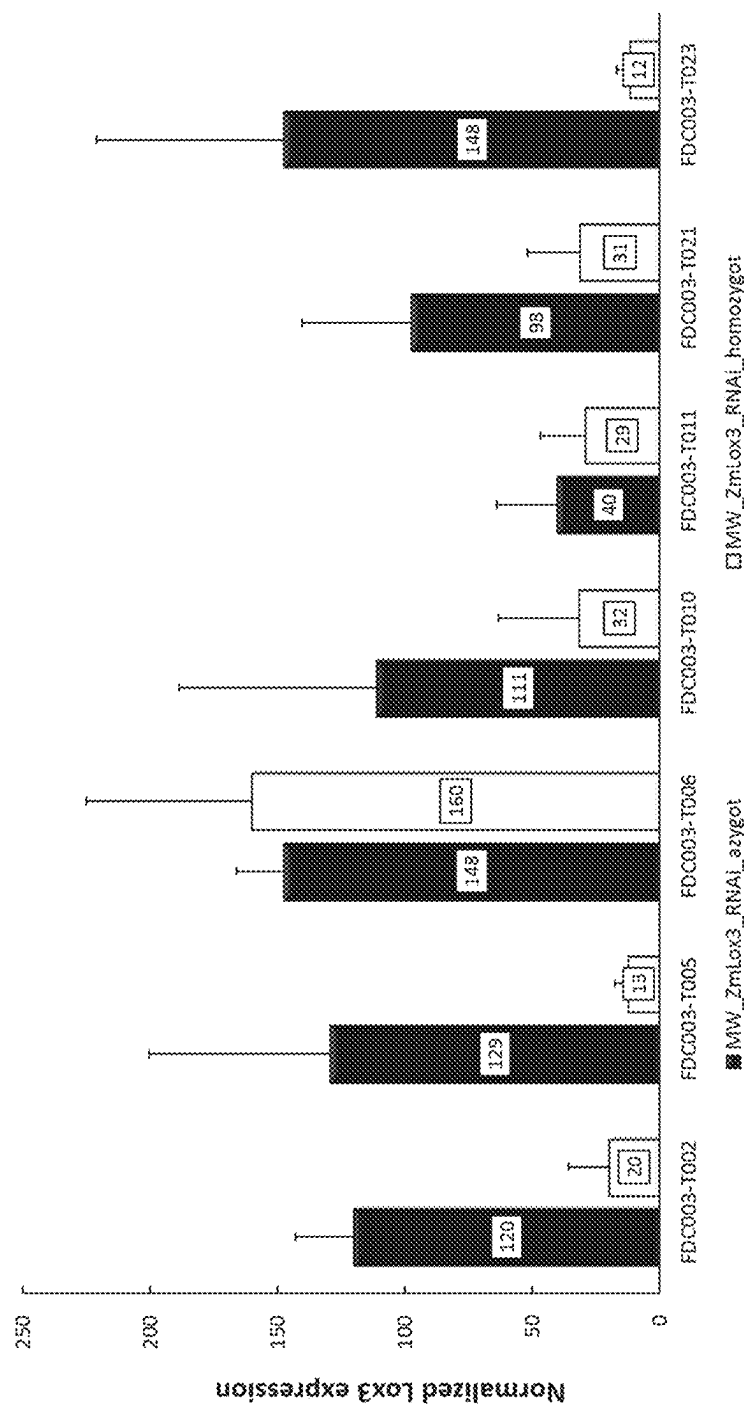
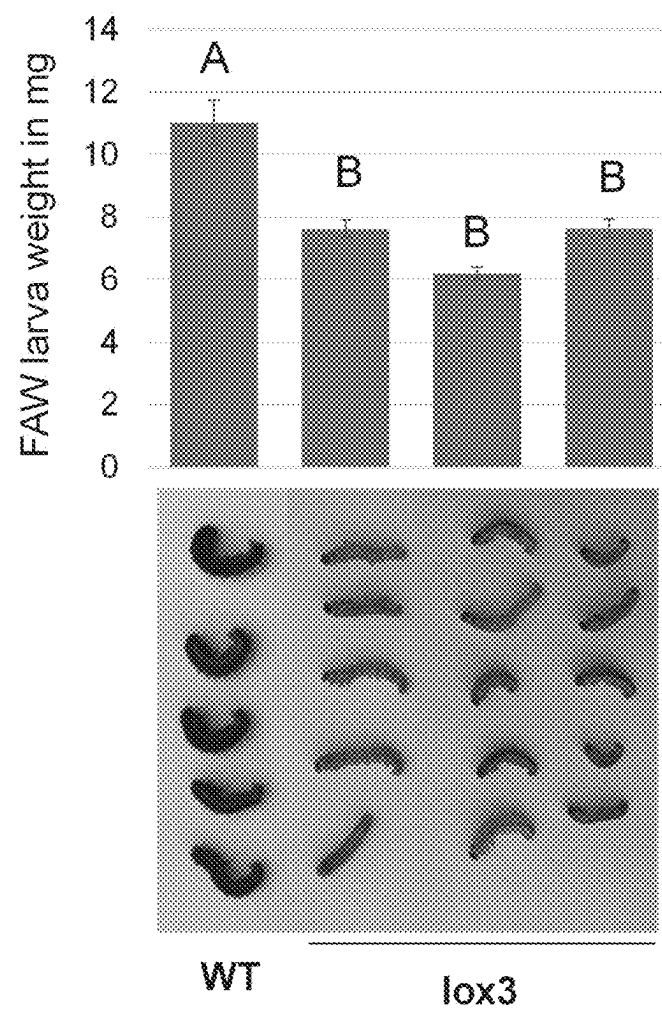


Figure 2

Figure 3



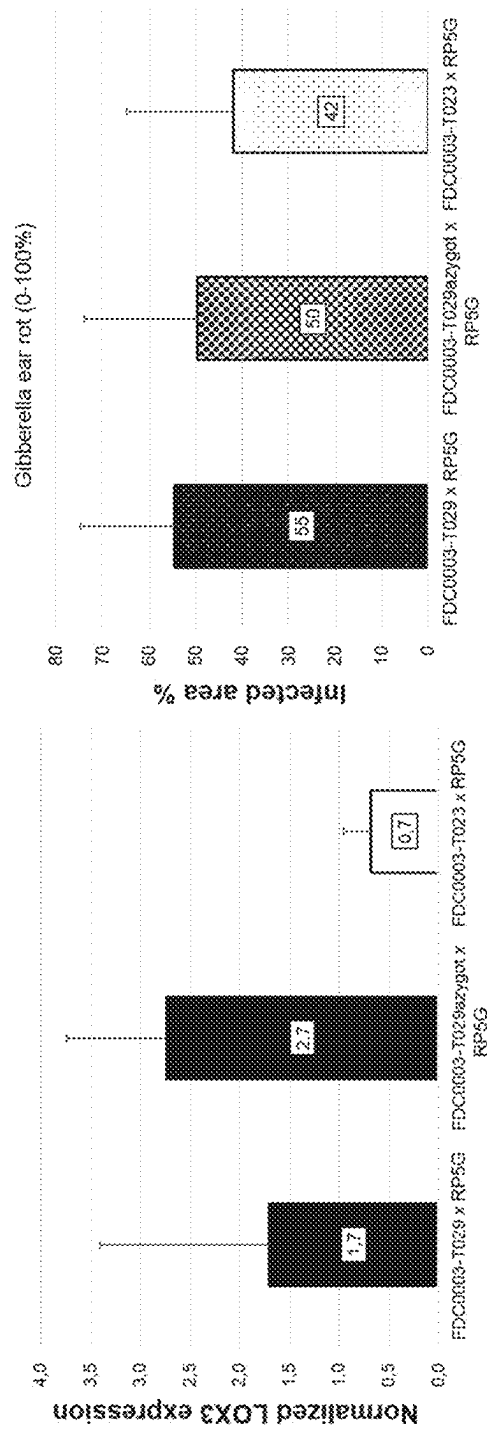


Figure 4

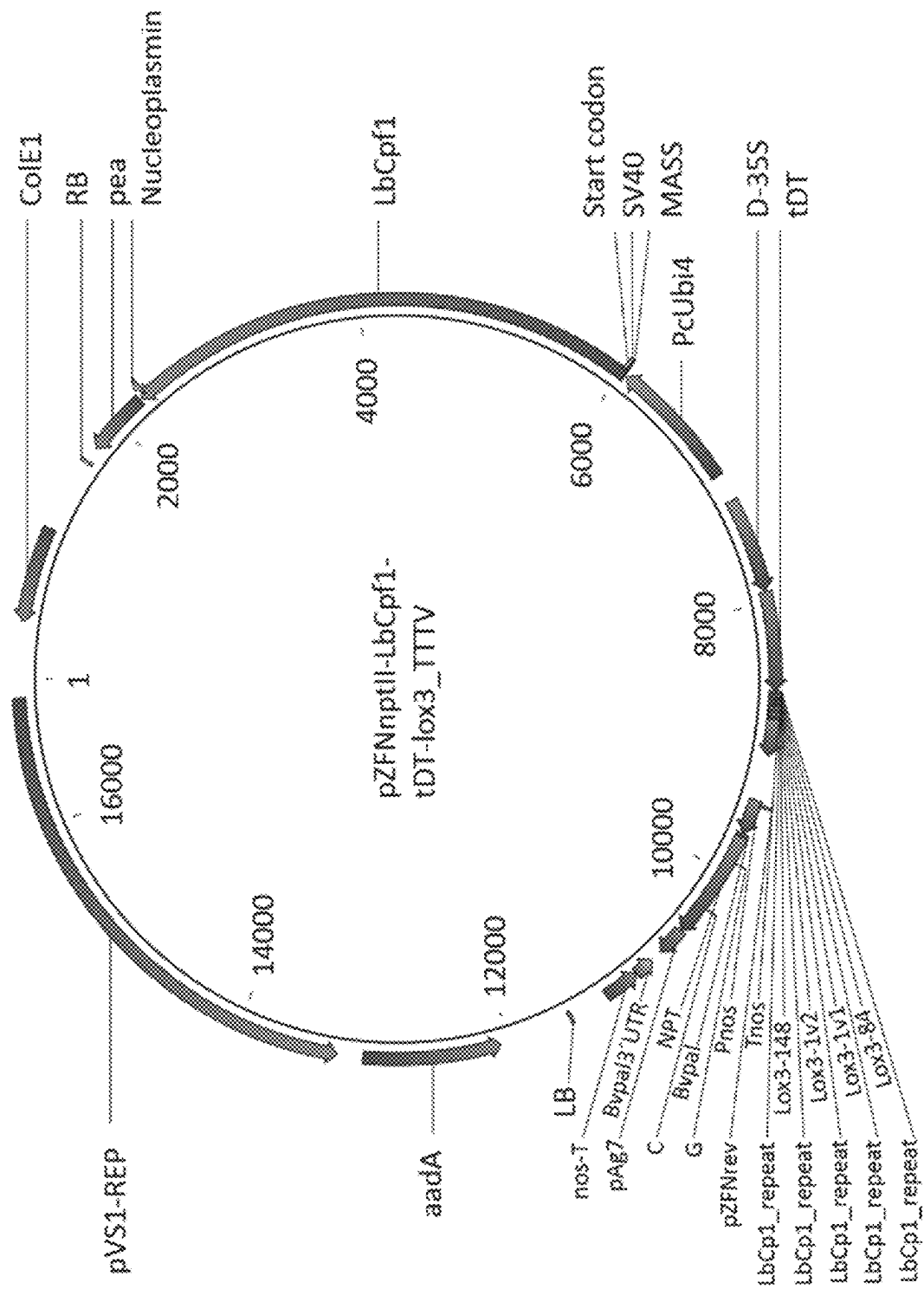


Figure 5

Figure 6

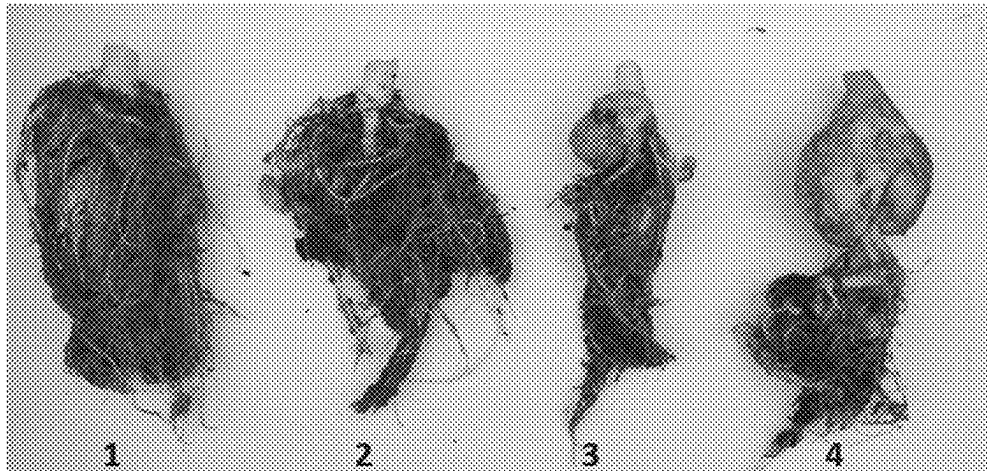


Figure 7

Entry Name	FV_G ON	FG_G ON	FV_M UR	FG_M UR	Gene	Plant_2020	Gene	Plant_2021	2020		2021	
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PH207	18,67	30,00	58,50	32,78		WVP20-73701/MIX		WVP20-73701/MIX				
PH207	37,50	56,00	42,22	38,89		WVP20-73701/MIX		WVP20-73701/MIX	22,92	42,56	43,92	35,36
PH207m0 34b			16,11	8,86	LOX3-MUT	WVP20-73214/005	LOX3-MUT	WVP20-73212/007				
PH207m0 34b	17,40		35,50	12,50	LOX3-MUT	WVP20-73216/005	LOX3-MUT	WVP20-73212/007				
PH207m0 34b	18,33	0,00	8,00	4,11	LOX3-MUT	WVP20-73217/003	LOX3-MUT	WVP20-73212/007				
PH207m0 34b	9,60	10,00	23,00	9,33	LOX3-MUT	WVP20-73217/006	LOX3-MUT	WVP20-73212/007	15,11	5,00	20,65	8,70
PH207w0 34b	36,50	9,63	14,17	25,00	LOX3-WT	WVP20-73214/006	LOX3-WT	WVP20-73212/007				
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PH207w0 34b	9,00	36,67		11,50	LOX3-WT	WVP20-73217/007	LOX3-WT	WVP20-73212/007	26,13	33,23		
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PH207m0 34h	27,80	54,38	n.d.	n.d.	LOX3- MUT	WVP20- 73270/002	28,44	45,72	
PH207w0 34h	57,00	46,80	n.d.	n.d.	LOX3- WT	WVP20- 73267/002			
PH207w0 34h	28,11	28,00	n.d.	n.d.	LOX3- WT	WVP20- 73267/003			
PH207w0 34h	37,78	71,67	n.d.	n.d.	LOX3- WT	WVP20- 73268/002			
PH207w0 34h	32,40	45,50	n.d.	n.d.	LOX3- WT	WVP20- 73270/004	38,82	47,99	
PH207m0 34d	n.d.	n.d.	0,00	65,83	LOX3- MUT	WVN18- 76855/003	n.d.	n.d.	
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PH207w0 34d	n.d.	n.d.	57,00	45,00	LOX3- WT	WVN18- 76857/007	n.d.	n.d.	
PH207w0 34d	n.d.	n.d.	41,67	46,67	LOX3- WT	WVN18- 76858/008	n.d.	55,57	45,75

Figure 7 cont'd

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**LOX3 GENE MODULATION AND
ARMYWORM TOLERANCE****CROSS-REFERENCE TO RELATED
APPLICATIONS**

This application claims the benefit of, and priority to, European Patent Application No. 22153142.9, filed Jan. 25, 2022 and U.S. Provisional Application No. 63/299,628, filed on Jan. 14, 2022. These applications are herein incorporated by reference in their entirety.

SEQUENCE LISTING

The instant application contains a Sequence Listing which has been submitted in XML format via Patent Center and is hereby incorporated by reference in its entirety. Said XML copy, created on Dec. 21, 2022, is named 245761000200.xml, and is 274,027 bytes in size.

TECHNICAL FIELD

The present application provides a new technology to confer or enhance insect resistance and, optionally also resistance to fungal pathogens in plants. In particular, the present invention provides a method for conferring or increasing resistance or tolerance to insect and optionally also to fungal pathogens in maize and oil seed rape (OSR) by targeting the endogenous Lox3 gene. By introducing either a gene silencing construct, a genome editing system or a genome modification, which leads to a targeted knock-down or knock-out of the Lox3 gene endogenous to the plant, a new or increased resistance to insect and, optionally fungal pathogens can be created. Further provided are resistant or tolerant maize or oil seed rape plants, cells, tissues, organs, or seeds, which are obtained or obtainable by the method according to the present invention. Expression constructs and vectors for the different approaches described herein are also provided as well as the use of such constructs and methods to confer or increase insect and optionally fungal resistance in plants.

BACKGROUND

In nature, there is a large variety of organisms causing disease in plants and/or negatively affecting plant health otherwise. These pathogens can be subdivided into (i) infectious organisms, which include fungi, oomycetes, bacteria, viruses, viroids, virus-like organisms, phytoplasmas, protozoa, nematodes, and parasitic plants and (ii) ectoparasites, such as insects, mites, vertebrates, and other pests negatively affecting plant health by eating plant tissue.

Monitoring and securing plant health in both natural and cultivated plant populations are paramount tasks for a reliable supply of food products and a large number of commodities on a global scale. Taken together, pests and diseases in plants cause up to 40% yield losses every year. Accordingly, there are numerous examples throughout history demonstrating the severe effects of plant disease on society.

For instance, the Great Famine of Ireland from 1845 to 1852 with a total number of deaths of approximately 1 million and the Highland potato famine from 1846 to 1856. The proximate cause of these famines was potato blight, a serious potato and tomato disease caused by the oomycete *Phytophthora infestans*.

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From 2010 to 2011, 750,000 hectares of trees in the western United States were lost to an infestation by the mountain pine beetle. This infestation was (at least in part) driven by droughts as insects generally thrive in a warmer climate. Droughts also make plant tissue even more nutritious for insects since the lack of water concentrates the contained amino acids. Accordingly, the threat of insects on agricultural crops increases as global warming proceeds.

Insects are among the most relevant organisms attacking and causing damage to agricultural and horticultural crops. The damage they cause is two-fold, namely (i) the direct injuries they cause to a plant for instance by eating plant tissue, and (ii) the indirect injuries caused by fungal, bacterial, or viral infections they transmit. Currently, a global average of 15% of crops is exclusively lost to insects. At the same time, crop yields need to be increased by at least 40% in order to be able to reliably feed a population of approximately 9 billion people projected to inhabit the Earth in 2050.

Within the class of insects, a particularly relevant species causing harm to plants is the fall armyworm (*Spodoptera frugiperda*) within the order Lepidoptera, which is the larval life stage of a fall armyworm moth. It shows large-scale invasive behaviour and it is regarded as a pest that can damage and even destroy a wide variety of crops causing large economic damage. It is one of the most damaging pests in corn (*Zea mays*). Geographically, the fall armyworm is distributed in eastern and central North America and in South America. Due to its susceptibility to colder temperatures, it can only survive the winter in the southern-most regions of the United States, namely Texas and Florida. However, seasonally it will spread across the eastern United States and up to southern Canada. A CLIMEX model of its potential global distribution indicated much of the potential distribution range in Europe, South Africa, China, and Australia. In recent years, the fall armyworm has already been found in at least 28 African countries (e.g., in Ghana, Togo, Benin, Nigeria, and Sao Tome), in Asia (e.g., in the Chinese province of Yunnan and at least 25 other Chinese provinces, in India, in Sri Lanka, and in Bangladesh), and across the Australian continent (e.g., in Western Australia, Queensland, the Northern Territory, New South Wales, North Queensland, and the Torres Strait Islands).

The fall armyworm's global distribution bears the potential for further severe economic damage. For example, it caused significant damage to maize crops in Africa in 2016. Moreover, heavy infestation of fall armyworm was reported for plantations in Sri Lanka and has reached China's north-eastern corn belt in 2020 and China's Ministry of Agriculture and Rural Affairs has rated the situation as 'very grave'. The fall armyworm is also expected to severely impact Australia's wool industry as it feeds on all major grazing plants.

Another extremely relevant species causing major damage to plants is the corn leafhopper (*Dalbulus maidis*) within the order Hemiptera. This species is widely spread through most tropical and subtropical regions on earth including Southeast Asia and China, Australia, Africa, and both North and South America. The corn leafhopper is predominantly a pest of maize and its relatives with high economic importance. In addition to the direct damage caused by its herbivorous lifestyle, it also functions as a vector for several species-specific maize viruses, such as maize stripe virus (MSV), maize mosaic virus (MMV), and maize tenuivirus (MSV). The latter two are pathogenic viruses, which might

reduce crop yields by 9 to 90%. It has even been suggested that the spreading of *Dalbulus maidis* together with MMV and MSTV to the New.

World contributed to the collapse of the Mayan civilization. Infestation with *Dalbulus maidis* will lead to physical damage the plant as the insect breaks through the vascular tissue of the plant in order to feed on the exuding sap. Eventually, this damage will cause yellowing of leaves, wilting, stem weakness, and finally death. The feeding behaviour alone of *Dalbulus maidis* might cause 10 to 15% crop loss.

Another exemplary insect species of major importance for plant damage is the globally distributed green peach aphid (*Myzus persicae*), which is the most significant aphid pest of peach trees and is known to attack more than 240 plant species from 64 different families. As a result of an infestation by the green peach aphid, peach trees exhibit decreased growth, shriveling of the leaves and death of various tissues. Prolonged infestation can lead to a drastic decrease in yield of various root and foliage crops. Furthermore, the green peach aphid can be a serious pest problem for oil seed rape crop. For example, a drastic infestation of oil seed rape by the green peach aphid in the southeast of Romania was reported in autumn 2018. In this instance, a high pest density of 243 aphids per oil seed rape leaf was reported. Moreover, the green peach aphid acts as a vector for plant viruses, such as pepper potyviruses, potato virus Y (PVY), tobacco etch virus (TEV), and cucumber mosaic virus (CMV), which can be passed on to many different food crops.

The green belly stink bug (*Dichelops melacanthus*) is a key pest in corn and wheat, two of the most important crop plants on a global scale. It is distributed in nearly all of South America and attacks at least 29 plant species. Besides corn and wheat, other affected crops of major economic importance include soybean, oats, and triticale. In the early life stages of plants, the damage caused by *Dichelops melacanthus* can be particularly severe as the insect physically damages the vascular tissue in order to feed on sap and thereby possibly introduces salivary toxic enzymes into the stem base of the plants. Eventually, this leads to withering of leaves, wilting, and finally to death of the plant.

Fungi and fungi-like organisms constitute the largest number of plant pathogens and are responsible for a wide range of plant diseases, which have been reported to lead up to 100% crop loss. For instance, most vegetable diseases are due to fungal infections. Causes for fungal infections include spreading through water and wind and through contaminated soils, animals, seedlings, and other plant materials. Fungi enter plants through naturally occurring openings, such as stomata and wounds caused by e.g., pruning, harvesting, insects, hail, and other causes for mechanical damage.

Leptosphaeria maculans (anamorph: *Phoma lingam*) is a globally distributed fungal pathogen of the phylum Ascomycota, which is the causal agent of *phoma* stem canker or blackleg disease in *Brassica* crops. The fungus can directly penetrate plant roots. Symptoms of blackleg disease include basal stem cankers, small grey lesions on leaves, and root rot. Basal stem cankers are the main cause for drastic crop yield losses. *Leptosphaeria maculans* infects a variety of different *Brassica* species including oilseed rape (*Brassica napus*) and cabbage (*Brassica oleracea*). It is especially virulent in oilseed rape. Infections can lead to decrease in crop yields by about 10 to 20%. The release of ascospores by *Leptosphaeria maculans* typically occurs from September to November at moderate temperatures between 8° C.

and 15° C. and a relatively high humidity. During this time, agricultural oilseed rape is the most vulnerable.

Another economically relevant fungal plant pathogen is the soil-borne Plasmodiophora *brassicae*, which belongs to the group Phytomyxea and causes clubroot in a large number of plants from the family Brassicaceae. Symptoms of clubroot include gall formation and distortion on latent roots giving rise to the shape of a club or spindle. In cabbages, these effects on the roots lead to underdeveloped heads or even an overall failure to head at all, which is often followed by decline in vigor and death of the plant. Other symptoms include wilting, yellowing and stunted growth. In the late 19th century, severe epidemics of clubroot lead to the loss of major parts of cabbage crops in St. Petersburg. Presently, clubroot is still a disease of great economic relevance affecting approximately 10% of cultured areas. Clubroot infections can affect entire fields and thus significantly reduce crop yields and even result in no crop yield at all. On the field, the pathogen can survive for years as resting spores.

Hence, there is an urgent and rapidly growing demand for efficient strategies to minimize and confine the economic damage caused by plant pathogens.

The most commonly used form of protection against insects is different insecticides. In southern regions, insecticides have to be applied every day to corn in order to be able to manage fall armyworm infestation. In 2020, a biopesticide—namely a caterpillar-specific virus—was approved under emergency regulations in Australia in order to control the fall armyworm. Different Parasitoids (e.g., the wasp *Trichogramma pretiosum*) are also used. The use of insecticides has many disadvantages. For instance, many insecticides non-selectively harm or even kill other species in addition to the targeted ones. A prominent example of this phenomenon is the observed decline of pollinators, such as bees due to colony collapse disorder (CCD). Even sub-lethal amounts of insecticides can affect bee foraging behaviour. The loss of pollinators results in a reduction of crop yields. Besides that, birds may be killed when consuming plants or insects that were in contact with insecticides. Populations of insectivorous birds also decline due to the collapse of their prey populations. Especially the spraying of insecticides on corn and wheat in Europe is believed to have caused a decline in flying insects of about 80%, which in turn has reduced the European bird population by one to two thirds. Runoff and percolation of (improperly applied) insecticides can negatively affect the quality of water sources and harm the natural ecology, which has an indirect effect on human populations through biomagnification and bioaccumulation.

Alternatively, in order to manage insect infestation different agricultural techniques, such as e.g., planting early, avoiding staggered planting, and inter-cropping are applied. These strategies are extremely cost- and time-intensive, cannot universally be applied due to ecological limitations and thus bear several risks. For example, inter-cropping has successfully been applied against fall army worm infestations in small-scale greenhouse, garden and field experiments. However, at larger commercial scales the pest damage to only a very small portion of crop plants could be reduced using this method.

Fungal plant diseases may be controlled employing fungicides and various other agricultural techniques. Typically applied fungicides include EBI and MBC fungicides, which can decrease instances of disease in crop populations. However, the use of fungicides bears similar disadvantages as described above for insecticides. Additionally, some fungicides also negatively affect the overall growth of the crop

plants. Cultural methods include stubble and crop rotation. Removing stubble has been shown to decrease the risk of *Leptosphaeria maculans* infection in *Brassica* species. In canola crops, crop rotation has been shown to reduce black-leg.

Based on the difficulties and disadvantages of agricultural and chemical methods for reducing infestations and infections, considerable effort is put into genetically engineering and optimizing the crop plants themselves in order to make them more tolerant towards environmental conditions and pests.

In case of maize and other crop plants, the lipoxygenase (Lox) pathway is known to play a role regarding resistance to pathogens. However, previous studies show that the actual function of the Lox pathway with respect to resistance towards certain pathogens is highly unpredictable due to an extremely high degree of host- and pathogen specificity to the extent that within the same host species, Lox-derived metabolism can even have completely opposite effects on pathogen resistance in a pathogen species-specific manner.

In the Lox pathway, various highly specialized forms of lipoxygenases catalyse the synthesis of hydroperoxy polyunsaturated fatty acids, which are substrates to at least seven different enzyme families. For example, plant oxylipins produced via the Lox pathway have been demonstrated to function as environmentally and developmentally regulated defence- and development signals. It has also been demonstrated that these molecules function as powerful regulators of sporogenesis and mycotoxin biosynthesis in fungi. Specifically, it could be demonstrated that fatty acid hydroperoxides derived from 9-Lox induce condition and mycotoxin production.

In 2007, Gao et al. (MPMI, vol. 20, No. 8, 2007, pp. 922-933) generated maize mutants lacking functional 9-Lox by disruption of ZmLox3 (gene coding for 9-Lox) via transposon mutagenesis. Accordingly, decreased levels of 9-Lox-derived hydroperoxides were observed. Consequently, in kernels of the lox3 mutants reduced condition and reduced production of mycotoxin fumonisin B1 by *Fusarium verticillioides* were observed as compared to the wild type. Additionally, lox3 mutants demonstrated reduced disease severity of various fungi-derived diseases, such as anthracnose leaf blight (*Colletotrichum graminicola*), southern leaf blight (*Cochliobolus heterostrophus*), and stalk rots (*Fusarium verticillioides* and *Colletotrichum graminicola*). They conclude from these findings that 9-LOX-based metabolism apparently is required for fungal pathogenesis, including disease development and production of spores and mycotoxins.

However, in 2008, Gao et al. (MPMI, vol. 21, No. 1, 2008, pp. 98-109) demonstrated that lox3-4 knockout mutants of maize displayed increased attractiveness to root-knot nematodes (*Meloidogyne incognita*). They observed that in these lox3-4 knockout mutants a phenylalanine ammonia lyase (PAL) gene was not inducible in a root-knot nematode-dependent manner. This suggests a PAL-mediated metabolism to be of importance for root-knot nematode resistance. Moreover, in lox3-4 knockout mutants a root-specific increase in jasmonic acid, ethylene, and salicylic acid and overexpression of the respective biosynthetic genes was observed. Therefore, the ZmLox3-mediated metabolic pathway apparently is required for the three major defence signalling pathways for the resistance against nematodes.

Therefore, there is a great need in defining new molecular mechanisms to increase tolerance or resistance of major crop plants towards their cognate and specific pathogens, by specifically studying and controlling endogenous signalling

pathways to optimize plant defence against pathogens and thus yields. By defining new ways to influence and control Lox3-pathways in different crop plants in response to some specific insect and fungal pathogens triggering a specific response in the respective target plant, these objects could be achieved by the methods as presented and disclosed below.

SUMMARY OF THE INVENTION

In a first aspect, the present invention relates to a method for conferring or increasing resistance or tolerance to an insect and, optionally a fungal pathogen to/in a plant comprising the steps of:

- (i) providing at least one plant cell;
- (ii) introducing into the at least one plant cell at least one gene silencing construct, at least one genome editing system or a genome modification, which leads to a targeted knock-down or a knock-out of a Lox3 gene endogenous to the plant;
- (iii) obtaining at least one modified plant cell having reduced or abolished expression of the Lox3 gene; and
- (iv) obtaining at least one plant cell, tissue, organ, plant, or seed having reduced or abolished expression of the Lox3 gene, optionally after an additional step of regenerating the plant tissue, organ, plant or seed from the at least one modified cell.

In one embodiment of the method described above, the method is for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* species, *Cercospora zeina* and *Cercospora zeae-maydis* to/in maize (*Zea mays*).

In another embodiment of the method described above, the method is for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of green peach aphid (*Myzus persicae*), diamond-back moth (*Plutella xylostella*), cabbage stem flea beetle (*Psylliodes chrysocephala*), crucifer flea beetle (*Phyllotreta cruciferae*), striped flea beetle (*Phyllotreta striolata*), hop flea beetle (*Psylliodes punctulata*), rape stem weevil (*Ceutorhynchus picitarsis*) and cabbage stem weevil (*Ceutorhynchus quadridens*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Phoma lingam* and *Plasmodiophora brassicae* to/in oilseed rape (*Brassica napus*).

In one embodiment of the method described above, the Lox3 gene is represented by a nucleic acid sequence of SEQ ID NO: 6, 7, 9, 10, 12, 13, 15, 16, 87 or 88 or a nucleic acid sequence having a sequence identity of at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 6, 7, 9, 10, 12, 13, 15, 16, 87 or 88.

In another embodiment of the method described above, the Lox3 gene encodes an amino acid sequence of SEQ ID NO: 8, 11, 14, 17 or 89 or an amino acid sequence having a sequence identity of at least 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%,

95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 8, 11, 14, 17 or 89.

In a further embodiment of the method described above, the Lox3 gene is represented by a nucleic acid sequence of SEQ ID NO: 75, 76, 77, 78, 83, 84, 85 or 86 or a nucleic acid sequence having a sequence identity of at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 75, 76, 77, 78, 83, 84, 85 or 86.

In yet another embodiment of the method described above, the Lox3 gene encodes an amino acid sequence of SEQ ID NO: 79, 80, 81 or 82 or an amino acid sequence having a sequence identity of at least 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 79, 80, 81 or 82.

In an embodiment of the method described above, in step (ii) a construct is introduced into the at least one plant cell, which targets the Lox3 gene for gene silencing.

In one embodiment of the method described above, the construct is or the construct encodes an RNAi construct comprising a sense and an antisense sequence targeting the Lox3 gene, the RNAi construct forming an RNA hairpin upon transcription.

In one embodiment of the method described above, the method is for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* species, *Cercospora zeina* and *Cercospora zea-maydis* to/in maize (*Zea mays*) and the sense sequence is encoded by a nucleic acid sequence of SEQ ID NO: 1, or a nucleic acid sequence having a sequence identity of at least 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 1.

In another embodiment of the method described above, the method is for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* species, *Cercospora zeina* and *Cercospora zea-maydis* to/in maize (*Zea mays*) and the antisense sequence is encoded by a nucleic acid sequence of SEQ ID NO: 2, or a nucleic acid sequence having a sequence identity of at least 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 2.

In yet another embodiment of the method described above, the method is for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* spe-

cies, *Cercospora zeina* and *Cercospora zea-maydis* to/in maize (*Zea mays*) and the RNA hairpin has an intervening intron loop sequence comprising a nucleic acid sequence of SEQ ID NO: 3, or a nucleic acid sequence having a sequence identity of at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 3.

In a further embodiment of the method described above, the method is for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* species, *Cercospora zeina* and *Cercospora zea-maydis* to/in maize (*Zea mays*) and the construct comprises a nucleic acid sequence of SEQ ID NO: 4, or a nucleic acid sequence having a sequence identity of at least 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 4.

In yet a further embodiment of the method described above, the method is for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* species, *Cercospora zeina* and *Cercospora zea-maydis* to/in maize (*Zea mays*) and a vector is introduced into the plant cell, which vector comprises or consists of a nucleic acid sequence of SEQ ID NO: 5, or a nucleic acid sequence having a sequence identity of at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 5.

In one embodiment of the method described above, the construct is introduced into the at least one plant cell by transformation or transfection mediated by biolistic bombardment, *Agrobacterium*-mediated transformation, micro- or nanoparticle delivery, chemical transfection, or a combination thereof.

In a further embodiment of the method described above, in step (ii) at least one genome editing system is introduced into the at least one cell, which targets the Lox3 gene, wherein the at least one genome editing system comprises

- (a) at least one site-specific nuclease or site-specific nickase, and optionally, in case a CRISPR system is used, at least one guide molecule or a sequence encoding the same, and
- (b) optionally, at least one repair template, or a sequence encoding the same.

In one embodiment of the method described above, the at least one genome editing system is selected from a CRISPR/Cas system, preferably from a CRISPR/MAD7 system, a CRISPR/Cpf1 (CRISPR/Cas12a) system, a CRISPR/MAD2 system, a CRISPR/Cas9 system, a CRISPR/CasX system, a CRISPR/CasY system, a CRISPR/Cas13 system, or a CRISPR/Csm system, or the at least one genome editing system, is selected from a zinc finger nuclease system, or a

transcription activator-like nuclease system, or a meganuclease system, or any combination, variant, or an active fragment thereof.

In another embodiment of the method described above, the at least one genome editing system is introduced into the at least one cell by transformation or transfection mediated by biolistic bombardment, *Agrobacterium*-mediated transformation, micro- or nanoparticle delivery, chemical transfection, or a combination thereof.

In yet another embodiment of the method described above, the method is for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of green peach aphid (*Myzus persicae*), diamond-back moth (*Plutella xylostella*), cabbage stem flea beetle (*Psylliodes chrysocephala*), crucifer flea beetle (*Phyllotreta cruciferae*), striped flea beetle (*Phyllotreta striolata*), hop flea beetle (*Psylliodes punctulata*), rape stem weevil (*Ceutorhynchus piciparsis*) and cabbage stem weevil (*Ceutorhynchus quadridens*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Phoma lingam* and *Plasmidiophora brassicae* to/in oilseed rape (*Brassica napus*), and the at least one genome editing system comprises a crRNA encoded by a nucleic acid sequence of any of SEQ ID NOS: 46 to 49, or a nucleic acid sequence having a sequence identity of at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of any of SEQ ID NOS: 46 to 49.

In one embodiment of the method described above, the genome editing system is encoded by a plasmid of the nucleic acid sequence of SEQ ID NO: 50 or a nucleic acid sequence having a sequence identity of at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 50.

In another embodiment of the method described above, in step (ii) a mutagenesis is performed on a single or on a plurality of cell(s) by applying chemicals or radiation.

In one embodiment of the method described above, an alkylating agent, in particular ethyl methanesulfonate is applied to the single or the plurality of cell(s) to induce mutagenesis.

In another embodiment of the method described above, one or more mutations in the Lox3 gene are inserted and identified by TILLING in step (ii).

In a further embodiment of the method described above, one or more cell(s) with knock-down or knock-out mutations in the Lox3 gene are selected in step (ii).

In another aspect, the present invention relates to a maize cell, maize tissue, maize organ, maize plant or maize seed obtained or obtainable by a method according to any of the embodiments described above.

In a further aspect the present invention relates to an oilseed rape cell, oilseed rape tissue, oilseed rape organ, oilseed rape plant or oilseed rape seed obtained or obtainable by a method according to any of the embodiments described above.

In yet a further aspect, the present invention relates to an expression construct, which targets the Lox3 gene in maize for gene silencing, wherein the construct encodes an RNAi construct comprising a sense and an antisense sequence targeting the Lox3 gene endogenous to a maize plant, which RNAi construct forms an RNA hairpin upon transcription.

In a further aspect, the present invention relates to an expression construct, which targets the Lox3 gene in oilseed rape for gene silencing, wherein the construct encodes an RNAi construct comprising a sense and an antisense sequence targeting the Lox3 gene endogenous to an oilseed rape plant, which RNAi construct forms an RNA hairpin upon transcription.

In one embodiment of the expression construct described above, the sense sequence is encoded by a nucleic acid sequence of SEQ ID NO: 1, or a nucleic acid sequence having a sequence identity of at least 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 1.

In another embodiment of the expression construct described above, the antisense sequence is encoded by a nucleic acid sequence of SEQ ID NO: 2, or a nucleic acid sequence having a sequence identity of at least 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 2.

In yet another embodiment of the expression construct described above, the RNA hairpin has an intervening intron loop sequence comprising a nucleic acid sequence of SEQ ID NO: 3, or a nucleic acid sequence having a sequence identity of at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 3.

In one embodiment of the expression construct described above, the construct comprises a nucleic acid sequence of SEQ ID NO: 4, or a nucleic acid sequence having a sequence identity of at least 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 4.

In another aspect, the present invention relates to a vector comprising or consisting of a nucleic acid sequence of SEQ ID NO: 5 or a nucleic acid sequence having a sequence identity of at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 5.

In yet another aspect, the present invention relates to an RNAi hairpin construct conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* species, *Cercospora zeina* and *Cercospora zea-maydis* to/in maize (*Zea mays*), wherein the RNAi hairpin construct comprises a nucleic acid sequence of SEQ ID NO: 4, or a nucleic acid sequence having a sequence identity of at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 4.

In a further aspect, the present invention relates to an expression construct encoding a genome editing system, which targets the Lox3 gene in maize, wherein the genome editing system comprises

(a) at least one site-specific nuclease or site-specific nickase, and optionally, in case a CRISPR system is used, at least one guide molecule or a sequence encoding the same, and

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(b) optionally, at least one repair template, or a sequence encoding the same.

In another aspect the present invention relates to an expression construct encoding a genome editing system, which targets the Lox3 gene in oilseed rape, wherein the genome editing system comprises

- (a) at least one site-specific nuclease or site-specific nickase, and optionally, in case a CRISPR system is used, at least one guide molecule or a sequence encoding the same, and
- (b) optionally, at least one repair template, or a sequence encoding the same.

In one embodiment of the expression construct described above, the at least one genome editing system is selected from a CRISPR/Cas system, preferably from a CRISPR/MAD7 system, a CRISPR/Cpf1 (CRISPR/Cas12a) system, a CRISPR/MAD2 system, a CRISPR/Cas9 system, a CRISPR/CasX system, a CRISPR/CasY system, a CRISPR/Cas13 system, or a CRISPR/Csm system, or the at least one genome editing system is selected from a zinc finger nuclease system, or a transcription activator-like nuclease system, or a meganuclease system, or any combination, variant, or an active fragment thereof.

In another embodiment of the expression construct described above, the expression construct comprises a crRNA encoded by a nucleic acid sequence of any of SEQ ID NOs: 46 to 49, or a nucleic acid sequence having a sequence identity of at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of any of SEQ ID NO: 46 to 49.

In a further embodiment of the expression construct described above, the genome editing system is encoded by a plasmid of the nucleic acid sequence of SEQ ID NO: 50 or a nucleic acid sequence having a sequence identity of at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 50.

In another aspect, the present invention relates to a vector encoding an expression construct according to any of the embodiments described above.

In further aspect, the present invention relates to a maize cell, maize tissue, maize organ, maize plant or maize seed comprising an expression construct or a vector according to any of the embodiments described above.

In yet a further aspect, the present invention relates to an oilseed rape cell, oilseed rape tissue, oilseed rape organ, oilseed rape plant or oilseed rape seed comprising an expression construct or a vector according to any of the embodiments described above.

In one aspect the present invention relates to the use of at least one gene silencing construct, at least one genome editing system or a genome modification, which leads to a targeted knock-down or a knock-out of the endogenous a Lox3 gene, for conferring or increasing resistance or tolerance to one or more insect(s) and, optionally one or more fungal pathogen(s) to a plant.

In one aspect the present invention relates to the use of at least one gene silencing construct, at least one genome editing system or a genome modification, which leads to a targeted knock-down or a knock-out of the endogenous a Lox3 gene, for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug

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(*Dichelops melacanthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* species, *Cercospora zeina* and *Cercospora zeae-maydis* to/in maize (*Zea mays*).

In another aspect the present invention relates to the use of at least one gene silencing construct, at least one genome editing system or a genome modification, which leads to a targeted knock-down or a knock-out of the endogenous a Lox3 gene, for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of green peach aphid (*Myzus persicae*), diamond-back moth (*Plutella xylostella*), cabbage stem flea beetle (*Psylliodes chrysocephala*), crucifer flea beetle (*Phyllotreta cruciferae*), striped flea beetle (*Phyllotreta striolata*), hop flea beetle (*Psylliodes punctulata*), rape stem weevil (*Ceutorhynchus piciparsis*) and cabbage stem weevil (*Ceutorhynchus quadridens*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Phoma* lingam and *Plasmidiophora brassicae* to/in oilseed rape (*Brassica napus*).

In a further aspect, the present invention relates to the use of a construct, the construct being or encoding an RNAi construct comprising a sense and an antisense sequence targeting the endogenous Lox3 gene of a maize plant, wherein the RNAi construct forms an RNA hairpin upon transcription, for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* species, *Cercospora zeina* and *Cercospora zeae-maydis* to/in maize (*Zea mays*).

In yet another aspect, the present invention relates to the use of a construct, the construct being or encoding an RNAi construct comprising a sense and an antisense sequence targeting the endogenous Lox3 gene of an oilseed rape plant, wherein the RNAi construct forms an RNA hairpin upon transcription, for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of green peach aphid (*Myzus persicae*), diamond-back moth (*Plutella xylostella*), cabbage stem flea beetle (*Psylliodes chrysocephala*), crucifer flea beetle (*Phyllotreta cruciferae*), striped flea beetle (*Phyllotreta striolata*), hop flea beetle (*Psylliodes punctulata*), rape stem weevil (*Ceutorhynchus piciparsis*) and cabbage stem weevil (*Ceutorhynchus quadridens*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Phoma* lingam and *Plasmidiophora brassicae* to/in oilseed rape (*Brassica napus*).

In one embodiment of the use described above the sense sequence is encoded by a nucleic acid sequence of SEQ ID NO: 1, or a nucleic acid sequence having a sequence identity of at least 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 1.

In another embodiment of the use described above, the antisense sequence is encoded by a nucleic acid sequence of SEQ ID NO: 2, or a nucleic acid sequence having a sequence

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identity of at least 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 2.

In a further embodiment of the use described above, the RNA hairpin has an intervening intron loop sequence comprising a nucleic acid sequence of SEQ ID NO: 3, or a nucleic acid sequence having a sequence identity of at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or at least 99% to the sequence of SEQ ID NO: 3.

In one embodiment of the use described above, the construct comprises a nucleic acid sequence of SEQ ID NO: 4, or a nucleic acid sequence having a sequence identity of at least 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 4.

In another aspect, the present invention relates to the use of a vector, which vector comprises or consists of a nucleic acid sequence of SEQ ID NO: 5, or a nucleic acid sequence having a sequence identity of at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 5 for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* species, *Cercospora zeina* and *Cercospora zeae-maydis* to/in maize (*Zea mays*).

In yet another aspect, the present invention relates to a use of a genome editing system, which targets the endogenous Lox3 gene in a maize plant, wherein the genome editing system comprises

(a) at least one site-specific nuclease or site-specific nickase, and optionally, in case a CRISPR system is used, at least one guide molecule or a sequence encoding the same, and

(b) optionally, at least one repair template, or a sequence encoding the same

for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* species, *Cercospora zeina* and *Cercospora zeae-maydis* to/in maize (*Zea mays*).

In a further aspect, the present invention relates to a use of a genome editing system, which targets the endogenous Lox3 gene in an oilseed rape plant, wherein the genome editing system comprises

(a) at least one site-specific nuclease or site-specific nickase, and optionally, in case a CRISPR system is used, at least one guide molecule or a sequence encoding the same, and

(b) optionally, at least one repair template, or a sequence encoding the same

for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of green peach aphid (*Myzus persicae*), diamondback moth (*Plutella*

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xylostella), cabbage stem flea beetle (*Psylliodes chrysocephala*), crucifer flea beetle (*Phyllotreta cruciferae*), striped flea beetle (*Phyllotreta striolata*), hop flea beetle (*Psylliodes punctulata*), rape stem weevil (*Ceutorhynchus picipitarsis*) and cabbage stem weevil (*Ceutorhynchus quadridens*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Phoma lingam* and *Plasmodiophora brassicae* to/in oilseed rape (*Brassica napus*).

In one aspect, the present invention relates to a use of an expression construct encoding a genome editing system, which targets the endogenous Lox3 gene in a maize plant, wherein the genome editing system comprises

(a) at least one site-specific nuclease or site-specific nickase, and optionally, in case a CRISPR system is used, at least one guide molecule or a sequence encoding the same, and

(b) optionally, at least one repair template, or a sequence encoding the same,

wherein the at least one genome editing system is selected from a CRISPR/Cas system, preferably from a CRISPR/MAD7 system, a CRISPR/Cpf1 (CRISPR/Cas12a) system, a CRISPR/MAD2 system, a CRISPR/Cas9 system, a CRISPR/CasX system, a CRISPR/CasY system, a CRISPR/Cas13 system, or a CRISPR/Csm system, or the at least one genome editing system is selected from a zinc finger nuclease system, or a transcription activator-like nuclease system, or a meganuclease system, or any combination, variant, or an active fragment thereof for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* species, *Cercospora zeina* and *Cercospora zeae-maydis* to/in maize (*Zea mays*).

In another aspect, the present invention relates to a use of an expression construct encoding a genome editing system, which targets the endogenous Lox3 gene in an oilseed rape plant, wherein the genome editing system comprises

(a) at least one site-specific nuclease or site-specific nickase, and optionally, in case a CRISPR system is used, at least one guide molecule or a sequence encoding the same, and

(b) optionally, at least one repair template, or a sequence encoding the same,

wherein the at least one genome editing system is selected from a CRISPR/Cas system, preferably from a CRISPR/MAD7 system, a CRISPR/Cpf1 (CRISPR/Cas12a) system, a CRISPR/MAD2 system, a CRISPR/Cas9 system, a CRISPR/CasX system, a CRISPR/CasY system, a CRISPR/Cas13 system, or a CRISPR/Csm system, or the at least one genome editing system is selected from a zinc finger nuclease system, or a transcription activator-like nuclease system, or a meganuclease system, or any combination, variant, or an active fragment thereof for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of green peach aphid (*Myzus persicae*), diamondback moth (*Plutella xylostella*), cabbage stem flea beetle (*Psylliodes chrysocephala*), crucifer flea beetle (*Phyllotreta cruciferae*), striped flea beetle (*Phyllotreta striolata*), hop flea beetle (*Psylliodes punctulata*), rape stem weevil (*Ceutorhynchus picipitarsis*) and cabbage stem weevil (*Ceutorhynchus quadridens*) and, optionally conferring or

increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Phoma lingam* and *Plasmodiophora brassicae* to/in oilseed rape (*Brassica napus*).

In one embodiment of the use described above, the expression construct comprises a crRNA encoded by a nucleic acid sequence of any of SEQ ID NOs: 46 to 49, or a nucleic acid sequence having a sequence identity of at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of any of SEQ ID NO: 46 to 49.

In another embodiment of the use described above, the genome editing system is encoded by a plasmid of the nucleic acid sequence of SEQ ID NO: 50 or a nucleic acid sequence having a sequence identity of at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 50.

In another aspect, the present invention relates to a use of a vector encoding an genome editing system as defined above for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper

(*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* species, *Cercospora zeina* and *Cercospora zea-maydis* to/in maize (*Zea mays*).

In a further aspect, the present invention relates to a use of a vector encoding an genome editing system as defined above for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of green peach aphid (*Myzus persicae*), diamondback moth (*Plutella xylostella*), cabbage stem flea beetle (*Psylliodes chrysocephala*), crucifer flea beetle (*Phyllotreta cruciferae*), striped flea beetle (*Phyllotreta striolata*), hop flea beetle (*Psylliodes punctulata*), rape stem weevil (*Ceutorhynchus piciparsis*) and cabbage stem weevil (*Ceutorhynchus quadridens*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Phoma lingam* and *Plasmodiophora brassicae* to/in oilseed rape (*Brassica napus*).

Brief Description of Sequences

SEQ ID NO.	Description
1	RNAi target region sense strand before RGA2intronII
2	RNAi target region antisense strand after the RGA2intronII
3	RGA2intronII sequence from <i>Triticum aestivum</i> —second intron of RGA2 (resistance gene analog; gene involved in leaf rust resistance in wheat) according to Loutre et al., 2009, The Plant Journal 60, 1043-1054
4	Sequence of RNAi cassette (RNAi target region sense strand before RGA2intronII + RGA2intronII + RNAi target region antisense strand after the RGA2intronII)
5	Sequence of a complete exemplary RNAi vector, here starting with ColE1 ori
6	Zm00001eb054040_B73AGP05_T001_genomic (ZmLox3)—Genomic sequence of Lox3 from <i>Zea mays</i>
7	Zm00001eb054040_B73AGP05_T001_CDS (ZmLox3)—Coding sequence of Lox3 from <i>Zea mays</i>
8	Zm00001eb054040_B73AGP05_T001_pro (ZmLox3)—Amino acid sequence of Lox3 from <i>Zea mays</i>
9	ZmLox3_gDNA_A188—genomic sequence of Lox3 from <i>Zea mays</i> line A188
10	ZmLox3_CDS_A188—coding sequence of Lox3 from <i>Zea mays</i> line A188
11	ZmLox3_protein_A188—amino acid sequence of Lox3 from <i>Zea mays</i> line A188
12	SL57_Zm00001d033623(LOX3)_gDNA—tropical corn line SL57—genomic sequence of Lox3 from tropical corn line SL57 (<i>Zea mays</i>)
13	SL57_Zm00001d033623(LOX3)_CDS—tropical corn line SL57—coding sequence of Lox3 from tropical corn line SL57 (<i>Zea mays</i>)
14	SL57_Zm00001d033623(LOX3)_protein—tropical corn line SL57—amino acid sequence of Lox3 from tropical corn line SL57 (<i>Zea mays</i>)
15	ZmLox3_PH207_genomic—genomic sequence of Lox3 from <i>Zea mays</i> line PH207
16	ZmLox3_PH207_CDS—coding sequence of Lox3 from <i>Zea mays</i> line PH207
17	ZmLox3_PH207_protein—amino acid sequence of Lox3 from <i>Zea mays</i> line PH207
18	Primer for Lox3 qRT-PCR: S3460—Corn—Primer for expression analysis of Lox3 in <i>Zea mays</i>
19	Primer for Lox3 qRT-PCR: S3461—Corn—Primer for expression analysis of Lox3 in <i>Zea mays</i>
20	Primer for Ef-Tu qRT-PCR: S3428—Primer for expression analysis of the housekeeping gene Elongation factor 1-alpha
21	Primer for Ef-Tu qRT-PCR: S3429—Primer for expression analysis of the housekeeping gene Elongation factor 1-alpha
22	Elongation factor 1-alpha (EF1) gene from A188—coding sequence of Elongation factor 1-alpha (EF1) gene from <i>Zea mays</i> line A188
23	Elongation factor 1-alpha (EF1) protein from A188—amino acid sequence of Elongation factor 1-alpha (EF1) gene from <i>Zea mays</i> line A188

-continued

SEQ ID NO.	Description
24	ZmPLT5 PRT—amino acid sequence of PLETHORA transcription factor (PLT) from <i>Zea mays</i>
25	ZmPLT5 CDS—coding sequence of PLETHORA transcription factor (PLT) from <i>Zea mays</i>
26	RBP8 PRT—amino acid sequence of regeneration booster protein
27	RBP8 CDS—coding sequence of regeneration booster protein
28	LbCpf1-RR PRT—amino acid sequence of Cpf1 from <i>Lachnospiraceae bacterium</i> RR
29	LbCpf1-RR CDS—coding sequence of Cpf1 from <i>Lachnospiraceae bacterium</i> RR
30	tdTomato PRT—amino acid sequence of fluorescent protein tdTomato
31	tdTomato CDS—coding sequence of fluorescent protein tdTomato
32	Protospacer m7GEP336 DNA—DNA sequence of protospacer m7GEP336
33	Protospacer m7GEP337 DNA—DNA sequence of protospacer m7GEP337
34	Protospacer m7GEP338 DNA—DNA sequence of protospacer m7GEP338
35	Protospacer m7GEP339 DNA—DNA sequence of protospacer m7GEP339
36	Zm00008a004913_pro.pro—amino acid sequence of Lox3 from <i>Zea mays</i> mutant Zm00008a004913
37	WVP18-09358-016_pro—amino acid sequence of Lox3 from <i>Zea mays</i> mutant WVP18-09358-016
38	WVP18-09344-014_pro.pro—amino acid sequence of Lox3 from <i>Zea mays</i> mutant WVP18-09344-014
39	WVP18-09309-014_339-009_pro.pro—amino acid sequence of Lox3 from <i>Zea mays</i> mutant WVP18-09309-014
40	WVP18-09307-014_pro.pro—amino acid sequence of Lox3 from <i>Zea mays</i> mutant WVP18-09307-014
41	Zm00008a004913_CDS—coding sequence of Lox3 from <i>Zea mays</i> mutant Zm00008a004913
42	WVP18-09358-016_CDS.seq—coding sequence of Lox3 from <i>Zea mays</i> mutant WVP18-09358-016
43	WVP18-09344-014_CDS.seq—coding sequence of Lox3 from <i>Zea mays</i> mutant WVP18-09344-014
44	WVP18-09309-014_339-009_CDS.seq—coding sequence of Lox3 from <i>Zea mays</i> mutant WVP18-09309-014
45	WVP18-09307-014_CDS.seq—coding sequence of Lox3 from <i>Zea mays</i> mutant WVP18-09307-014
46	crRNA1—DNA sequencing coding for crRNA1 for targeting the OSR lox3 gene
47	crRNA2.1—DNA sequencing coding for crRNA2.1 for targeting the OSR lox3 gene
48	crRNA2.2—DNA sequencing coding for crRNA2.2 for targeting the OSR lox3 gene
49	crRNA3—DNA sequencing coding for crRNA3 for targeting the OSR lox3 gene
50	pZFNptII-LbCpf1-tDT-lox3_TTTV—plasmid sequence of pZFNptII-LbCpf1-tDT-lox3_TTTV
51	cuaxxxxxxf02x—forward primer cuaxxxxxxf02x for analyzing the presence of the transgene
52	cuaxxxxxxr01x—reverse primer cuaxxxxxxr01x for analyzing the presence of the transgene
53	nptIIxxx01—forward primer nptIIxxx01 for analyzing the presence of the transgene
54	nptIIxxr01—reverse primer nptIIxxr01 for analyzing the presence of the transgene
55	tDTxxxf04—forward primer tDTxxxf04 for analyzing the presence of the transgene
56	tDTxxxr01—reverse primer tDTxxxr01 for analyzing the presence of the transgene
57	CruaxxxMGB—probe CruaxxxMGB for analyzing the presence of the transgene
58	nptIIxxMGB—probe nptIIxxMGB for analyzing the presence of the transgene
59	tDTxxMGB—probe tDTxxMGB for analyzing the presence of the transgene
60	Fwmfor nuclease—forward primer LbCpf1 for nuclease test
61	LbCpf1-Rv—reverse primer LbCpf1 for nuclease test DNA-LbCpf1
62	Probe—probe for nuclease test
63	IR106_lox3_F1—primer IR106_lox3_F1 for editing profile analysis
64	IR106_lox3_F2—primer IR106_lox3_F2 for editing profile analysis
65	IR106_lox3_F3—primer IR106_lox3_F3 for editing profile analysis
66	IR106_lox3_F4—primer IR106_lox3_F4 for editing profile analysis
67	IR106_lox3_R1—primer IR106_lox3_R1 for editing profile analysis
68	IR106_lox3_R2—primer IR106_lox3_R2 for editing profile analysis

SEQ ID NO.	Description
69	IR106_lox3_R3—primer IR106_lox3_R3 for editing profile analysis
70	IR106_lox3_R4—primer IR106_lox3_R4 for editing profile analysis
71	946R—primer 946R for amplicon sequencing
72	1145F—primer 1145F for amplicon sequencing
73	346F—primer 346F for amplicon sequencing
74	1948R—primer 1948R for amplicon sequencing
75	OSR Lox3 genomic DNA BnaC08g37760D_At_ortholog:AT1G17420—genomic DNA sequence of Lox3 from oil seed rape (<i>Brassica napus</i>)
76	OSR Lox3 genomic DNA BnaA08g23120D_At_ortholog:AT1G17420—genomic DNA sequence of Lox3 from oil seed rape (<i>Brassica napus</i>)
77	OSR Lox3 genomic DNA BnaA09g45010D_At_ortholog:AT1G17420—genomic DNA sequence of Lox3 from oil seed rape (<i>Brassica napus</i>)
78	OSR Lox3 genomic DNA BnaC08g48320D_At_ortholog:AT1G17420—genomic DNA sequence of Lox3 from oil seed rape (<i>Brassica napus</i>)
79	OSR Lox3 PRT BnaC08g37760D_At_ortholog:AT1G17420—amino acid sequence of Lox3 from oil seed rape (<i>Brassica napus</i>)
80	OSR Lox3 PRT BnaA08g23120D_At_ortholog:AT1G17420—amino acid sequence of Lox3 from oil seed rape (<i>Brassica napus</i>)
81	OSR Lox3 PRT BnaA09g45010D_At_ortholog:AT1G17420—amino acid sequence of Lox3 from oil seed rape (<i>Brassica napus</i>)
82	OSR Lox3 PRT BnaC08g48320D_At_ortholog:AT1G17420—amino acid sequence of Lox3 from oil seed rape (<i>Brassica napus</i>)
83	OSR Lox3 CDS BnaC08g37760D_At_ortholog:AT1G17420—coding sequence of Lox3 from oil seed rape (<i>Brassica napus</i>)
84	OSR Lox3 CDS BnaA09g45010D_At_ortholog:AT1G17420—coding sequence of Lox3 from oil seed rape (<i>Brassica napus</i>)
85	OSR Lox3 CDS BnaA09g45010D_At_ortholog:AT1G17420—coding sequence of Lox3 from oil seed rape (<i>Brassica napus</i>)
86	OSR Lox3 CDS BnaC08g48320D_At_ortholog:AT1G17420—coding sequence of Lox3 from oil seed rape (<i>Brassica napus</i>)
87	Zm00001eb054040_B73AGP05_T002_CDS—coding sequence of an alternative transcript of Lox3 from <i>Zea mays</i>
88	Zm00001eb054040_B73AGP05_T002_genomic—genomic sequence coding for an alternative transcript of Lox3 from <i>Zea mays</i>
89	Zm00001eb054040_B73AGP05_T002_pro—amino acid sequence of an alternative splice variant of Lox3 from <i>Zea mays</i>

Definitions

A plant exhibits “resistance” or “tolerance” to an insect or fungal pathogen, when symptoms of infestation are reduced or not observed at all when the plant is exposed to the pathogen under conditions allowing infestation. Indirectly, resistance and tolerance can also be observed by looking at the performance of the pathogen on resistant or tolerant plants compared to plants with no resistance or tolerance. For example, the amount, size, or weight of pathogen can be observed and compared. Larval weight of insects may for example and not limitation be reduced in resistant or tolerant plants compared to a control. Typically, a resistant or tolerant plant shows reduced yield loss compared to a non-resistant plant infested with the same pathogen under the same conditions. Ideally, in a resistant plant, the yield loss caused by infestation is completely compensated, meaning that the plant produces as much yield as a plant, which was not exposed to the pathogen at all but grown under the same conditions.

“Insect pathogens” or “fungal pathogens” are any insects or fungi, which invade and infest crop plants and cause damage and ultimately yield loss in crop plants. For each crop plant, a number of insect or fungal pathogens are known to the skilled person. In some cases, fungal diseases are favoured by the mechanical damage caused by invading insects so that damage can be exaggerated. Therefore, resistance or tolerance to insect pests can also provide a degree of fungal resistance.

The terms “RNA interference” or “RNAi” or “RNA silencing” or “gene silencing” as used herein interchangeably refer to a gene down-regulation (or knock-down)

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mechanism meanwhile demonstrated to exist in all eukaryotes. The mechanism was first recognized in plants where it was called “post-transcriptional gene silencing” or “PTGS”. In RNAi, small RNAs function to guide specific effector proteins to a target nucleotide sequence by complementary base pairing resulting in degradation of the target. A “gene silencing construct” or “RNAi construct” usually comprises so called “sense” and “antisense” sequences. The sense and antisense sequences are complementary sequences, which are present in reverse orientation in a nucleic acid sequence. If a nucleic acid construct comprises a sense and a corresponding antisense sequence, the two complementary sequences form an RNA double strand upon transcription, which results in an “RNA hairpin”. In an RNA hairpin, sense sequences and corresponding antisense sequences, together form a double strand and are separated by an “intervening intron loop sequence” forming the loop of the hairpin structure. An “RNAi construct” may also comprise more than one sense and antisense pair and form several loops.

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A “genome editing system” refers to a combination of a site-specific nuclease or site-specific nickase or a functional active fragment or variant thereof together with the cognate guide RNA (or pegRNA or crRNA) guiding the relevant CRISPR nuclease to its target site to be cleaved. A “site-specific nuclease” refers to a nuclease or an active fragment thereof, which is capable to specifically recognize and cleave DNA at a certain target site. Such nucleases typically produce a double strand break (DSB), which is then repaired by nonhomologous end-joining (NHEJ) or homologous recombination (HR). The nucleases include zinc-finger nucleases, transcription activator-like effector nucleases, engineered homing endonucleases, recombinases, transpos-

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ases and meganucleases and CRISPR nucleases and/or any combination, variant or active fragment thereof. The genome editing system may be a CRISPR/Cas system including CRISPR/Cas9 systems, CRISPR/Cas13 systems, CRISPR/Cpf1 (CRISPR/Cas12a) systems, CRISPR/C2C2 systems, CRISPR/CasX systems, CRISPR/CasY systems, CRISPR/Cmr systems, CRISPR/Csm systems, CRISPR/MAD2 systems, CRISPR/MAD7 systems, CRISPR/CasZ systems, or catalytically active fragment thereof. The “guide molecule”, in particular the “guide RNA” (gRNA) may be a trans-activating CRISPR RNA (tracrRNA) plus a synthetic CRISPR RNA (crRNA) or a single guide RNA (sgRNA), which comprises the sequence information targeting the genomic sequence for cleavage by the nuclease.

A “CRISPR nuclease”, as used herein, is any nuclease which has been identified in a naturally occurring CRISPR system, which has subsequently been isolated from its natural context, and which preferably has been modified or combined into a recombinant construct of interest to be suitable as tool for targeted genome engineering. Any CRISPR nuclease can be used and optionally reprogrammed or additionally mutated to be suitable for the various embodiments according to the present invention as long as the original wild-type CRISPR nuclease provides for DNA recognition, i.e., binding properties. Said DNA recognition can be PAM (protospacer adjacent motif) dependent. CRISPR nucleases having optimized and engineered PAM recognition patterns can be used and created for a specific application. The expansion of the PAM recognition code can be suitable to target site-specific effector complexes to a target site of interest, independent of the original PAM specificity of the wild-type CRISPR-based nuclease. Cpf1 variants can comprise at least one of a S542R, K548V, N552R, or K607R mutation, preferably mutation S542R/K607R or S542R/K548V/N552R in AsCpf1 from *Acidaminococcus*. Furthermore, modified Cas or Cpf1 variants or any other modified CRISPR effector variants, e.g., Cas9 variants, can be used according to the methods of the present invention as part of a base editing complex, e.g., BE3, VQR-BE3, EQR-BE3, VRER-BE3, SaBE3, SaKKH-BE3 (see Kim et al., Nat. Biotech., 2017, doi: 10.1038/nbt.3803). Therefore, according to the present invention, artificially modified CRISPR nucleases are envisaged, which might indeed not be any “nucleases” in the sense of double-strand cleaving enzymes, but which are nickases or nuclease-dead variants, which still have inherent DNA recognition and thus binding ability. Suitable Cpf1-based effectors for use in the methods of the present invention are derived from *Lachnospiraceae* bacterium (LbCpf1, e.g., NCBI Reference Sequence: WP_051666128.1), or from *Francisella tularensis* (FnCpf1, e.g., UniProtKB/Swiss-Prot: A0Q7Q2.1). Variants of Cpf1 are known (cf. Gao et al., BioRxiv, dx.doi.org/10.1101/091611). Variants of AsCpf1 with the mutations S542R/K607R and S542R/K548V/N552R that can cleave target sites with TYCV/CCCC and TATV PAMs, respectively, with enhanced activities in vitro and in vivo are thus envisaged as site-specific effectors according to the present invention. Genome-wide assessment of off-target activity indicated that these variants retain a high level of DNA targeting specificity, which can be further improved by introducing mutations in non-PAM-interacting domains. Together, these variants increase the targeting range of AsCpf1 to one cleavage site for every ~8.7 base pairs (bp) in non-repetitive regions of the human genome, providing a useful addition to the CRISPR/Cas genome engineering toolbox (see Gao et al., supra).

The terms “SDN-1”, “SDN-2”, and “SDN-3” as used herein are abbreviations for the platform technique “site-directed nuclease” 1, 2, or 3, respectively, as caused by any site directed nuclease of interest, including, for example, Meganucleases, Zinc-Finger Nucleases (ZFNs), Transcription Activator Like Effector Nucleases (TALENs), and CRISPR nucleases. SDN-1 produces a double-stranded or single-stranded break in the genome of a plant without the addition of foreign DNA. For SDN-2 and SDN-3, an exogenous nucleotide template is provided to the cell during the gene editing. For SDN-2, however, no recombinant foreign DNA is inserted into the genome of a target cell, but the endogenous repair process copies, for example, a mutation as present in the template to induce a (point) mutation. In contrast, SDN-3 mechanism uses the introduced template during repair of the DNA break so that genetic material is introduced into the genomic material.

A “genome modification” in the context of the present invention refers to any change of a (nucleic acid) sequence that results in at least one difference in the (nucleic acid) sequence distinguishing it from the original sequence. In particular, a modification can be achieved by insertion or addition of one or more nucleotide(s), or substitution or deletion of one or more nucleotide(s) of the original sequence or any combination of these.

A “knock-down” of a gene refers to an experimental technique by which the expression of the gene is reduced. A reduced expression can e.g., be achieved by gene silencing. A “knock-out”, on the other hand, leads to an abolished expression, i.e., the gene is not expressed at all. This can for example and not limitation be achieved by replacing or interrupting the sequence of the target gene by genome editing.

A nucleic acid molecule or gene that is “endogenous” to a cell or organism refers to a nucleic acid molecule that naturally occurs in the genome of this cell or organism. On the other hand, a nucleic acid molecule that is “exogenous” to a cell or organism refers to a nucleic acid molecule that does not naturally occur in this cell or organism but has been introduced by a transgenic event.

“Regenerating” a plant, tissue, organ or seed is done by culturing a modified or edited cell in a way that may include steps of de-differentiation and differentiation to obtain specialized tissue or a whole plant, which carries the modification or edit, preferably in every cell. Techniques for regeneration of a plant are well known to the skilled person.

A “nucleic acid construct”, “construct” or “expression construct” refers to a nucleic acid molecule encoding or comprising one or more genetic elements, which upon introduction into a target cell can be transcribed and/or translated into a functional form, e.g., RNA(s) or polypeptide(s) or protein(s). A nucleic acid construct may also comprise regulatory sequences such as promoter and terminator sequences facilitating expression of the genetic element(s) as well as spacers and introns. The genetic elements of the present invention can also be encoded on a set of constructs, which constructs can be introduced into a cell simultaneously or consecutively.

A (nucleic acid) construct of the present invention “targets” a genomic sequence or a gene when it contains sequence information, which allows recognition of the genomic sequence or gene and can thus interfere with the sequence, e.g., by site-specific cleavage or silencing. The targeting can be affected either by direct interaction with the genomic sequence itself or by interaction with the transcript of the genomic sequence. For example, if the nucleic acid construct of the present invention comprises or encodes a

sense and a corresponding antisense sequence targeting a genomic sequence, an RNA silencing or RNA interference (RNAi) mechanism is activated upon transcription of the construct, which leads to the destruction of the transcript of the genomic target sequence and thus suppresses expression of the target. In another case, the nucleic acid construct may encode a site-specific nuclease and a guide RNA, which results in cleavage of the target sequence.

“Transformation” or “Transfection” of a plant cell with a construct or set of constructs refers to any established technique to introduce nucleic acid molecules into a cell, such as biolistic approaches (e.g., particle bombardment), microinjection, permeabilising the cell membrane with various treatments such as electroporation or PEG treatment or *Agrobacterium tumefaciens* mediated transformation. Generally, incorporating (a) nucleic acid construct(s), for example by way of transformation, may be accomplished with techniques that are basically known to the person skilled in the art. For example, the nucleic acid construct can be incorporated into the plant cells by infecting a plant tissue or a plant cell with *Agrobacterium tumefaciens* containing the nucleic acid sequence to be transferred in its plasmid that can be integrated into the plant genome. Incorporating by means of a biolistic transfer is another option, wherein the nucleic acid construct to be incorporated into the plant cell is applied to gold particles or tungsten particles, which are then shot into the cells at a high speed. Another option known to the person skilled in the art for incorporating a nucleic acid construct into a plant cell, is the protoplast transformation, wherein either polyethylene glycol is added to the protoplasts in the presence of the nucleic acid molecules to be incorporated, or the protoplasts are exposed to a short current impulse, so that the protoplast membrane transiently becomes permeable for the nucleic acid construct(s).

The term “vector” refers to an element used for introducing a nucleic acid construct or set of nucleic acid constructs into a cellular system. The vector may be a plasmid or plasmid vector, cosmid, artificial yeast artificial chromosomes (YAC), bacterial artificial chromosome (BAC) or P1 artificial chromosomes (PACs), phagemid, bacterial phage-based vector, an isolated single-stranded or double-stranded nucleic acid sequence, comprising DNA and RNA sequences in linear or circular form, or a mixture thereof, for introduction or transformation into a plant, plant cell, tissue, organ or material according to the present disclosure.

The terms “plant” or “plant cell” or “part of a plant” as used herein refer to a plant organism, a plant organ, differentiated and undifferentiated plant tissues, plant cells, seeds, and derivatives and progeny thereof. Plant cells include without limitation, for example, cells from seeds, from mature and immature cells or organs, including embryos, meristematic tissues, seedlings, callus tissues in different differentiation states, leaves, flowers, roots, shoots, male or female gametophytes, sporophytes, pollen, pollen tubes and microspores and protoplasts.

“Mutagenesis” refers to a technique, by which modifications or mutations are introduced into a nucleic acid sequence in a random or non-site-specific way. For example, mutations can be induced by certain chemicals such as EMS (ethyl methanesulfonate) or ENU (N-ethyl-N-nitrosourea) or physically, e.g., by irradiation with UV or gamma rays. “Site-specific modifications”, on the other hand, rely on the action of site-specific effectors such as nucleases, nickases, recombinases, transposases, base editors. These tools recognize a certain target sequence and allow to introduce a modification at a specific location within the target sequence.

“TILLING” (Targeting Induced Local Lesions in Genomes) is a process, which allows to identify mutations in a specific gene after an (unspecific) mutagenesis has been performed. Mutagenesis may for example and not limitation be performed using a chemical mutagen such as EMS. Then, a sensitive DNA screening technique is used to identify single base mutations. Methods for performing TILLING are known to the skilled person.

Whenever the present disclosure relates to the percentage of identity of nucleic acid or amino acid sequences to each other these values define those values as obtained by using the EMBOSS Water Pairwise Sequence Alignments (nucleotide) programme (www.ebi.ac.uk/Tools/psa/emboss_water/nucleotide.html) nucleic acids or the EM-BOSS Water Pairwise Sequence Alignments (protein) programme (www.ebi.ac.uk/Tools/psa/emboss_water/) for amino acid sequences. Alignments or sequence comparisons as used herein refer to an alignment over the whole length of two sequences compared to each other. Those tools provided by the European Molecular Biology Laboratory (EMBL) European Bioinformatics Institute (EBI) for local sequence alignments use a modified Smith-Waterman algorithm (see www.ebi.ac.uk/Tools/psa/and Smith, T. F. & Waterman, M. S. “Identification of common molecular subsequences” *Journal of Molecular Biology*, 1981 147 (1): 195-197). When conducting an alignment, the default parameters defined by the EMBL-EBI are used. Those parameters are (i) for amino acid sequences: Matrix=BLOSUM62, gap open penalty=10 and gap extend penalty=0.5 or (ii) for nucleic acid sequences: Matrix=DNAfull, gap open penalty=10 and gap extend penalty=0.5. The skilled person is well aware of the fact that, for example, a sequence encoding a protein can be “codon-optimized” if the respective sequence is to be used in another organism in comparison to the original organism a molecule originates from.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 shows the binary vector used for RNAi mediated downregulation of ZmLx3. An RNAi hairpin construct directed against the ZmLx3 gene was inserted into a binary vector for corn transformation. The T-DNA of the binary vector was transformed into the corn genotype A188. The intron between the d35S promoter and the RNAi target region is the first intron of the maize polyubiquitin gene (ZmUbi; Christensen, A. H., Quail, P. H. Ubiquitin promoter-based vectors for high-level expression of selectable and/or screenable marker genes in monocotyledonous plants. *Transgenic Research* 5, 213-218 (1996). <https://doi.org/10.1007/BF01969712>).

FIG. 2 shows the reduction of ZmLx3 expression in transgenic ZmLx3_RNAi lines. Leaf samples of homozygous GM (genetically modified) plants were analyzed by qRT-PCR for expression of ZmLx3 gene and compared with the gene expression of non-transgenic, azygous plants. The lines FDC003-T002, FDC003-T005, FDC003-T010, FDC003-T021 and FDC003-T023 show a strong reduction of Lox3 expression. The Lox3 expression was normalized against the expression of the house-keeping gene EF1. Mean of 3 and 5 plants are shown. Azygous lines n=3 plants. GM lines n=5 plants.

FIG. 3 shows enhanced insect resistance of ZmLx3_RNAi lines. Fall armyworm larva weight (in mg) 10 days after fed on A188 corn leaves (WT (A)) and fed on

three different ZmLox3_RNAi lines (B1: FDC003-T002, B2: FDC003-T011 and B3: FDC003-T023). Data are from 3 independent experiments pulled together (mean $\pm$ SE). Different letters represent groups that are significantly different (ANOVA and Tukey's honest significant difference post hoc, n=75; P<0.05).

FIG. 4 shows enhanced fungal resistance of ZmLox3_RNAi line FDC0003-T023 Left: Disease scores of *Gibberella* ear rot infected corn cobs of hybrids from FDC0003-T029, FDC0003-T029 azygous and FDC0003-T023 tester line RP5G. Infected area of 40-50 heads of each hybrid was measured 40 days after infection of corn plants with *Fusarium graminearum* in the greenhouse. Right: Lox3 expression of leaves from the hybrids FDC003-T029, FDC0003-T029 azygous and FDC0003-T023 with tester line RP5G at the end of the fungal assay. The Lox3 expression was determined with qRT-PCR and is reduced in the hybrid FDC0003-T023, which shows enhanced fungal resistance and insect resistance.

FIG. 5 shows the binary vector used for generation of Lox3 SDN-1 knock-down in tropical corn by genome editing.

FIG. 6 shows the scoring scheme of the clubroot test performed in Example 11.

FIG. 7 shows infection rates of *F. graminearum* (FG) and *F. verticilloides* (FV) in homozygous mutant class for PH207m034b and PH207m034h.

DETAILED DESCRIPTION

The present invention provides means and methods to confer resistance or tolerance to several relevant insect pathogens and, optionally—in addition—fungal pathogens, in plants. In particular, highly damaging diseases in corn (Zm) and oil seed rape (OSR) are addressed in the following disclosure. In some cases, fungal diseases are favoured by the mechanical damage caused by invading insects so that damage can be exaggerated. It was thus shown that resistance or tolerance to insect pests can also provide a degree of fungal resistance or tolerance. Insect infestation in crop plants is often managed using insecticide treatments and Bt traits, i.e., genetically modified (GM) plants, which produce insecticidal toxins. The present invention provides a novel approach to manage insect pests using a non-GM trait. Through a product development by TILLING, a non-GM trait increasing insect tolerance could be achieved. This can be used in addition to pesticides to reduce their use and/or to protect the crop against insects when pesticides cannot be applied (e.g., when it rains). Additionally, a knock-down using gene silencing and a knock-out using genome editing techniques could be achieved.

In a first aspect, the present invention relates to a method for conferring or increasing resistance or tolerance to an insect and, optionally a fungal pathogen to/in a plant comprising the steps of:

- (i) providing at least one plant cell;
- (ii) introducing into the at least one plant cell at least one gene silencing construct, at least one genome editing system or a genome modification, which leads to a targeted knock-down or a knock-out of a Lox3 gene endogenous to the plant;
- (iii) obtaining at least one modified plant cell having reduced or abolished expression of the Lox3 gene; and
- (iv) obtaining at least one plant cell, tissue, organ, plant or seed having reduced or abolished expression of the

Lox3 gene, optionally after an additional step of regenerating the plant tissue, organ, plant or seed from the at least one modified cell.

It was established in the context of the present invention that reduced or abolished expression of the endogenous Lox3 gene in plants confers or increases resistance to insect pests and can at the same time also provide a degree of resistance to fungal pathogens. To reduce or abolish expression of the endogenous Lox3 gene, three different approaches may be taken: Lox3 expression may be reduced by gene silencing using an RNAi hairpin construct, which contains as sense and antisense sequences sections of the endogenous Lox3 gene. Alternatively, a genome editing system can be used, which introduces a double strand break (DSB) at or in the endogenous Lox3 sequence resulting in a disruption or (partial) replacement of the Lox3 locus. Finally, it is also possible to introduce mutations into the endogenous Lox3 by random mutagenesis and identify such mutations, which provide the desired knock-down or knock-out effect.

In a preferred embodiment, the method described above is for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melancthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* species, *Cercospora zeina* and *Cercospora zeae-maydis* to/in maize (*Zea mays*).

The above-mentioned pathogens cause a lot of damage to corn and reduce yields significantly. Using the approaches described herein, resistance or tolerance to these pathogens can be conferred or increased. Another important crop plant, oil seed rape, may also suffer from certain insect or fungal pests.

In another preferred embodiment, the method described above therefore is for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of green peach aphid (*Myzus persicae*), diamond-back moth (*Plutella xylostella*), cabbage stem flea beetle (*Psylliodes chrysocephala*), crucifer flea beetle (*Phyllotreta cruciferae*), striped flea beetle (*Phyllotreta striolata*), hop flea beetle (*Psylliodes punctulata*), rape stem weevil (*Ceutorhynchus picitarsis*) and cabbage stem weevil (*Ceutorhynchus quadridens*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Phoma lingam* and *Plasmodiophora brassicae* to/in oilseed rape (*Brassica napus*).

The Lox3 gene in corn (*Zea mays*) is represented by the genomic DNA sequence of SEQ ID NO: 6, while the coding sequence is represented by the sequence of SEQ ID NO: 7. For the corn lines A188, a tropical corn line, the line PH207 and an alternative transcript, the genomic sequences and coding sequences are respectively represented by the sequences of SEQ ID NOS: 9 (A188, genomic DNA), 10 (A188, coding sequence), 12 (tropical corn line, genomic DNA), 13 (tropical corn line, coding sequence), 15 (PH207, genomic DNA), 16 (PH207, coding sequence), 87 (alternative transcript, coding sequence) and 88 (alternative transcript, genomic sequence).

In one embodiment of the method described above, the Lox3 gene is therefore represented by a nucleic acid sequence of SEQ ID NO: 6, 7, 9, 10, 12, 13, 15, 16, 87 or 88 or a nucleic acid sequence having a sequence identity of

at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 6, 7, 9, 10, 12, 13, 15, 16, 87 or 88.

The protein (amino acid) sequences of Lox3 in corn (*Zea mays*) translated from the above nucleic acid sequences are represented by the sequences of SEQ ID NOS: 8 (ZmLox3 protein), 11 (A188, ZmLox3 protein), 14 (tropical corn line, ZmLox3 protein), 17 (PH207, ZmLox3 protein) and 89 (alternative transcript, ZmLox3 protein).

In one embodiment of the method described above, the Lox3 gene therefore encodes an amino acid sequence of SEQ ID NO: 8, 11, 14, 17 or 89 an amino acid sequence having a sequence identity of at least 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 8, 11, 14, 17 or 89.

The Lox3 gene in oil seed rape (*Brassica napus*) is represented by the genomic DNA sequences of SEQ ID NOS: 75, 76, 77 and 78 while the coding sequences are represented by the sequences of SEQ ID NOS: 83, 84, 85 and 86.

In one embodiment of the method described above, the Lox3 gene is therefore represented by a nucleic acid sequence of SEQ ID NO: 75, 76, 77, 78, 83, 84, 85 or 86 or a nucleic acid sequence having a sequence identity of at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 75, 76, 77, 78, 83, 84, 85 or 86.

The protein (amino acid) sequences of Lox3 in oil seed rape (*Brassica napus*) translated from the above nucleic acid sequences are represented by the sequences of SEQ ID NOS: 79, 80, 81 and 82.

In one embodiment, the Lox3 gene therefore encodes an amino acid sequence of SEQ ID NO: 79, 80, 81 or 82 or an amino acid sequence having a sequence identity of at least 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 79, 80, 81 or 82.

In a further embodiment of the method according to any of the embodiments described above, in step (ii) a construct is introduced into the at least one plant cell, which targets the Lox3 gene for gene silencing.

RNAi techniques for targeted gene silencing are well known in the art. To this end, an RNAi construct is introduced into a plant cell, which contains sequence information of the genomic target to be silenced in the cell. The construct is preferably introduced in form of a DNA sequence, which is then transcribed into functional RNA in the cell. In particular, the RNAi construct encodes sense and antisense sequences, which represents (a fragment of) the genomic target. The complementary sense and antisense sequences, which are present in reverse orientation in the construct form an RNA double strand upon transcription, which results in an RNA hairpin with an intervening intron loop sequence. The presence of the RNAi construct ultimately results in a reduced expression of the target, i.e., a knock-down.

In one embodiment of the method described above, the construct encodes an RNAi construct comprising a sense and an antisense sequence targeting the Lox3 gene, the RNAi construct forming an RNA hairpin upon transcription.

Preferably, the sense and antisense sequences have a length of 40 to 500 nucleotides, more preferably 100 to 300 nucleotides.

In one embodiment of the method described in any of the embodiment described above, the method is for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* species, *Cercospora zeina* and *Cercospora zeae-maydis* to/in maize (*Zea mays*) and the sense sequence is encoded by a nucleic acid sequence of SEQ ID NO: 1, or a nucleic acid sequence having a sequence identity of at least 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 1.

In another embodiment of the method described in any of the embodiment described above, the method is for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* species, *Cercospora zeina* and *Cercospora zeae-maydis* to/in maize (*Zea mays*) and the antisense sequence is encoded by a nucleic acid sequence of SEQ ID NO: 2, or a nucleic acid sequence having a sequence identity of at least 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 2.

In yet another embodiment of the method described in any of the embodiment described above, the method is for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* species, *Cercospora zeina* and *Cercospora zeae-maydis* to/in maize (*Zea mays*) and the RNA hairpin has an intervening intron loop sequence comprising a nucleic acid sequence of SEQ ID NO: 3, or a nucleic acid sequence having a sequence identity of at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 3.

In one embodiment of the method described in any of the embodiment described above, the method is for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* species, *Cercospora zeina* and *Cercospora zeae-maydis* to/in maize (*Zea mays*) and the construct comprises a nucleic acid sequence of SEQ ID NO: 4, or a nucleic acid sequence having a sequence identity of at least 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%,

89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 4.

The RNAi construct may preferably be introduced on a vector encoding the RNAi hairpin, which is formed upon transcription.

In one embodiment of the method according to any of the embodiments described above, the method is for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* species, *Cercospora zeina* and *Cercospora zeae-maydis* to/in maize (*Zea mays*) and a vector is introduced into the plant cell, which vector comprises or consists of a nucleic acid sequence of SEQ ID NO: 5, or a nucleic acid sequence having a sequence identity of at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 5.

The introduction of the construct into the plant cell may for example and not limitation be achieved by means of transformation or transfection. Besides transformation methods based on biological approaches, like *Agrobacterium* transformation or viral vector mediated plant transformation, methods based on physical delivery methods, like particle bombardment or microinjection, have evolved as prominent techniques for importing genetic material into a plant cell or tissue of interest. Helenius et al. ("Gene delivery into intact plants using the Helios™ Gene Gun", Plant Molecular Biology Reporter, 2000, 18 (3): 287-288) discloses a particle bombardment as physical method for transferring material into a plant cell. Currently, there are a variety of plant transformation methods to introduce genetic material in the form of a genetic construct into a plant cell of interest, comprising biological and physical means known to the skilled person on the field of plant biotechnology, which can be applied. A common biological means is transformation with *Agrobacterium* spp. which has been used for decades for a variety of different plant materials. Viral vector mediated plant transformation represents a further strategy for introducing genetic material into a cell of interest. Physical means finding application in plant biology are particle bombardment, also named biolistic transfection or microparticle-mediated gene transfer, which refers to a physical delivery method for transferring a coated microparticle or nanoparticle comprising a nucleic acid or a genetic construct of interest into a target cell or tissue. Physical introduction means are suitable to introduce nucleic acids, i.e., RNA and/or DNA, and proteins. Likewise, transformation or transfection methods exist for specifically introducing a nucleic acid or an amino acid construct of interest into a plant cell, including electroporation, microinjection, nanoparticles, and cell-penetrating peptides (CPPs). Furthermore, chemical-based transfection methods exist to introduce genetic constructs and/or nucleic acids and/or proteins, comprising inter alia transfection with calcium phosphate, transfection using liposomes, e.g., cationic liposomes, or transfection with cationic polymers, including DEAD-dextran or polyethylenimine, or combinations thereof. Every delivery method has to be specifically fine-tuned and opti-

mized so that a construct of interest can be introduced into a specific compartment of a target cell of interest in a fully functional and active way.

In one embodiment of the method described above, the construct is introduced into the at least one plant cell by transformation or transfection mediated by biolistic bombardment, *Agrobacterium*-mediated transformation, micro- or nanoparticle delivery, chemical transfection, or a combination thereof.

As already mentioned above, as an alternative to gene silencing, the knock-down or knock-out of the endogenous Lox3 gene to confer or enhance resistance or tolerance can also be effected by genome editing.

In one embodiment of the method described above, in step (ii) at least one genome editing system is introduced into the at least one cell, which targets the Lox3 gene, wherein the at least one genome editing system comprises

- (a) at least one site-specific nuclease or site-specific nickase, and optionally, in case a CRISPR system is used, at least one guide molecule or a sequence encoding the same, and
- (b) optionally, at least one repair template, or a sequence encoding the same.

Genome editing techniques allow to introduce a double strand break at one or more predetermined target site(s), i.e., by or within the endogenous Lox3 thereby disrupting the Lox3 locus and optionally inserting an exogenous sequence or replacing an endogenous sequence. The double strand break is introduced by a site-specific nuclease such as meganucleases, zinc finger nucleases (ZFNs), transcription activator-like effector nucleases (TALENs), or the clustered regularly interspaced short palindromic repeat (CRISPR) nucleases. The nucleases cause double strand breaks (DSBs) at specific cleaving sites, which are repaired by nonhomologous end-joining (NHEJ) or homologous recombination (HR). The use of a repair template guides the cellular repair process so that the results of the repair are error-free and predictable. A repair template preferably comprises symmetric or asymmetric homology arms, which are complementary to the sequences flanking a double strand break and therefore allow to insert a sequence or close the break in a controlled manner.

In one embodiment of the method described above, the at least one genome editing system is selected from a CRISPR/Cas system, preferably from a CRISPR/MAD7 system, a CRISPR/Cpf1 (CRISPR/Cas12a) system, a CRISPR/MAD2 system, a CRISPR/Cas9 system, a CRISPR/CasX system, a CRISPR/CasY system, a CRISPR/Cas13 system, or a CRISPR/Csm system, or the at least one genome editing system is selected from a zinc finger nuclease system, or a transcription activator-like nuclease system, or a meganuclease system, or any combination, variant, or an active fragment thereof.

A preferred Cpf1 (Cas12a) nuclease to be used in the method of the present invention is encoded by the nucleic acid sequence of SEQ ID NO: 29, or a nucleic acid sequence having a sequence identity of at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 29.

The preferred Cpf1 (Cas12a) nuclease is further represented by the amino acid sequence of SEQ ID NO: 28 or an amino acid sequence having a sequence identity of at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%,

90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 28.

The genome editing system can be introduced by transformation or transfection mediated by biolistic bombardment, *Agrobacterium*-mediated transformation, micro- or nanoparticle delivery, chemical transfection, or a combination thereof as explained in more detail above in the context of the RNAi construct.

A CRISPR system includes the use of a guide RNA (gRNA) or CRISPR (crRNA), which guides the CRISPR nuclease to the target site by sequence recognition.

In one embodiment of the method described in any of the embodiments above, the method is for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* species, *Cercospora zeina* and *Cercospora zea-maydis* to/in maize (*Zea mays*), in particular a tropical maize line, and the at least one genome editing system comprises a protospacer having a nucleic acid sequence of SEQ ID NO: 32, 33, 34 or 35, or a nucleic acid sequence having 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of any of SEQ ID NOs: 32, 33, 34 or 35.

In one embodiment of the method described in any of the embodiments above, the method is for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of green peach aphid (*Myzus persicae*), diamondback moth (*Plutella xylostella*), cabbage stem flea beetle (*Psylliodes chrysocephala*), crucifer flea beetle (*Phyllotreta cruciferae*), striped flea beetle (*Phyllotreta striolata*), hop flea beetle (*Psylliodes punctulata*), rape stem weevil (*Ceutorhynchus piciparsis*) and cabbage stem weevil (*Ceutorhynchus quadridens*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Phoma lingam* and *Plasmidiophora brassicae* to/in oilseed rape (*Brassica napus*), and the at least one genome editing system comprises a crRNA encoded by a nucleic acid sequence of any of SEQ ID NOs: 46 to 49, or a nucleic acid sequence having a sequence identity of at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of any of SEQ ID NOs: 46 to 49.

In one embodiment of this method, the genome editing system is encoded by a plasmid of the nucleic acid sequence of SEQ ID NO: 50 or a nucleic acid sequence having a sequence identity of at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 50.

A further alternative to introduce a modification, which results in reduced or abolished expression of endogenous Lox3, is by mutagenesis. From a range of mutants, the ones with the desired knock-down or knock-out can then be identified, e.g., by TILLING.

In one embodiment of the method described above, in step (ii) a mutagenesis is performed on a single or on a plurality of cell(s) by applying chemicals or radiation.

Preferably, an alkylating agent, in particular ethyl methanesulfonate is applied to the single or the plurality of cell(s) to induce mutagenesis.

In one embodiment of the method described above, one or more mutations in the Lox3 gene are inserted and identified by TILLING in step (ii).

In another embodiment of the method described above, one or more cell(s) with knock-down or knock-out mutations in the Lox3 gene are selected in step (ii).

In one embodiment of the method described above, the maize mutant selected in step (ii) comprises a Lox3 with an amino acid sequence selected from the sequences of SEQ ID NOs: 36 to 40 or an amino acid sequence having a sequence identity of at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of any of SEQ ID NOs: 36 to 40.

In another embodiment of the method described above, the maize mutant selected in step (ii) comprises a Lox3 encoded by a nucleic acid sequence selected from the sequences of SEQ ID NOs: 41 to 45 or a nucleic acid sequence having a sequence identity of at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of any of SEQ ID NOs: 41 to 45.

In another aspect, the present invention relates to a maize cell, maize tissue, maize organ, maize plant or maize seed obtained or obtainable by a method according to any of the embodiments described above.

In a further aspect, the present invention relates to an oilseed rape cell, oilseed rape tissue, oilseed rape organ, oilseed rape plant or oilseed rape seed obtained or obtainable by a method according to any of the embodiments described above.

The present invention also provides expression constructs for specifically targeting the Lox3 gene in maize and in oil seed rape for gene silencing. To obtain such a construct, sequences from the respective endogenous Lox3 gene are used as sense and antisense sequences, which form an RNA hairpin upon transcription in the cell.

In one aspect, the present invention relates to an expression construct, which targets the Lox3 gene in maize for gene silencing, wherein the construct encodes an RNAi construct comprising a sense and an antisense sequence targeting the Lox3 gene endogenous to a maize plant, which RNAi construct forms an RNA hairpin upon transcription.

In another aspect, the present invention relates to an expression construct, which targets the Lox3 gene in oilseed rape for gene silencing, wherein the construct encodes an RNAi construct comprising a sense and an antisense sequence targeting the Lox3 gene endogenous to an oilseed rape plant, which RNAi construct forms an RNA hairpin upon transcription.

Preferably, the sense and antisense sequences have a length of 40 to 500 nucleotides, more preferably 100 to 300 nucleotides.

In one embodiment of the expression construct described above, the sense sequence is encoded by a nucleic acid sequence of SEQ ID NO: 1, or a nucleic acid sequence having a sequence identity of at least 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 1.

In another embodiment of the expression construct described above, the antisense sequence is encoded by a nucleic acid sequence of SEQ ID NO: 2, or a nucleic acid

sequence having a sequence identity of at least 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 2.

In yet another embodiment of the expression construct described above, the RNA hairpin has an intervening intron loop sequence comprising a nucleic acid sequence of SEQ ID NO: 3, or a nucleic acid sequence having a sequence identity of at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 3.

In a further embodiment of the expression construct described above, the construct comprises a nucleic acid sequence of SEQ ID NO: 4, or a nucleic acid sequence having a sequence identity of at least 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 4.

In another aspect, the present invention relates to a vector comprising or consisting of a nucleic acid sequence of SEQ ID NO: 5 or a nucleic acid sequence having a sequence identity of at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 5.

In yet another aspect, the present invention relates to an RNAi hairpin construct conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* species, *Cercospora zeina* and *Cercospora zeae-maydis* to/in maize (*Zea mays*), wherein the RNAi hairpin construct comprises a nucleic acid sequence of SEQ ID NO: 4, or a nucleic acid sequence having a sequence identity of at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 4.

The present invention also provides constructs, which target the endogenous Lox3 gene in maize and oilseed rape for genome editing. Such genome editing systems comprise a site-specific nuclease, in case of a CRISPR based system, a guide RNA and, optionally a repair template.

In one aspect the present invention relates to an expression construct encoding a genome editing system, which targets the Lox3 gene in maize, wherein the genome editing system comprises

- (a) at least one site-specific nuclease or site-specific nickase, and optionally, in case a CRISPR system is used, at least one guide molecule or a sequence encoding the same, and
- (b) optionally, at least one repair template, or a sequence encoding the same.

In another aspect, the present invention relates to an expression construct encoding a genome editing system, which targets the Lox3 gene in oilseed rape, wherein the genome editing system comprises

- (a) at least one site-specific nuclease or site-specific nickase, and optionally, in case a CRISPR system is used, at least one guide molecule or a sequence encoding the same, and

- (b) optionally, at least one repair template, or a sequence encoding the same.

In one embodiment of the expression construct described above, the at least one genome editing system is selected from a CRISPR/Cas system, preferably from a CRISPR/MAD7 system, a CRISPR/Cpf1 (CRISPR/Cas12a) system, a CRISPR/MAD2 system, a CRISPR/Cas9 system, a CRISPR/CasX system, a CRISPR/CasY system, a CRISPR/Cas13 system, or a CRISPR/Csm system, or the at least one genome editing system is selected from a zinc finger nuclease system, or a transcription activator-like nuclease system, or a meganuclease system, or any combination, variant, or an active fragment thereof.

In one embodiment of the expression construct described above, the expression construct comprises a crRNA encoded by a nucleic acid sequence of any of SEQ ID NOS: 46 to 49, or a nucleic acid sequence having a sequence identity of at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of any of SEQ ID NO: 46 to 49.

In another embodiment of the expression construct described above, the genome editing system is encoded by a plasmid of the nucleic acid sequence of SEQ ID NO: 50 or a nucleic acid sequence having a sequence identity of at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 50.

In one embodiment of the expression construct described above, the expression construct comprises a protospacer having a nucleic acid sequence of SEQ ID NO: 32, 33, 34 or 35, or a nucleic acid sequence having 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of any of SEQ ID NOS: 32, 33, 34 or 35.

In yet another aspect, the present invention relates to a vector encoding an expression construct according to any of the embodiments described above.

The present invention also provides a maize cell, maize tissue, maize organ, maize plant or maize seed or an oilseed rape cell, oilseed rape tissue, oilseed rape organ, oilseed rape plant or oilseed rape seed comprising an expression construct or a vector according to any of the embodiments described above.

Furthermore, the present invention provides a use of an expression construct according to any of the embodiments described above for conferring or increasing resistance or tolerance to an insect and, optionally a fungal pathogen to/in a plant, in particular for use in a method according to any of the embodiments described above.

In one aspect, the present invention relates to a use of at least one gene silencing construct, at least one genome editing system or a genome modification, which leads to a targeted knock-down or a knock-out of the endogenous Lox3 gene, for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* species, *Cercospora zeina* and *Cercospora zeae-maydis* to/in maize (*Zea mays*).

In another aspect, the present invention relates to a use of at least one gene silencing construct, at least one genome editing system or a genome modification, which leads to a targeted knock-down or a knock-out of the endogenous a Lox3 gene, for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of green peach aphid (*Myzus persicae*), diamond-back moth (*Plutella xylostella*), cabbage stem flea beetle (*Psylliodes chrysocephala*), crucifer flea beetle (*Phyllotreta cruciferae*), striped flea beetle (*Phyllotreta striolata*), hop flea beetle (*Psylliodes punctulata*), rape stem weevil (*Ceutorhynchus picipitarsis*) and cabbage stem weevil (*Ceutorhynchus quadridens*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Phoma lingam* and *Plasmiodiophora brassicae* to/in oilseed rape (*Brassica napus*).

According to one approach described in detail above, an RNAi or gene silencing construct may be used for conferring or increasing resistance or tolerance to an insect and, optionally a fungal pathogen to/in a plant.

In one aspect, the present invention therefore relates to the use of a construct, the construct being or encoding an RNAi construct comprising a sense and an antisense sequence targeting the endogenous Lox3 gene of a maize plant, wherein the RNAi construct forms an RNA hairpin upon transcription, for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* species, *Cercospora zeina* and *Cercospora zeae-maydis* to/in maize (*Zea mays*).

In another aspect, the present invention relates to the use of a construct, the construct being or encoding an RNAi construct comprising a sense and an antisense sequence targeting the endogenous Lox3 gene of an oilseed rape plant, wherein the RNAi construct forms an RNA hairpin upon transcription, for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of green peach aphid (*Myzus persicae*), diamond-back moth (*Plutella xylostella*), cabbage stem flea beetle (*Psylliodes chrysocephala*), crucifer flea beetle (*Phyllotreta cruciferae*), striped flea beetle (*Phyllotreta striolata*), hop flea beetle (*Psylliodes punctulata*), rape stem weevil (*Ceutorhynchus picipitarsis*) and cabbage stem weevil (*Ceutorhynchus quadridens*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Phoma lingam* and *Plasmiodiophora brassicae* to/in oilseed rape (*Brassica napus*).

In one embodiment of the use described above, the sense sequence is encoded by a nucleic acid sequence of SEQ ID NO: 1, or a nucleic acid sequence having a sequence identity of at least 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 1.

In another embodiment of the use according to any of the embodiments described above, the antisense sequence is encoded by a nucleic acid sequence of SEQ ID NO: 2, or a nucleic acid sequence having a sequence identity of at least 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 2.

In yet another embodiment of the use according to any of the embodiments described above, the RNA hairpin has an intervening intron loop sequence comprising a nucleic acid sequence of SEQ ID NO: 3, or a nucleic acid sequence having a sequence identity of at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or at least 99% to the sequence of SEQ ID NO: 3.

In one embodiment of the use described above, the construct comprises a nucleic acid sequence of SEQ ID NO: 4, or a nucleic acid sequence having a sequence identity of at least 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 4.

The present invention also relates to the use of a vector, which vector comprises or consists of a nucleic acid sequence of SEQ ID NO: 5, or a nucleic acid sequence having a sequence identity of at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 5 for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* species, *Cercospora zeina* and *Cercospora zeae-maydis* to/in maize (*Zea mays*).

In another aspect, the present invention also relates to the use of a genome editing system according to any of the embodiments described above.

In one embodiment of the use described above, the genome editing system targets the endogenous Lox3 gene in a maize plant, wherein the genome editing system comprises

- (a) at least one site-specific nuclease or site-specific nickase, and optionally, in case a CRISPR system is used, at least one guide molecule or a sequence encoding the same, and
- (b) optionally, at least one repair template, or a sequence encoding the same for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* species, *Cercospora zeina* and *Cercospora zeae-maydis* to/in maize (*Zea mays*).

In one embodiment of the use described above, the genome editing system comprises a protospacer having a nucleic acid sequence of SEQ ID NO: 32, 33, 34 or 35, or a nucleic acid sequence having 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of any of SEQ ID NOs: 32, 33, 34 or 35.

In another embodiment of the use described above, the genome editing system targets the endogenous Lox3 gene in an oilseed rape plant, wherein the genome editing system comprises

(a) at least one site-specific nuclease or site-specific nickase, and optionally, in case a CRISPR system is used, at least one guide molecule or a sequence encoding the same, and

(b) optionally, at least one repair template, or a sequence encoding the same

for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of green peach aphid (*Myzus persicae*), diamondback moth (*Plutella xylostella*), cabbage stem flea beetle (*Psylliodes chrysocephala*), crucifer flea beetle (*Phyllotreta cruciferae*), striped flea beetle (*Phyllotreta striolata*), hop flea beetle (*Psylliodes punctulata*), rape stem weevil (*Ceutorhynchus pictarsis*) and cabbage stem weevil (*Ceutorhynchus quadridens*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Phoma lingam* and *Plasmodiophora brassicae* to/in oilseed rape (*Brassica napus*).

The present invention also relates to the use of an expression construct encoding a genome editing system, which targets the endogenous Lox3 gene in a maize plant, wherein the genome editing system comprises

(a) at least one site-specific nuclease or site-specific nickase, and optionally, in case a CRISPR system is used, at least one guide molecule or a sequence encoding the same, and

(b) optionally, at least one repair template, or a sequence encoding the same,

wherein the at least one genome editing system is selected from a CRISPR/Cas system, preferably from a CRISPR/MAD7 system, a CRISPR/Cpf1 (CRISPR/Cas12a) system, a CRISPR/MAD2 system, a CRISPR/Cas9 system, a CRISPR/CasX system, a CRISPR/CasY system, a CRISPR/Cas13 system, or a CRISPR/Csm system, or wherein the at least one genome editing system is selected from a zinc finger nuclease system, or a transcription activator-like nuclease system, or a meganuclease system, or any combination, variant, or an active fragment thereof for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* species, *Cercospora zeina* and *Cercospora zeae-maydis* to/in maize (*Zea mays*).

Furthermore, the present invention also relates to a use of an expression construct encoding a genome editing system, which targets the endogenous Lox3 gene in an oilseed rape plant, wherein the genome editing system comprises

(a) at least one site-specific nuclease or site-specific nickase, and optionally, in case a CRISPR system is used, at least one guide molecule or a sequence encoding the same, and

(b) optionally, at least one repair template, or a sequence encoding the same,

wherein the at least one genome editing system is selected from a CRISPR/Cas system, preferably from a CRISPR/MAD7 system, a CRISPR/Cpf1 (CRISPR/Cas12a) system, a CRISPR/MAD2 system, a CRISPR/Cas9 system, a CRISPR/CasX system, a CRISPR/CasY system, a CRISPR/Cas13 system, or a CRISPR/Csm system, or wherein the at least one genome editing system is selected from a zinc finger nuclease system, or a transcription activator-like nuclease system, or a meganuclease system, or any combi-

nation, variant, or an active fragment thereof for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of green peach aphid (*Myzus persicae*), diamondback moth (*Plutella xylostella*), cabbage stem flea beetle (*Psylliodes chrysocephala*), crucifer flea beetle (*Phyllotreta cruciferae*), striped flea beetle (*Phyllotreta striolata*), hop flea beetle (*Psylliodes punctulata*), rape stem weevil (*Ceutorhynchus pictarsis*) and cabbage stem weevil (*Ceutorhynchus quadridens*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Phoma lingam* and *Plasmodiophora brassicae* to/in oilseed rape (*Brassica napus*).

In one embodiment of the use described above, the expression construct comprises a crRNA encoded by a nucleic acid sequence of any of SEQ ID NOS: 46 to 49, or a nucleic acid sequence having a sequence identity of at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of any of SEQ ID NO: 46 to 49.

In another embodiment of the use described above, the genome editing system is encoded by a plasmid of the nucleic acid sequence of SEQ ID NO: 50 or a nucleic acid sequence having a sequence identity of at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 50.

The present invention also relates to the use a vector encoding an genome editing system as defined above for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* species, *Cercospora zeina* and *Cercospora zeae-maydis* to/in maize (*Zea mays*).

Finally, the present invention also relates to the use of a vector encoding an genome editing system as defined above for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of green peach aphid (*Myzus persicae*), diamondback moth (*Plutella xylostella*), cabbage stem flea beetle (*Psylliodes chrysocephala*), crucifer flea beetle (*Phyllotreta cruciferae*), striped flea beetle (*Phyllotreta striolata*), hop flea beetle (*Psylliodes punctulata*), rape stem weevil (*Ceutorhynchus pictarsis*) and cabbage stem weevil (*Ceutorhynchus quadridens*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Phoma lingam* and *Plasmodiophora brassicae* to/in oilseed rape (*Brassica napus*).

Example 1: Generation of RNAi Corn Lines with Lox3 Knock-Down

An RNAi hairpin construct directed against the ZmLox3 gene was constructed (FIG. 1, SEQ ID NOS: 1 to 5) and transformed into corn genotype A188 by *Agrobacterium tumefaciens* mediated transformation. Transgenic TO plants were identified by PCR and transferred into the greenhouse for T1 seed production by selfing. A segregation analysis was performed with T1 plants. Homozygous ZmLox3_RNAi plants and non-transgenic, azygous plants

were selected and T2 seeds were produced. The T2 seeds were used for the insect resistance assays.

Downregulation of ZmLox3 gene (SEQ ID NO: 6-16, 87-89) in the ZmLox3_RNAi lines was shown by qRT-PCR after isolation of RNA from leaves of homozygous GM and non-transgenic, azygous plants (FIG. 2). The expression of ZmLox3 was measured using the primer S3460 (SEQ ID NO: 18) and primer S3461 (SEQ ID NO: 19). Lox3 expression was normalized against the expression of the house-keeping gene Elongation factor 1-alpha (EF1, SEQ ID NO: 22 and 23) using the primer S3428 (SEQ ID NO: 20) and the primer S3429 (SEQ ID NO: 21).

The expression analysis of corn leaves revealed that the lines FDC003-T002, FDC003-T005, FDC003-T010, FDC003-T021 and FDC003-T023 show a strong reduction of Lox3 expression (FIG. 2).

Example 2: Determine Insect Resistance by Measuring Larval Weight of Fall Army Worm Feeding on Lox3 RNAi Lines

The purpose of this experiment was to evaluate if Lox3 gene down-regulation is linked to fall armyworm tolerance in corn. To test this, fall armyworm (*Spodoptera frugiperda*) larval performance was chosen as a measure of antibiosis-based resistance.

Three ZmLox3_RNAi corn lines (FDC003-T002, FDC003-T011 and FDC003-T023) were evaluated using unmodified A188 corn as control. Three independent insect resistance experiments were performed. For each experiment, five biological replicates were used for A188 and for each ZmLox3_RNAi lines. Three newly hatched fall armyworm larva were added into individual Falcon tubes containing corn leaves. Fresh leaves were provided daily and after ten days the larva weight was scored.

All tested lines conferred a significant increase of resistance against fall armyworm (FIG. 3). These data showed that the ZmLox3 gene is a susceptibility gene for fall armyworm and that down-regulation of ZmLox3 expression can improve resistance.

Example 3: Determine Fungal Resistance of Lox3 RNAi Lines

Since stalk and ear rot caused by *Fusarium* spp. are favored by the mechanical damage caused by invading insects, the Lox3 mediated insect resistance may have an effect on fungal resistance.

The insect resistant line FDC003-T023 was crossed with the tester line RP5G and compared with test crosses of FDC0003-T029 and FDC0003-T029 azygous with the tester line RP5G, which did not show LOX3 reduction as determined by qRT-PCR (FIG. 4, left). The FDC003-T023 hybrid showed 13 and 8% less *Fusarium* infection compared to the FDC0003-T029 and FDC0003-T029 azygous hybrids (FIG. 4, right), which revealed that an insect resistant RNAi line also has higher fungal resistance.

Example 4: Generation of Lox3 SDN1 Knock-Down in Tropical Corn by Genome Editing

Lox3 was knocked out in tropical corn genotypes by means of a site-directed nuclease.

1. Tropical corn lines were grown in soil from seed for five weeks and immature tassels were harvested as previously described (WO2021170785).

2. Tassel material was bombarded using the following vectors:

- a. TGCD035 (ZmPLT5, SEQ ID Nos: 24 and 25)
- b. GEZM145 (RBP8, SEQ ID Nos: 26 and 27)
- c. GEZM152 (gRNA, protospacer m7GEP336: GCACGTTCTTGCGCATGAGCA, SEQ ID NO: 32)
- d. GEZM153 (gRNA, protospacer m7GEP337: GCGC-CACCGTCGTTGACAGCA, SEQ ID NO: 33)
- e. GEZM154 (gRNA, protospacer m7GEP338: CTGTGCAGACAACGGCAACCG, SEQ ID NO: 34)
- f. GEZM155 (gRNA, protospacer m7GEP339: AGCTTCTCCACCTCCCAGTCG, SEQ ID NO: 35)
- g. GEMT121 (LbCpf1-RR and tdTomato, SEQ ID Nos: 28-31)

3. Using the standard regeneration protocol for tropical corn lines, 12 plants were sampled and analyzed by amplicon sequencing using the following primers:

TABLE 1

Primer	Sequence	SEQ ID NO
946R	GCTCTCTCGGCCCACTTTT	71
1145F	TGTTCTGCGCCAGGCTACA	72
346F	TCGCGAACACACCGGTCGTA	73
1948R	ACTCACATGCCTCGCCCTCA	74

Three plants were identified with edits in the Lox3 gene. All three plants were edited in multiple locations (Table 2).

TABLE 2

	Target 1	Target 2	Target 3
GEZM152T012	100%: 3&12 bp deletions	100%: 3&12 bp deletions	50%: 15 bp deletion
GEZM152T013	50%: 5 bp deletion	50%: 5 bp deletion	50%: 23 bp deletion
GEZM152T014	100%: 3 & 11 bp deletions	100%: 3 & 11 bp deletions	50%: 15 bp deletion

4. Plants were transferred from media to soil and kept in short day conditions (in a lean-in Conviron) for as long as possible to trigger tassel and ear development before ultimately moving them to standard greenhouse conditions.

Example 5: Determine Tropical Corn Insect Resistance by Measuring Larval Weight of Fall Armyworm Feeding on Lox3 SDN1 Tropical Corn Edited Lines

The purpose of this experiment was to evaluate if Lox3 gene knock-out is linked to fall armyworm tolerance in tropical corn (SL57). To test this, fall armyworm (*Spodoptera frugiperda*) larval performance was chosen as a measure of antibiosis-based resistance.

Three ZmLox3_SDN-1 tropical corn lines (GEZM152T012, GEZM152T013, GEZM152T014) were evaluated using unmodified SL57 tropical corn as control. Three independent insect resistance experiments were performed. For each experiment, five biological replicates were used for the unmodified SL57 tropical corn and for each ZmLox3_SDN-1 lines. Three newly hatched fall armyworm larva were added into individual Falcon tubes containing

corn leaves. Fresh leaves were provided daily and after ten days the larva weight was scored.

Example 6: Generation of Lox3 TILLING Mutants

EMS mutagenesis: A mutagenized population (EMS treatment) from PH207 was developed. The exonic region 7 of Zm00008a004913 was screened and 5 positive mutants harboring amino acid changes and stop codons were detected (SEQ ID NOS: 36-40, Table 3). In Zm00008a004913 cDNA of Mutant WVP18-09344-014 C at position 1876 replaced by T (see also SEQ ID NO: 43), leading to an amino acid exchange from P to S at position 569 (see also SEQ ID NO: 37); in Zm00008a004913 cDNA of Mutant WVP18-09309-014 and WVP18-09339-009 G at position 2004 replaced by A (see also SEQ ID NO: 44), leading to an amino acid exchange from W to STOP at position 668 (see also SEQ ID NO: 39); in Zm00008a004913 cDNA of Mutant WVP18-09358-016 G at position 2079 replaced by A (see also SEQ ID NO: 42), leading to an amino acid exchange from W to STOP at position 693 (see also SEQ ID NO: 37); in Zm00008a004913 cDNA of Mutant WVP18-09307-014 G at position 2287 replaced by A (see also SEQ ID NO: 45), leading to an amino acid exchange from G to S at position 763 (see also SEQ ID NO: 40). After selfing of these mutants to produce two homozygote classes (homozygote wildtype and mutant), they have been evaluated in the field for resistance to two *Fusarium* species (*F. graminearum*, *F. verticilloides*, see Example 7).

TABLE 3

Name	Gen- Name	Code	Pos CDS	Base WT	Base Mut	Position in AS sequence	AS WT	AS Mut
WP18-09344-014	LOX3	PH207m034b	1876	C	T	569	P	S
WP18-09309-014	LOX3	PH207m034d	2004	G	A	668	W	STOP
WVP18-09339-009	LOX3	PH207m034d	2004	G	A	668	W	STOP
WVP18-09358-016	LOX3	PH207m034f	2079	G	A	693	W	STOP
WVP18-09307-014	LOX3	PH207m034h	2287	G	A	763	G	S

Example 7: Determine Fungal Resistance of Lox3 TILLING Mutants

After selfing of these mutants to produce two homozygote classes (homozygote wildtype and mutant), they have been evaluated in the field for resistance to two *Fusarium* species (*F. graminearum* (FG), *F. verticilloides* (FV)).

In 2020 plants were grown in rows with 20 plants per row in the location GON and in 2021 in MUR. Ten plants of each experimental unit were inoculated four to six days after the experimental unit was flowering, excluding the first plant of the row to avoid possible border effect. Rows were declared as flowering when at least 50% of the plants in the row presented extruded silks. Female flowering (FF) dates were collected for each row daily.

1 ml of an inoculum suspension with a concentration of 15.000 conidia ml⁻¹ (*F. graminearum* (FG)) and 1,000,000

conidia ml⁻¹ (*F. verticilloides* (FV)) was inoculated into the maize silk channel with a self-refilling syringe. The experiments for the two different species were kept separate.

Approximately 50 days after inoculation, cobs were dehusked and all plants were visually assessed for *Fusarium* symptoms by estimating the percentage of the ear covered by mycelium. The remnant not-inoculated plants of each experimental unit were used as a control of the proportion of naturally infected cobs. The arithmetic mean of the 10 assessed inoculated were employed for further analyses.

The homozygous mutant class for PH207m034b and PH207m034h showed lower infection of *F. graminearum* (FG) and *F. verticilloides* (FV) (FIG. 7).

Example 8: Determine Insect Resistance of Lox3 TILLING Mutants

The purpose of this experiment was to evaluate if Lox3 Tilling stop-mutations is linked to fall armyworm tolerance in temperate corn (line PH207). To test this, fall armyworm (*Spodoptera frugiperda*) larval performance was chosen as a measure of antibiosis-based resistance.

Two different Stop-Mutants for Lox3, the according wildtypes derived from a heterozygote mutant plant and the line PH207 which is the parent of the mutant population were evaluated. Three independent insect resistance experiments were performed and for each experiment, five biological replicates were used. Three newly hatched fall armyworm larva were added into individual Falcon tubes containing

corn leaves. Fresh leaves were provided daily and after ten days the larva weight was scored.

Example 9: Generation of Lox3 SDN1 Knock-Down in OSR by Genome Editing

1. Agro-based transformation, the nuclease was provided in the same construct together with the gRNAs (FIG. 5: construct map, plasmid sequence: pZFNptII-LbCpf1-tDT-lox3_TTTV, SEQ ID NO: 50); four crRNAs were used in total to target the OSR lox3 gene (SEQ ID Nos: 46-49); BnLox3 sequences SEQ ID Nos: 75-86.

a. Agro-based transformation was adopted to generate GE plants using various different starting materials of seedlings between 1 to 10 days old.

The oil seed rape transformation protocol includes the following steps:

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1. Seed sterilization with 70% ethanol and bleach
2. Seed germination on germination medium
3. *Agrobacterium* culture of a strain containing the construct of interest
4. Preparation of *Agrobacterium* for inoculation
5. Preparation of plant material and co-culture
6. Recovery of callus
7. Selection
8. Shoot growth
9. Rooting
2. Molecular analyses: Next Generation Sequencing (NGS), digital droplet PCR (ddPCR) and Decomposition Regression to Identify Variations for Editing Events (DRIVE)
 - a. The SDN-1 gene edited plants were first analyzed for the presence of the transgene by qPCR using the settings shown in tables 5 and 6:

TABLE 5

	primer	work stock concentration (uM)	SEQ ID NO
Primers	cruaxxxxxxf02x	20	51
	cruaxxxxxxr01x	20	52
	nptIIxxx01	20	53
	nptIIxxx01	20	54
	tDTxxx04	20	55
probes	tDTxxx01	20	56
	CruaxxxMGB	10	57
	nptIIxxxMGB	10	58
	tDTxxxMGB	10	59

qPCR parameters:

TABLE 6

50° C.	2 min	40 cycles
95° C.	5 min	
95° C.	15 s	
60° C.	30 s	

- b. The transgenic plants were then analyzed by qPCR using the primers sets SEQ ID 60, 61 and 62 (probe) for nuclease test.
- c. The editing profile was then obtained by amplifying and sequencing the PCR product using the primers sets as shown in table 7 and then run Drive analysis on OMICS to get editing information.

TABLE 7

Primer	SEQ ID NO:
IR106_lox3_F1	CCTCTGACCTCCAAAGACCCTTA
IR106_lox3_F2	CCATTACTTGTTTGGTCGGCGT
IR106_lox3_F3	GCCATCACTTGTTTCTCGGC
IR106_lox3_F4	CCGTCACCTTGTTTCTCGGC
IR106_lox3_R1	TCAAAGGCTAATATACTGACACGT
IR106_lox3_R2	GGCAAACGCGTTTCCTTAATCA
IR106_lox3_R3	ATAACGCTGATGTGCTAAGC
IR106_lox3_R4	GTTAATAACGCTGATGCTAAGC

Example 10: Determine OSR Insect Resistance by Measuring Leaf Damage from Diamondback Moth and by Measuring Aphid Reproduction from Insect Feeding on Lox3 SDN1 Edited Line

The purpose of this experiment was to evaluate if Lox3 gene knock-out is linked to diamondback moth and green

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peach aphid tolerance in OSR (Palma). To test this, diamondback moth (*Plutella xylostella*) feeding damage and green peach aphid (*Myzus persicae*) reproduction were chosen as a measure of plant resistance.

One Lox3_SDN-1 OSR line was evaluated using unmodified Palma OSR as control. Three independent insect resistance experiments were performed and for each experiment, fourteen biological replicates were used.

For the test using diamondback moth, four newly hatched larva were placed on each one-week-old OSR seedlings. Four days later, the percentage of plant damage (=percentage of leaf eaten by the larva) was visually scored using a 0 to 100% scale.

For the test using green peach aphid, one adult aphid was placed on each one-week-old OSR seedlings. Seven days later, the number of aphids on each seedling was recorded (adult aphid+progeny).

Example 11: Determine Fungal Resistance of Lox3 Edited Lines

Clubroot Test

Infected oil seed rape roots were collected in the field. 100 g of infected roots were crushed with 400 ml water. The solution was then roughly filtered. Two different isolates were used in the screening, isolate A (Mendel-based varieties show still resistance) and C (Mendel-based varieties are highly susceptible).

OSR plants to be tested were sown in sand and were picked after 10 to 14 days. Roots were washed and incubated in the spore solution for 15 to 20 minutes. Plants were then removed and planted into a soil/sand substrate. Afterwards the spore solution was additionally pipetted directly on to the plants. Afterwards they were grown under greenhouse conditions for approximately 6 weeks.

After washing of the roots, the gall formation was scored from 1, meaning no gall formation to 4 strong gall formation (FIG. 6). A disease index (DI) was calculated (E. Diederichsen and M. D. Sacristan, Plant Breeding 115, 5-10 (1996), Disease response of resynthesized *Brassica napus* L. lines carrying different combinations of resistance to *Plasmodiophora brassicae* Wor.) by using the following formula:

$$DI = \frac{\sum (0n_0 + 1n_1 + 2n_2 + 3n_3)}{3N} \times 100$$

The values n0 to n3 correspond to the number of plants per class and N being the total number of tested plants. Genotypes with a DI<10 were assessed to be highly resistant. A DI between 10-25 leads to an assessment as resistant and genotypes with a DI>25 are rated as susceptible.

Blackleg Test

OSR seedlings were germinated in soil in a controlled environment chamber at 20-24° C. for seven days. Each fully expanded cotyledon was wounded twice by puncturing it with a needle. The wounded sites were inoculated with a 107 pycnidiospore solution obtained from a German field isolate (pycnidia harvested from media grown fungus, suspended in sterile water and quantified). In order to delay cotyledon senescence, true leaves were cut at the petiole insertion. After about 10 days the symptoms were scored on a scale of 1-6 (1-3=resistant, 4-6=susceptible) based on the increasing injury of the cotyledon and the sporulation of the fungus.

LIST OF EMBODIMENTS

- The following is a non-exhaustive list of embodiments.
- Item 1. A method for conferring or increasing resistance or tolerance to an insect and, optionally a fungal pathogen to/in a plant comprising the steps of:
- (i) providing at least one plant cell;
 - (ii) introducing into the at least one plant cell at least one gene silencing construct, at least one genome editing system or a genome modification, which leads to a targeted knock-down or a knock-out of a Lox3 gene endogenous to the plant;
 - (iii) obtaining at least one modified plant cell having reduced or abolished expression of the Lox3 gene; and
 - (iv) obtaining at least one plant cell, tissue, organ, plant or seed having reduced or abolished expression of the Lox3 gene, optionally after an additional step of regenerating the plant tissue, organ, plant or seed from the at least one modified cell.
- Item 2. The method of item 1, wherein the method is for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* species, *Cercospora zeina* and *Cercospora zeae-maydis* to/in maize (*Zea mays*).
- Item 3. The method of item 1, wherein the method is for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of green peach aphid (*Myzus persicae*), diamondback moth (*Plutella xylostella*), cabbage stem flea beetle (*Psylliodes chrysocephala*), crucifer flea beetle (*Phyllotreta cruciferae*), striped flea beetle (*Phyllotreta striolata*), hop flea beetle (*Psylliodes punctulata*), rape stem weevil (*Ceutorhynchus pictaris*) and cabbage stem weevil (*Ceutorhynchus quadridens*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Phoma lingam* and *Plasmodiophora brassicae* to/in oilseed rape (*Brassica napus*).
- Item 4. The method according to item 2, wherein the Lox3 gene is represented by a nucleic acid sequence of SEQ ID NO: 6, 7, 9, 10, 12, 13, 15, 16, 87 or 88 or a nucleic acid sequence having a sequence identity of at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 6, 7, 9, 10, 12, 13, 15, 16, 87 or 88.
- Item 5. The method according to item 2 or 4, wherein the Lox3 gene encodes an amino acid sequence of SEQ ID NO: 8, 11, 14, 17 or 89 an amino acid sequence having a sequence identity of at least 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 8, 11, 14, 17 or 89.
- Item 6. The method according to item 3, wherein the Lox3 gene is represented by a nucleic acid sequence of SEQ ID NO: 75, 76, 77, 78, 83, 84, 85 or 86 or a nucleic acid sequence having a sequence identity of at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%,

- 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 75, 76, 77, 78, 83, 84, 85 or 86.
- Item 7. The method according to item 3 or 6, wherein the Lox3 gene encodes an amino acid sequence of SEQ ID NO: 79, 80, 81 or 82 or an amino acid sequence having a sequence identity of at least 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 79, 80, 81 or 82.
- Item 8. The method according to any of the preceding items, wherein in step (ii) a construct is introduced into the at least one plant cell, which targets the Lox3 gene for gene silencing.
- Item 9. The method according to item 8, wherein the construct is or wherein the construct encodes an RNAi construct comprising a sense and an antisense sequence targeting the Lox3 gene, the RNAi construct forming an RNA hairpin upon transcription.
- Item 10. The method according to item 9, wherein the method is for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* species, *Cercospora zeina* and *Cercospora zeae-maydis* to/in maize (*Zea mays*) and wherein the sense sequence is encoded by a nucleic acid sequence of SEQ ID NO: 1, or a nucleic acid sequence having a sequence identity of at least 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 1.
- Item 11. The method according to item 9 or 10, wherein the method is for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* species, *Cercospora zeina* and *Cercospora zeae-maydis* to/in maize (*Zea mays*) and wherein the antisense sequence is encoded by a nucleic acid sequence of SEQ ID NO: 2, or a nucleic acid sequence having a sequence identity of at least 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 2.
- Item 12. The method according to any of items 9 to 11, wherein the method is for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* species, *Cercospora zeina* and *Cercospora zeae-maydis* to/in maize (*Zea mays*) and wherein the RNA hairpin has an intervening intron loop sequence comprising a nucleic acid sequence of SEQ ID

- NO: 3, or a nucleic acid sequence having a sequence identity of at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 3.
- Item 13. The method according to item 8, wherein the method is for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* species, *Cercospora zeina* and *Cercospora zeae-maydis* to/in maize (*Zea mays*) and wherein the construct comprises a nucleic acid sequence of SEQ ID NO: 4, or a nucleic acid sequence having a sequence identity of at least 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 4.
- Item 14. The method according to item 8, wherein the method is for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* species, *Cercospora zeina* and *Cercospora zeae-maydis* to/in maize (*Zea mays*) and wherein a vector is introduced into the plant cell, which vector comprises or consists of a nucleic acid sequence of SEQ ID NO: 5, or a nucleic acid sequence having a sequence identity of at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 5.
- Item 15. The method according to any of items 8 to 14, wherein the construct is introduced into the at least one plant cell by transformation or transfection mediated by biolistic bombardment, *Agrobacterium*-mediated transformation, micro- or nanoparticle delivery, chemical transfection, or a combination thereof.
- Item 16. The method according to any of items 1 to 7, wherein in step (ii) at least one genome editing system is introduced into the at least one cell, which targets the Lox3 gene, wherein the at least one genome editing system comprises
- at least one site-specific nuclease or site-specific nickase, and optionally, in case a CRISPR system is used, at least one guide molecule or a sequence encoding the same, and
 - optionally, at least one repair template, or a sequence encoding the same.
- Item 17. The method according to item 16, wherein the at least one genome editing system is selected from a CRISPR/Cas system, preferably from a CRISPR/MAD7 system, a CRISPR/Cpf1 (CRISPR/Cas12a) system, a CRISPR/MAD2 system, a CRISPR/Cas9 system, a CRISPR/CasX system, a CRISPR/CasY system, a CRISPR/Cas13 system, or a CRISPR/Csm system, or wherein the at least one genome editing system is selected

- from a zinc finger nuclease system, or a transcription activator-like nuclease system, or a meganuclease system, or any combination, variant, or an active fragment thereof.
- Item 18. The method of item 16 or 17 wherein the at least one genome editing system is introduced into the at least one maize cell by transformation or transfection mediated by biolistic bombardment, *Agrobacterium*-mediated transformation, micro- or nanoparticle delivery, chemical transfection, or a combination thereof.
- Item 19. The method of any of items 16 to 18, wherein the method is for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of green peach aphid (*Myzus persicae*), diamondback moth (*Plutella xylostella*), cabbage stem flea beetle (*Psylliodes chrysocephala*), crucifer flea beetle (*Phyllotreta cruciferae*), striped flea beetle (*Phyllotreta striolata*), hop flea beetle (*Psylliodes punctulata*), rape stem weevil (*Ceutorhynchus picitarsis*) and cabbage stem weevil (*Ceutorhynchus quadridens*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Phoma lingam* and *Plasmodiophora brassicae* to/in oilseed rape (*Brassica napus*), wherein the at least one genome editing system comprises a crRNA encoded by a nucleic acid sequence of any of SEQ ID NOs: 46 to 49, or a nucleic acid sequence having a sequence identity of at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of any of SEQ ID NOs: 46 to 49.
- Item 20. The method of item 19, wherein the genome editing system is encoded by a plasmid of the nucleic acid sequence of SEQ ID NO: 50 or a nucleic acid sequence having a sequence identity of at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 50.
- Item 21. The method according to any of items 1 to 7, wherein in step (ii) a mutagenesis is performed on a single or on a plurality of cell(s) by applying chemicals or radiation.
- Item 22. The method according to item 21, wherein an alkylating agent, in particular ethyl methanesulfonate is applied to the single or the plurality of cell(s) to induce mutagenesis.
- Item 23. The method according to item 21 or 22, wherein one or more mutations in the Lox3 gene are inserted and identified by TILLING in step (ii).
- Item 24. The method according to any of items 21 to 23, wherein one or more cell(s) with knock-down or knock-out mutations in the Lox3 gene are selected in step (ii).
- Item 25. A maize cell, maize tissue, maize organ, maize plant or maize seed obtained or obtainable by a method according to any of items 1, 2, 4, 5, 8 to 18 or 21 to 24.
- Item 26. An oilseed rape cell, oilseed rape tissue, oilseed rape organ, oilseed rape plant or oilseed rape seed obtained or obtainable by a method according to any of items 1, 3, 6 to 9 or 15 to 24.
- Item 27. An expression construct, which targets the Lox3 gene in maize for gene silencing, wherein the construct encodes an RNAi construct comprising a sense and an antisense sequence targeting the Lox3 gene endogenous to a maize plant, which RNAi construct forms an RNA hairpin upon transcription.

- Item 28. An expression construct, which targets the Lox3 gene in oilseed rape for gene silencing, wherein the construct encodes an RNAi construct comprising a sense and an antisense sequence targeting the Lox3 gene endogenous to an oilseed rape plant, which RNAi construct forms an RNA hairpin upon transcription.
- Item 29. The expression construct according to item 27, wherein the sense sequence is encoded by a nucleic acid sequence of SEQ ID NO: 1, or a nucleic acid sequence having a sequence identity of at least 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 1.
- Item 30. The expression construct according to item 27 or 29, wherein the antisense sequence is encoded by a nucleic acid sequence of SEQ ID NO: 2, or a nucleic acid sequence having a sequence identity of at least 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 2.
- Item 31. The expression construct according to any of items 27, 29 or 30, wherein the RNA hairpin has an intervening intron loop sequence comprising a nucleic acid sequence of SEQ ID NO: 3, or a nucleic acid sequence having a sequence identity of at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 3.
- Item 32. The expression construct according to item 27, wherein the construct comprises a nucleic acid sequence of SEQ ID NO: 4, or a nucleic acid sequence having a sequence identity of at least 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 4.
- Item 33. A vector comprising or consisting of a nucleic acid sequence of SEQ ID NO: 5 or a nucleic acid sequence having a sequence identity of at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 5.
- Item 34. An RNAi hairpin construct conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* species, *Cercospora zeina* and *Cercospora zeae-maydis* to/in maize (*Zea mays*), wherein the RNAi hairpin construct comprises a nucleic acid sequence of SEQ ID NO: 4, or a nucleic acid sequence having a sequence identity of at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 4.
- Item 35. An expression construct encoding a genome editing system, which targets the Lox3 gene in maize, wherein the genome editing system comprises
- (a) at least one site-specific nuclease or site-specific nickase, and optionally, in case a CRISPR system is used, at least one guide molecule or a sequence encoding the same, and
 - (b) optionally, at least one repair template, or a sequence encoding the same.

- Item 36. An expression construct encoding a genome editing system, which targets the Lox3 gene in oilseed rape, wherein the genome editing system comprises
- (a) at least one site-specific nuclease or site-specific nickase, and optionally, in case a CRISPR system is used, at least one guide molecule or a sequence encoding the same, and
 - (b) optionally, at least one repair template, or a sequence encoding the same.
- Item 37. The expression construct according to item 35 or 36, wherein the at least one genome editing system is selected from a CRISPR/Cas system, preferably from a CRISPR/MAD7 system, a CRISPR/Cpf1 (CRISPR/Cas12a) system, a CRISPR/MAD2 system, a CRISPR/Cas9 system, a CRISPR/CasX system, a CRISPR/CasY system, a CRISPR/Cas13 system, or a CRISPR/Csm system, or wherein the at least one genome editing system is selected from a zinc finger nuclease system, or a transcription activator-like nuclease system, or a meganuclease system, or any combination, variant, or an active fragment thereof.
- Item 38. The expression construct according to item 36 or 37, wherein the expression construct comprises a crRNA encoded by a nucleic acid sequence of any of SEQ ID NOs: 46 to 49, or a nucleic acid sequence having a sequence identity of at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of any of SEQ ID NO: 46 to 49.
- Item 39. The expression construct according to item 36 or 37, wherein the genome editing system is encoded by a plasmid of the nucleic acid sequence of SEQ ID NO: 50 or a nucleic acid sequence having a sequence identity of at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 50.
- Item 40. A vector encoding an expression construct according to any of items 35 to 39.
- Item 41. A maize cell, maize tissue, maize organ, maize plant, or maize seed comprising an expression construct or a vector according to any of items 35, 37 or 40.
- Item 42. An oilseed rape cell, oilseed rape tissue, oilseed rape organ, oilseed rape plant or oilseed rape seed comprising an expression construct or a vector according to any of items 36 to 39 or 40.
- Item 43. A use of at least one gene silencing construct, at least one genome editing system or a genome modification, which leads to a targeted knock-down or a knock-out of the endogenous a Lox3 gene, for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* species, *Cercospora zeina* and *Cercospora zeae-maydis* to/in maize (*Zea mays*).
- Item 44. A use of at least one gene silencing construct, at least one genome editing system or a genome modification, which leads to a targeted knock-down or a knock-out of the endogenous a Lox3 gene, for conferring or increasing resistance or tolerance to one or more insect(s)

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selected from the group consisting of green peach aphid (*Myzus persicae*), diamondback moth (*Plutella xylostella*), cabbage stem flea beetle (*Psylliodes chrysocephala*), crucifer flea beetle (*Phyllotreta cruciferae*), striped flea beetle (*Phyllotreta striolata*), hop flea beetle (*Psylliodes punctulata*), rape stem weevil (*Ceutorhynchus picipitarsis*) and cabbage stem weevil (*Ceutorhynchus quadridens*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Phoma lingam* and *Plasmodiophora brassicae* to/in oilseed rape (*Brassica napus*).

Item 45. A use of a construct, the construct being or encoding an RNAi construct comprising a sense and an antisense sequence targeting the endogenous Lox3 gene of a maize plant, wherein the RNAi construct forms an RNA hairpin upon transcription, for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* species, *Cercospora zeina* and *Cercospora zeae-maydis* to/in maize (*Zea mays*).

Item 46. A use of a construct, the construct being or encoding an RNAi construct comprising a sense and an antisense sequence targeting the endogenous Lox3 gene of an oilseed rape plant, wherein the RNAi construct forms an RNA hairpin upon transcription, for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of green peach aphid (*Myzus persicae*), diamondback moth (*Plutella xylostella*), cabbage stem flea beetle (*Psylliodes chrysocephala*), crucifer flea beetle (*Phyllotreta cruciferae*), striped flea beetle (*Phyllotreta striolata*), hop flea beetle (*Psylliodes punctulata*), rape stem weevil (*Ceutorhynchus picipitarsis*) and cabbage stem weevil (*Ceutorhynchus quadridens*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Phoma lingam* and *Plasmodiophora brassicae* to/in oilseed rape (*Brassica napus*).

Item 47. The use according to item 45, wherein the sense sequence is encoded by a nucleic acid sequence of SEQ ID NO: 1, or a nucleic acid sequence having a sequence identity of at least 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 1.

Item 48. The use according to item 45 or 47, wherein the antisense sequence is encoded by a nucleic acid sequence of SEQ ID NO: 2, or a nucleic acid sequence having a sequence identity of at least 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 2.

Item 49. The use according to any of items 45, 47 or 48, wherein the RNA hairpin has an intervening intron loop sequence comprising a nucleic acid sequence of SEQ ID NO: 3, or a nucleic acid sequence having a sequence identity of at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or at least 99% to the sequence of SEQ ID NO: 3.

Item 50. The use according to item 45 wherein the construct comprises a nucleic acid sequence of SEQ ID NO: 4, or

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a nucleic acid sequence having a sequence identity of at least 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 4.

Item 51. A use of a vector, which vector comprises or consists of a nucleic acid sequence of SEQ ID NO: 5, or a nucleic acid sequence having a sequence identity of at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 5 for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* species, *Cercospora zeina* and *Cercospora zeae-maydis* to/in maize (*Zea mays*).

Item 52. A use of a genome editing system, which targets the endogenous Lox3 gene in a maize plant, wherein the genome editing system comprises

(a) at least one site-specific nuclease or site-specific nickase, and optionally, in case a CRISPR system is used, at least one guide molecule or a sequence encoding the same, and

(b) optionally, at least one repair template, or a sequence encoding the same

for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* species, *Cercospora zeina* and *Cercospora zeae-maydis* to/in maize (*Zea mays*).

Item 53. A use of a genome editing system, which targets the endogenous Lox3 gene in an oilseed rape plant, wherein the genome editing system comprises

(a) at least one site-specific nuclease or site-specific nickase, and optionally, in case a CRISPR system is used, at least one guide molecule or a sequence encoding the same, and

(b) optionally, at least one repair template, or a sequence encoding the same

for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of green peach aphid (*Myzus persicae*), diamondback moth (*Plutella xylostella*), cabbage stem flea beetle (*Psylliodes chrysocephala*), crucifer flea beetle (*Phyllotreta cruciferae*), striped flea beetle (*Phyllotreta striolata*), hop flea beetle (*Psylliodes punctulata*), rape stem weevil (*Ceutorhynchus picipitarsis*) and cabbage stem weevil (*Ceutorhynchus quadridens*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen selected from the group consisting of *Phoma lingam* and *Plasmodiophora brassicae* to/in oilseed rape (*Brassica napus*).

Item 54. A use of an expression construct encoding a genome editing system, which targets the endogenous Lox3 gene in a maize plant, wherein the genome editing system comprises

- (a) at least one site-specific nuclease or site-specific nickase, and optionally, in case a CRISPR system is used, at least one guide molecule or a sequence encoding the same, and
- (b) optionally, at least one repair template, or a sequence encoding the same,

wherein the at least one genome editing system is selected from a CRISPR/Cas system, preferably from a CRISPR/MAD7 system, a CRISPR/Cpf1 (CRISPR/Cas12a) system, a CRISPR/MAD2 system, a CRISPR/Cas9 system, a CRISPR/CasX system, a CRISPR/CasY system, a CRISPR/Cas13 system, or a CRISPR/Csm system, or wherein the at least one genome editing system is selected from a zinc finger nuclease system, or a transcription activator-like nuclease system, or a meganuclease system, or any combination, variant, or an active fragment thereof for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* species, *Cercospora zeina* and *Cercospora zeae-maydis* to/in maize (*Zea mays*).

Item 55. A use of an expression construct encoding a genome editing system, which targets the endogenous Lox3 gene in an oilseed rape plant, wherein the genome editing system comprises

- (a) at least one site-specific nuclease or site-specific nickase, and optionally, in case a CRISPR system is used, at least one guide molecule or a sequence encoding the same, and
- (b) optionally, at least one repair template, or a sequence encoding the same,

wherein the at least one genome editing system is selected from a CRISPR/Cas system, preferably from a CRISPR/MAD7 system, a CRISPR/Cpf1 (CRISPR/Cas12a) system, a CRISPR/MAD2 system, a CRISPR/Cas9 system, a CRISPR/CasX system, a CRISPR/CasY system, a CRISPR/Cas13 system, or a CRISPR/Csm system, or wherein the at least one genome editing system is selected from a zinc finger nuclease system, or a transcription activator-like nuclease system, or a meganuclease system, or any combination, variant, or an active fragment thereof

for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of green peach aphid (*Myzus persicae*), diamondback moth (*Plutella xylostella*), cabbage stem flea beetle (*Psylliodes chrysocephala*), crucifer flea beetle (*Phyllotreta cruciferae*), striped flea beetle (*Phyllotreta striolata*), hop flea beetle (*Psylliodes punctulata*), rape stem weevil (*Ceutorhynchus piciparsis*) and cabbage stem weevil (*Ceutorhynchus quadridens*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Phoma lingam* and *Plasmodiophora brassicae* to/in oilseed rape (*Brassica napus*).

Item 56. Use according to item 55 wherein the expression construct comprises a crRNA encoded by a nucleic acid sequence of any of SEQ ID NOs: 46 to 49, or a nucleic acid sequence having a sequence identity of at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of any of SEQ ID NO: 46 to 49.

Item 57. Use according to item 55 or 56, wherein the genome editing system is encoded by a plasmid of the nucleic acid sequence of SEQ ID NO: 50 or a nucleic acid sequence having a sequence identity of at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or at least 99% to the sequence of SEQ ID NO: 50.

Item 58. A use of a vector encoding an genome editing system as defined in item 54 for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Fusarium* species, *Colletotrichum* species, in particular *Colletotrichum graminicola* and *Colletotrichum sublineolum*, *Diplodia* species, *Cercospora zeina* and *Cercospora zeae-maydis* to/in maize (*Zea mays*).

Item 59. A use of a vector encoding an genome editing system as defined in any of items 55 to 57 for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of green peach aphid (*Myzus persicae*), diamondback moth (*Plutella xylostella*), cabbage stem flea beetle (*Psylliodes chrysocephala*), crucifer flea beetle (*Phyllotreta cruciferae*), striped flea beetle (*Phyllotreta striolata*), hop flea beetle (*Psylliodes punctulata*), rape stem weevil (*Ceutorhynchus piciparsis*) and cabbage stem weevil (*Ceutorhynchus quadridens*) and, optionally conferring or increasing resistance or tolerance to one or more fungal pathogen(s) selected from the group consisting of *Phoma lingam* and *Plasmodiophora brassicae* to/in oilseed rape (*Brassica napus*).

SEQUENCE LISTING

Sequence total quantity: 89

SEQ ID NO: 1 moltype = DNA length = 201

FEATURE Location/Qualifiers

misc_feature 1..201

note = RNAi target region sense strand before RGA2intronII

source 1..201

mol_type = other DNA

organism = synthetic construct

SEQUENCE: 1

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SEQ ID NO: 2          moltype = DNA length = 201
FEATURE              Location/Qualifiers
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source                1..201
                      mol_type = other DNA
                      organism = synthetic construct

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source                1..312
                      mol_type = other DNA
                      organism = synthetic construct

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source                1..726
                      mol_type = other DNA
                      organism = synthetic construct

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SEQ ID NO: 5          moltype = DNA length = 12924
FEATURE              Location/Qualifiers
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                      note = Sequence of a complete artificial vector
source                1..12924
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 5
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SEQ ID NO: 8          moltype = AA length = 864
FEATURE              Location/Qualifiers
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FPLKTIITLDD VPGRGAVTFV ANSWVYPAGK YRYNRVFFSN DTYLPSQMPA ALKPYRDEL 180
RNLRGDDQQG PYQEHDVYR YDVYNDLGEP DGGNPRPILG GSADHPYPRR CRTGRKPTKT 240
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QDIINLYEGG IKLPKIQALE DMRKLFPLQL VKDLLPAGGD YLLKLPIQI IQEDKNWRT 360
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GDEELQAWWK EVREVGHDG KDAPWWPKMQ AVSELASACT TIIWIASALH AAVNFGQYPY 720
AGYLPNRPTV SRRRMPEPGS KEYEELERDP ERGFIHTITS QIQTIIGISL IEILSKHSSD 780
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SEQ ID NO: 9          moltype = DNA length = 4150
FEATURE              Location/Qualifiers
source                1..4150
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                     organism = Zea mays

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SEQ ID NO: 10      moltype = DNA length = 2595
FEATURE            Location/Qualifiers
source              1..2595
                    mol_type = genomic DNA
                    organism = Zea mays

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SEQ ID NO: 11      moltype = AA length = 864
FEATURE           Location/Qualifiers
source            1..864
                  mol_type = protein
                  organism = Zea mays

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                  organism = Zea mays

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 organism = Zea mays

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cgctcatcgca	acgaaccggc	acgtgagcgt	ggtgcatcca	gtgcacaaag	tgctgagccc	19620
acactaccgt	gacacgctga	acatcaacgc	cctggcacgc	cagacgctca	tcaacgccga	19680
cggcattctc	gagcgacacg	tgttcctg	aaagtacgcg	ctggggatgt	cctccgacgt	19740
gtacaagagc	tggaaatttca	cagcagcggc	ctctccagca	gacctcgtca	agaggtagct	19800
acatgtatag	tactaaactac	attgagcagt	aaacgcctat	agcagtgacg	agatggttct	19860
gacgggttta	tttgcctc	tgtgtgtg	tttcagaggt	gtggctgtgc	cggaaccagt	19920
gagcccttac	ggtgtccgg	tgtgatcaa	ggactaccct	tacgcgctgg	acgggctggt	19980
catctggtg	gcgatcgagc	ggtgggtcaa	ggagtacctg	gacgtctact	accccaacga	20040
cggcgagctc	cagcgcgacg	tggagctgca	ggcgtggtgg	aaggaggtgc	gcgaggaggc	20100
gcacggcgac	ctcaaggacc	gagactggtg	gcccaggatg	gacgcgctcc	agcggtggc	20160
cagggcgtgc	acgaccgtca	tctgggtagc	gtccgcgctg	cacgcggcgc	tcaacttcgg	20220
gcagtaccgc	tacgcgggtg	acctgccgaa	ccggccgacc	gtgagccggc	ggccgatgcc	20280
ggagccgggc	agcgacgact	acaagaagct	ggaggcgggg	cagaaggagg	cggacgcggt	20340
gttcatccgc	acatcacca	gccagttcca	gaccatcctg	ggcatctcgc	tcacgagat	20400
cctctccaa	cactcctcgc	acgaggtgta	cctcgccag	cgcgacgagc	ctgagcgctg	20460
gacgtcggac	gccaggcg	tggacgcgtt	cagaaggttc	ggaagccggc	tggtaggat	20520
cgagaagcgg	atcaggacga	tgaacgacag	cccgcgctg	aagaaccgga	aggggcccgt	20580
ggagatgcgc	tacatgctgc	tgtaccocaa	cacgtcggat	gtcaccggcg	agaagggcga	20640
ggggctcact	gccatgggca	ttcccaacag	catctccata	tga		20683

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SEQ ID NO: 16      moltype = DNA length = 5376
FEATURE           Location/Qualifiers
source            1..5376
                  mol_type = genomic DNA
                  organism = Zea mays

SEQUENCE: 16
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acgggtggtgc tcatgcgcaa gaacgtgctg gacctcaacg acttcggcgc caccgtcgtc 120
gacagcatca cgcagttcct cggcaaggcg gtcacctgcc agctcatcag ctccactctc 180
gtcgacgcga acaacggcaa ccgcgggcgg gtcggggcgg aggcgaacct ggagcagtgg 240
ctgacgagcc tgccgtcgct gacgacccgc gagtccaagt tcggcgtcac gttcgcactg 300
gaggtggaga agctgggagt gccggggggc gtcgtcgta agaacaacca cgccgcccag 360
ttcttctcca agacaatcac cctcgacgac gtgcccgccc gcggcgccgt cactctcgtc 420
gccaactcct ggggttacct cgcgggcaag taccgtaca acccgctctt cttctccaac 480
gatacgtacc tgccaagcca gatgccggcg gcgctgaagc cgtaccgcga cgacgagctc 540
cgcaacctcc gcggcgacga ccagcagggc cctaccagg agcacgaccg cgtgtaccgc 600
tacgacgtgt acaacgacct cggcgagccc gacggcgcca acccgcgccc catcctcggc 660
ggctccggcg accaccgcta cccgcgccgc tgcgcacgg gccgcaagcc caccaaaacc 720
gaacccaaact ccgagagccg actgtcgctg gtggagcaga tctacgtgcc gcgggacgag 780
cgcttcggcc acctcaagat gtcgcacttc ctgggctact ccatcaaggc catcaccgag 840
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gacatgcgca agctcttccc gctccagctc gtcaaggacc tctctcccgc cgccggggac 1020
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gtcagggggc ggctgtgttc tgggtcgcat atatatatgt gctcctatga ggacaagaac 1140
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cactcctccg acgaggtgta cctcggccag cgcgacaccc ccgagtgagc ctccgacgcc 2460
tggcgctgag cggcgctcaa gaggttcagc gacgcgctgg tcaagatcga gggcaagggtg 2520
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cggccggcca ccatgttctg gcaacgggtc gcggaccggc tgacggggaa gaacaaggag 2760
gcgtggaaag agggaaagat ccgcggcacg gtgaggctgg tcaagaagga ggtgctggac 2820
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ggcgctcgct tccagctcgt cagcgccacc gcggccgacc ccagcaacgg gagcccgccc 2940
aaggtcggga aggcggcgca cctggaggag gcgggtggtg cgtcgaagtc gacgacggac 3000
ggggagacgg tgtaccgggt gagcttcgag tgggacgggt cgcaggcgct cccggcgccc 3060
gtcctgggtc ggaacctgca gcacgccgag ttcttctcca agtcgctcac cctcgagggc 3120
gtccccggca ggggcacccg cgtcttcgtc gccaaactcg ggatctaccc gcacaactc 3180
tactccagg aacgcgtctt cctcgccaac gacacttatc tgccaagcaa aatgcctcgc 3240
gcattgggtc cttaccggca ggacgagctc aagattctcc gcggcgacga taactcctga 3300
ccatacaagg agcacgaccc cgtctaccgt tacgactact acaacgacct cggtagacca 3360
gacaagggtg aagaccatgc ccggcctgtc ctcgggggca gccaaagaaca cccgtatccc 3420
cgtcgtcgca ggaccggcgg gcgtccaaca gagacagacc ccaactcgga gagcaggctg 3480
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tcggacttcc tcgggtactc actgaaggcg atcatcgagg ctgtccttcc gacgtggggg 3600
acgttcgtcg acgatacgcc caaggagttc gattcggtcg aagacatcct tgggctctac 3660
gagccgggtc cagaggcgcc caacaaccca ctggtagcag aggtcaggaa gagaatcccc 3720
agcgagttcc tcagaagcat tctgcccatt ggtagccatg accacccctc gaagatgccc 3780
cttccaaata tcatcagatc agatgtgttg aaaaaggctc cagagtttaa gtttggctg 3840
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cgtctgacgg agtctccagc taaaagtacc ctggacccaa gtcaatacgg agaccatacg 3960
agcaagatca ccgaagctca catccagcat aacatggaag gcctgtcagt gcagaatgca 4020
ctgaagaaga acaggctctt catcctagac caccatgacc atttcatgcc gtacctcaac 4080
aagatcaacg agttggaggg gaacttcatc tacgccagca ggacctact gttcctgaag 4140
gacgatggca cgctgaagcc cctggccgtc gagctgagcc tgccccacc tgatggccag 4200
cagcacggcg cggtcagcaa ggtgtacacc ccagctcact ccggcgctga gggccacgtc 4260
tggaacttgc ccaaggctta tgctcgctg aacgactccg cctggcatca gctgatcagc 4320

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cactggtgta acacgcacgc ggtgatcgag ccgttcgtca tcgcaacgaa ccggcagctg 4380
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aacgcccttg cagccagac gctcatcaac gccgacggca tottcgagcg caccgtgttc 4500
cctgcaaaagt acgcgctggg gatgtcctcc gacgtgtaca agagctggaa ttccaacgag 4560
caggctctcc cagcagacct cgtcaagaga ggtgtggctg tgccggacca gtcgagcccc 4620
tacggtgtcc ggttgtgat caaggactac ccttacgccg tggacggctt ggtcatctgg 4680
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gacctcaagg accgagactg gtggccagg atggacgccg tccagcgctt ggccaggcg 4860
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aagcactctt ccagcaggt gtacctggc cagcgcgacg agcctgagcg ctggacgtcg 5160
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cggatcagga cgatgaacga cagcccgacg ttgaagaacc ggaagggggc ggtggagatg 5280
ccgtacatgc tctgtatccc caacacgtcg gatgcaccg gcgagaaggg cgagggggctc 5340
actgcatagg gcattcccaa cagcatctcc atatga 5376

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SEQ ID NO: 17      moltype = AA  length = 1791
FEATURE            Location/Qualifiers
source              1..1791
                    mol_type = protein
                    organism = Zea mays

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SEQUENCE: 17
MLSGIIDGLT GANKHARLKG TVVLMRKQNV DLNDFGATVV DSISEFLGKG VTCQLISSTL 60
VDANNNGNRG VGAEANLEQW LTSLSPLTTG ESKFGVTFDW EVEKLGVPGA VVVKNNHAAE 120
FPLKTIITLDD VPGGAVTFV ANSWVYPAGK YRYNRVFFSN DTYLPSQMPA ALKPYPDDDEL 180
RNLRGDDQQG PYQEHRVYR YDVYNDLGEP DGGNPRPILG GSADHPYPRR CRTGRKPTKT 240
DPNSESRLSL VEQIYVPRDE RFGHLKMSDF LGYSIKAITQ GIIPAVRTYV DTPGFEFDSF 300
QDIINLYEGG IKLPKIQALE DMKKLFPLQL VKDLLPAGGD YLLKLPIQI IQGTSRIPID 360
VRGRLLSGLH IYMCSYEDKN AWRTEEFAR EVLAGVNPV ITRLTEFPPK STLDPSKYGD 420
HTSTITAEHI EKNLEGLTVQ QALDGNRLYI LDHHRFMPF LIDVNNLEGN FIYATRTLFF 480
LRGDGRLAPL AIELSEPYID GDLTVAKSKV YTPASSGVEA WVWQLAKAYV AVNDSGWHQL 540
VSHWLNTHAV MEPFVIATNR QLSVTHPVHK LLSSHFRDTM TINALARQTL INGGGIFEMT 600
VFPKGKALGM SSVVYKSWNF TEQGLPADLV KRGVAVADPS SPYKVRLLIE DYPYASDGLA 660
IWHAIQWVG EYLAIIYPDD GALRGDEELQ AWWKEVREVG HGDHKDAPWW PKMQAVSELA 720
SACTTIIWIA SALHAANVFG QYPYAGYLPN RPTVSRRRMP EPGSKEYEEL ERDPERGFH 780
TITSQIQITII GISLIEILSK HSSDEVYLGQ RDTPEWTSDA RALAAPKRFS DALVKIEGKV 840
VGENRDPQLR NRNGPAEFYP MLLYPNTSDH SEAREERAAG GIHTHTHRTL PEKREARSEE 900
RPATMFHWGV ADRLTGKNKE AWMNEGKIRGT VRLVKKEVLD VGDFFNASLLD GVHRILGWDD 960
GVAFQLVSAT AADPSNGSRG KVGKAAHLEE AVVSLKSTTD GETVYRVSE WDGSGQVPGA 1020
VLVRNLQHAE FFLKSLTLEG VPGRGTVVVF ANSWIYPHNL YSQERVFFAN DTYLPSKMPA 1080
ALVPYRQDEL KILRGDDNPG PYKEHDRVYR YDYNDLGEP DKGEDHARPV LGGSQEHYPY 1140
RRCRTGRRPT ETDPNSESRL FLLNLNIYVP RDERFGHLKM SDFLGYSLKA IIEAVLPTLG 1200
TFVDDTPKEF EPFEDILGLY EPGPEAPNNP LVAEVRKRIP SEFLRSILPN GSHDHPKMP 1260
LPNIIRSDVL KKAPEFKPGW RTDEEFARET LAGVNPVLIK RLTEFPAKST LDPSQYGDHT 1320
SKITEAHIQH NMEGLSVQNA LKKNRLFILD HHDHFMPLYN KINELEGNFI YASRTLFLK 1380
DDGTLKPLAV ELSLPHPDGQ QHGAVSKVYT PAHSGAEGHV WQLAKAYACV NDSAWHQLIS 1440
HWLNTHAVIE PFVIATNRQL SVVHPVHKLL SPHYRDTLNI NALARQTLIN ADGIFERTVF 1500
PAKALGMSS DVKSWNFNE QALPADLVKR GVAVPDQSSP YGVRLLIKDY PYAVDGLVIW 1560
WAIERWVKEY LDVYYPNDGE LQRDVELQAW WKEVREEAHG DLKDRDWWRP MDAVQRLARA 1620
CTTVIWVASA LHAANVFGQY PYAGYLPNRP TVSRPMPPEP GSDDYKKLEA GQKEADAVFI 1680
RTITSQFQTI LGISLIEILS KHSDEVYLGQ QRDEPERWTS DARALDAFRF FGSRLVEIEK 1740
RIRTMNDSPT LKNRKGPVEM PYMLLYPNTS DVTGEKGEGL TAMGIPNSIS I 1791

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SEQ ID NO: 18      moltype = DNA  length = 18
FEATURE            Location/Qualifiers
misc_feature         1..18
                    note = Primer S3460 for Lox3 of Zea mays
source              1..18
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 18
cctcgctgcag gccacaaa 18

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SEQ ID NO: 19      moltype = DNA  length = 19
FEATURE            Location/Qualifiers
misc_feature         1..19
                    note = Primer S3461 for Lox3 of Zey mays
source              1..19
                    mol_type = other DNA
                    organism = synthetic construct

```

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SEQUENCE: 19
acgtgacgcc gaacttgga 19

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SEQ ID NO: 20      moltype = DNA  length = 25
FEATURE            Location/Qualifiers
misc_feature         1..25

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source          note = Primer S3428 for Elongation factor 1-alpha
                1..25
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 20
ggtaaagaga agtcccacat caaca                               25

SEQ ID NO: 21      moltype = DNA length = 23
FEATURE           Location/Qualifiers
misc_feature       1..23
                  note = Primer S3429 for Elongation factor 1-alpha
source            1..23
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 21
ctccttctca aacctctega tca                               23

SEQ ID NO: 22      moltype = DNA length = 1913
FEATURE           Location/Qualifiers
source            1..1913
                  mol_type = genomic DNA
                  organism = Zea mays

SEQUENCE: 22
ttgtagttca tgcgcagtaa gagttgtaac ctcataatat gaccaaattc ctgatctgct 60
ttatagagtt gttgtgccta attttgaatt cgtcttactg gctcaaaatt atgtgcgcac 120
acctattagt ttgtctctta ttttctatgc ctgctttatt tttttatatt acattgattg 180
ctctgtttta tatttgcttc tttattttta cactgcttat gtgtatcaag cttttttcaa 240
gctcataact ccggttacaa cgtagatct agtttctgat cccatctgat aaatttgttt 300
ctaactcgtt attcgtcttg ttaacagttc aatctcacct ccaaccatgg gtaaaagagaa 360
gtcccacatc aacatttggt ttattggcca cgttgactct ggcaagtcga ccaccacagg 420
acacctgata tacaagcttg gaggcattga caagcgtgtg atcgagaggt ttgagaagga 480
ggctgctgaa atgaacaagc ggctcctcaa gtacgcatgg gtgctcgaca agctcaaggc 540
tgagcgtgag agagggtatc ccattgatat cgtctgtgtg aagtttgaga ccaccaagta 600
ctactgcacg gtcattgatg cccctggaca ccgtgacttc atcaagaaca tgatcactgg 660
tacctcccag gctgactgtg ctgtccttat cattgactcc accactgggtg gttttgaggc 720
tggtatctcc aaggatggcc agaccctgta acatgctctc cttgcgttca cccttgagg 780
gaagcagatg atttgcctgt gcaacaagat ggatgcaacc actccaaat actccaaggc 840
acgttatgat gagatttgta aggaagtctc atcctacctc aagaaagtgt ggtacaaccc 900
tgataagatt gccttcgttc ccatttctgg ttttgagggc gacaacatga ttgagaggtc 960
caccacacct cactggtaaga aaggcccaac cctgcttgag gctcttgacc agatcaccca 1020
gcccagaggc ccttcggaca agccctgcg tctgcccctc caggatgtgt acaagattgg 1080
tggtattgga actgtaccgg ttggtcgtgt ggagactggt gtcatcaagc ctggtatggt 1140
agtcaccttt ggtcctgact gacctgactac tgaggtgaag tctgttgaga tgcaccatga 1200
ggcgcttcag gaggtctctc cggcgacaaa tgttggtctc aacgtgaaga acgttgctgt 1260
caaggatctc aagcgtgggt tctgtggctc caactccaag gatgacctg ccaaggaggc 1320
tgccagcttc acctccagc tcatcatcat gaaccacct gggcagattg gcaacggcta 1380
tgccccagtg ctggactgcc acacctccca catcgccgtc aagtttgctg agctcattac 1440
caagatcgac aggcgatctg gcaaggagct cgagaaggag cccaagtcc tgaagaacgg 1500
tgatgctggt atggtgaaaga tgattccac caagcctatg gtggtggaga cattctccgc 1560
gtatcctccc ctgggtaggt ttgcctgctc cgacatgagg cagacggttg ctggtggagt 1620
catcaagagt gtgagaaga aggacccaac cggcgccaag gtgaccaagg cggcgcccaa 1680
gaagaaatga tgcgatcccg tctctgcttt tgcagtgcct ggataacttg ccactaaagt 1740
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tgggtagcgc agcgtcggcc tcttcagggt tctgttgtct gttgtcttta ctatcagtaa 1860
tgctcctgag acctgtatct ctccttttgg cgccgctaaa tggaaatttc gaa          1913

SEQ ID NO: 23      moltype = AA length = 447
FEATURE           Location/Qualifiers
source            1..447
                  mol_type = protein
                  organism = Zea mays

SEQUENCE: 23
MGKEKSHINI VVIGHVDSGK STTGHLYIK LGGIDKRVIE RFEKEAAEMN KRSFKYAWVL 60
DKLKAERERG ITIDIALWKF ETTKYVCTVI DAPGHRDFIK NMITGTSQAD CAVLIIDSTT 120
GGFEAGISKD QOTREHALLA FTLGVKQMIC CCNKMDATTP KYSKARYDEI VKEVSSYLKK 180
VGYNPDKIAF VPISGFEGDN MIERSTNLWD YKGPTLLEAL DQITEPKRPS DKPLRLPLQD 240
VYKIGGIGTV PVGRVETGVI KPGMVVTFGP TGLTTEVKSV EMHHEALQEA LPGDNVGFNV 300
KNVAVKDLKR GFVASNSKDD PAKEAASFTS QVIIMNHPGQ IGNGYAPVLD CHTSHIAVKF 360
AELITKIDRR SGKELEKEPK FLKNGDAGMV KMIPTKPMVV ETFSAYPPLG RFAVRDMRQT 420
VAVGVKSIVE KKDPTGAKVT KAAAKKK                               447

SEQ ID NO: 24      moltype = AA length = 494
FEATURE           Location/Qualifiers
source            1..494
                  mol_type = protein
                  organism = Zea mays

SEQUENCE: 24
MDTSHHYHPW LNFSLAHCD LEEERGAAG ELAAIAGAAP PKLEDPLGG GVATGGPEAV 60
APAEMYDSDL KFIAAAGFLG GSAAAAATSP LSSLDQAGSK LALPAAAAAP APEQRKAVDS 120

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FGQRTSIYRG VTRHRWTGRY EAHLWDNSCR REGQSRKGRQ VYLGGYDKEE KAARAYDLAA 180
LKYWGSSTTT NFPVAEYEKE VEEKNMTRQ EFVASLRRKS SGFSRGASIY RGVTRHHQH 240
RWQARIGRVA GNKDLYLGTF STEEEAAEAY DIAAIKFRGL NAVTNFEISR YNVETIMSSN 300
LPVASMSSSS AAAAGGRSSK ALESPPSGSL DGGGMPVVE GSTAPPLFIP VKYDQQQOEY 360
LSMLALQHHH QQQQAGNLLQ GPLVGFGLY SSGVNLDPAN SHGTAAPSSM AHHCYANGTA 420
SASHEHQHH QMQQGENET QPQPQQSSSS CSSLPFATPV AFNGSYESSI TAAGPFGYSY 480
PNVAAPQTPI YGME 494

SEQ ID NO: 25      moltype = DNA length = 1485
FEATURE           Location/Qualifiers
source            1..1485
                  mol_type = genomic DNA
                  organism = Zea mays

SEQUENCE: 25
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ctcgaggagg aggagagggg cgcggccggc gagctggccg cgatagccgg cgccgcgcgg 120
cgcgcgaagc tggaggactt cctcggcggg ggcgtcgcca ccggtggtcc ggaggcggtg 180
ggcgcgcggc agatgtacga ctccggacctc aagttccatg ccgcgcgcgg gtcccttggc 240
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gaggcacatc tgtgggacaa cagctgccga cgcgaagggc agagccgcaa gggccgcaa 480
gtatatattg gtggtatga taaggaggag aaggctgcca gggcgatga tcttgagct 540
tgtgaagtact ggggttctag caccaccacc aactttccgg ttgctgagta tgagaaggag 600
gtcggaggaga tgaagaacat gacgcgacaa gagttgtgtt cttcccttcg aaggaagagc 660
agtggattct ctccgggtgc ttccatctac agaggtgtaa ccagacatca ccagcatgga 720
cgggtggcagg cgaggatcgg aaggggtggc ggtaacaagg acctctacct tgggacgttc 780
agcaccgagg aggaagctgc agaggcctac gacatagcgg ccatcaagtt cagaggcctg 840
aacgcctgca caaacttga gatcagccgg tacaacgtgg agaccataat gacgagcaac 900
cttcacgtcg cgacgatgct gtcgtcgctc gcggcggcgg cgggtggcgg gacgagcaag 960
gcgctggagt cccctccgtc cggctcgctt gacggcggcg gcggcatgcc agtctcgtaa 1020
ggcagcagcg caccgcgcgt gttcattccg gtgaagtacg accagcagca gcaggagtac 1080
ctgtcgatgc tcgcgttgca gcaccaccac cagcagcaac aagcagggaa cctgttgtag 1140
gggcgcgtag tagggttcgg cggcctctac tctccggggg tgaacctgga ttccgccaac 1200
tcccacggca cggcggctcc gtcgtcgatg gcccaccact gctacgcaa tggcaccgcg 1260
tccgcctcgc atgagcacca gcaccagcac cagatgcagc agggcggcga gaacgagacg 1320
cagcgcgagc cgcagcagag ctcacgagc tgctcctccc tgccattcgc caccocggtc 1380
gctttcaatg ggtcctatga aagctccatc acggcggcag gcccttttgg atactcctac 1440
ccaaatgtgg cagcctttca gacgcgcgatc tatggaatgg aatga 1485

SEQ ID NO: 26      moltype = AA length = 164
FEATURE           Location/Qualifiers
REGION            1..164
                  note = Regeneration booster P8
source            1..164
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 26
MVSKGEEVIK EFSGSTRWNP TAEQVKVLR E IYRAGLRTP SADQIRQISS QLKRYGKVEG 60
KNVFWFQNH KARERQKRL STVGSAGSAA GSGEFRHVKT LPLFPLHNE DSGSGLDL 120
NLELRSLATA AATSGSQRAS ALDLLELRL GFALGNESCG LHDN 164

SEQ ID NO: 27      moltype = DNA length = 498
FEATURE           Location/Qualifiers
misc_feature      1..498
                  note = Regeneratio booster P8 CDS
source            1..498
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 27
atgggtttcg agggggaaga agtcattaaa gagttttcag ggtctacgcg ctggaacctc 60
accgtgaac aagtcagggt tttgagggaa atatatattacc gcgcgggtct taggacgcca 120
tccgcggatc agatccgcca aatatcctca cagctgaaac ggtatggaaa ggtggaagga 180
aaaaacgtgt tttattgggt ccagaacctt aaagcgcggg aaagacaaaa aaagcggctc 240
agcaccgttg ggagcgtcgg ctcagcagcg ggttcagggt agttcaggca cgttaaaaca 300
ctgccttctg tccctttgca caataatgaa gactcaggga gcggtttgga cttggatctg 360
aaccttgagt tgcgcctgtc ggcgactgct gctgccactt cgggctcaca gcgggcgtct 420
gccctcgatc tggacttgga gctcagactc ggtttcgcgc ttggaatga gtcgtgtgga 480
ctgcacgaca actagtga 498

SEQ ID NO: 28      moltype = AA length = 1252
FEATURE           Location/Qualifiers
REGION            1..1252
                  note = LbCpf1-RR
source            1..1252
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 28
MAPKKRKVS KLEKFTNCYS LSKTLRFKAI PVGKTQENID NKRLLEDEK RAEDYKGVKK 60

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LLDRYYLSFI	NDVLHSIKLK	NLNNYISLFR	KKTRTEKENK	ELENLEINLR	KEIAKAFKGN	120
EGYKSLFKKD	IIETILPEFL	DDKDEIALVN	SFNGFTTAF	GFFDNRENMF	SEEAkstSIA	180
PRCINENLTR	YISNMDIFEK	VDAIFDKHEV	QEIKEKILNS	DYDVEDFFEG	EFFNFVLTQE	240
GIDVYNAIIG	GFVTESGEKI	KGLNEYINLY	NQKTKQKLPK	PKPLYKQVLS	DRESLSFYGE	300
GYSDEEVL	VFRNTLNKNS	EIFSSIKKLE	KLFKNFDEYS	SAGIFVKNGP	AISTISKDIF	360
GEWNVIRDKW	NAEYDDIHLK	KKAVVTEKYE	DDRRKSFKKI	GSFSLEQLQE	YADADLSVVE	420
KLKEIIIQKV	DEIYKVYSS	EKLFDADFVL	EKSLKKNDAV	VAIMKDLLDS	VKSFENYIKA	480
FFGEGKETNR	DESFGDFVL	AYDILLKVDH	IYDAIRNYVT	QKPYSKDKFK	LYFQNPQFMR	540
GWDDKDKETD	RATILRYGSK	YYLAIMDKKY	AKCLQKIDKD	DVNGNYEKIN	YKLLPGPNKM	600
LPRVFFSKKW	MAYYNPSEDI	QKIYKNGTFK	KGDMFNLNDC	HKLIDFFKDS	ISRYPKWSNA	660
YDFNFSETEK	YKDIAGFYRE	VEEQGYKVSF	ESASKKEVDK	LVEEGKLYMF	QIYNKDFSDK	720
SHGTPNLHTM	YFKLLFDENN	HGQIRLSGGA	ELFMRRASLK	KEELVVHPAN	SPIANKNPDN	780
PKKTTLSYD	VYKDKRFS	QYELHIPIAI	NKCPKNIFKI	NTEVRVLKX	DDNPYVIGID	840
RGERNLIIYV	VVDGKGNIVE	QYSLNEIINN	FNGIRIKTDY	HSLLDKKEKE	RFEARQNWTS	900
IEINIKELKAG	YISQVVKHIC	ELVEKYDAVI	ALEDLNSGFK	NSRVKVEKQV	YQKFEKMLID	960
KLNYMVDKKS	NPCATGGALK	GYQITNKFES	FKSMSTQNGF	IFYIPAWLTS	KIDPSTGFVN	1020
LLKTKYTSIA	DSKKFISSFD	RIMYVPEEDL	FEFALDYKNF	SRTDADYIKK	WKLYSYGNRI	1080
RIFRNPFKNN	VFDWEEVCLT	SAYKELFNKY	GINYQQGDIR	ALLCEQSDKA	FYSSFMALMS	1140
LMLQMRNSIT	GRTDVDFLIS	PVKNSDGI	FSRNYEAQEN	AILPKNADAN	GAYNIARKVL	1200
WAIGQFKKAE	DEKLDKVKIA	ISNKEWLEYA	QTSVKHKRPA	ATKAGQAKK	KK	1252

SEQ ID NO: 29 moltype = DNA length = 3759
 FEATURE Location/Qualifiers
 misc_feature 1..3759
 note = LbCpf1-RR CDS
 source 1..3759
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 29

atggcgccga	agaagaagcg	caagggtgtcc	aagctcgaga	agttcacgaa	ctgctactcc	60
ctctccaaga	ccctccgctt	caaggccatc	cccggtggca	agacccagga	gaacatcgac	120
aacaagcgcc	tcctggttga	ggacgagaag	agggcgagg	actacaaggg	cgtgaagaag	180
ctcctggacc	gctactacct	ctccttcatc	aacgacgtcc	tgacacagcat	caagctcaag	240
aacctgaaca	ctacatcttc	ctgtttccgc	aagaagacga	ggacggagaa	ggagaacaag	300
gagctcgaga	acctggagat	caacctccgc	aaggagatcg	ccaaggcgtt	caaggggcaac	360
gagggctaca	agagcctgtt	caagaaggac	atcatcgaga	cgatcctccc	ggagtctctg	420
gacgacaagg	acagatctgc	ctcgtgaac	tccttcaacg	gcttcaccac	ggcgttcacc	480
ggcttcttcg	acaaccgcga	gaacatgttc	agcgaggagg	ccaagtccac	gagcatcgcg	540
ttccgctgca	tcaacgagaa	cctgaccagg	tacatctcca	acatggacat	cttcgagaag	600
gtcgacgcca	ttctcgacaa	gcacgaggtg	caggagatca	aggagaagat	cctcaacagc	660
gactacgacg	tcgaggactt	cttcgagggc	gagttcttca	acttcgtcct	gacgcaggag	720
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gagattcttc	ccagctatca	gaagctcgag	aagctgttca	agaacttcga	cgagtactcc	1020
agcgccggca	ttctcgtcaa	gaacggcccg	gcgactctca	cgatcagcaa	ggatatcttc	1080
ggcgagtgga	acgtgatcag	ggacaagtg	aacgcggagt	acgacgacat	ccacctcaag	1140
aagaaggcgg	tggtcaccga	gaagtacgag	gacgaccgca	ggaagtccct	caagaagatc	1200
ggctccttca	gcttcgagca	gttcgaggag	tacgcccagc	cggacctctc	cgtggctcag	1260
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gagaagctgt	tcgacgcgca	cttcgtcttc	gagaagtcct	tgaagaagaa	cgacgcctgt	1380
gtcgcgatca	tgaaggacct	cctggactcc	gtgaagagct	tcgagaacta	catcaaggcg	1440
ttcttcggcg	agggcaagga	gacgaaccgc	gacgagtcct	tctacggcga	cttcgtcttc	1500
gcctacgaca	tcctcctgaa	ggtggaccac	atctacgacg	cgatcaggaa	ctacgtgacc	1560
cagaagccgt	acagcaagga	caagttcaag	ctgtacttcc	agaaccccc	gttcatgcgc	1620
ggctgggaca	aggacaagga	gacggactac	cgcgccacca	tccttcgcta	cggcgagcaag	1680
tactacctgg	ccatcatgga	caagaagtac	gcgaagtgc	tcagaagat	cgacaaggac	1740
gacgtcaacg	gcaactacga	gaagatcaac	tacaagctcc	tgccggggcc	caacaagatg	1800
ctgccgaggg	tggtctcttc	caagaagtg	atggcctact	acaaccccag	cgaggacatc	1860
cagaagatct	acaagaacgg	cacgttcaag	aaggcgacga	tggtcaacct	caacgactgc	1920
cacaagctga	tcgacttctt	caaggactcc	atcagccgct	accggaagt	gtccaacgcc	1980
tacgacttca	acttcagcga	gacagagaag	tacaaggaca	tcgcgggctt	ctacagggag	2040
gtcgaggagc	agggtctcaa	ggtgtccttc	gagtcggcca	gcaagaagga	ggtcgacaag	2100
ctcgtggagg	agggcaagct	gtacatgttc	cagatctaca	acaaggactt	ctccgacaag	2160
agccacggca	cgcccaacct	ccacaccatg	tacttcaagc	tcctgttcga	cgagaacaac	2220
cacggccaga	tcgcgctctc	cggcgggcgc	gagctgttca	tgaggagggc	gagcctcaag	2280
aaggagagcg	tgggtgtcca	ccccgctaac	agcccaatcg	cgaacaagaa	cccggaacaac	2340
ccaagaagaa	ccacgacctc	ctcctacgac	gtgtacaagg	acaagcgctt	cagcgaggac	2400
cagtacgagc	tgacatccc	gatcgccatc	aacaagtgc	ccaagaacat	cttcaagatc	2460
aacaccgagg	tcagggtgct	cctgaagcac	gacgacaacc	cctacgtgat	cggcatcgac	2520
cgcggcgaga	ggaacctctc	gtacatcgtg	gtcgtggacg	gcaagggcaa	catcgtggag	2580
cagtactccc	tgaacgagat	catcaacaac	ttcaacggca	tcggcatcaa	gacggactac	2640
cacagcctcc	tggaacaagaa	ggagaaggag	cgttcgagg	ccaggcagaa	ctggacctcc	2700
atcgagaaca	tcaaggagct	caaggcgggc	tacatcagcc	aggtcgtgca	caagatctgc	2760
gagctggtcg	agaagtacga	cgcctggtat	gcgctcgagg	acctgaactc	cggcttcaag	2820
aacagcaggg	tcaaggtgga	gaagcaggtc	taccagaagt	tcgagaagat	gctcatcgac	2880
aagctgaact	acatgggtgga	caagaagtcc	aaccggtgcg	ctacggggcg	cgcgctcaag	2940
ggctaccaga	tcaccaacaa	gttcgagagc	ttcaagtcca	tgagcaccga	gaacggcttc	3000

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atctttctaca tcccggcctg gctgacgtcc aagatcgacc ccagcaccgg cttegtcaac 3060
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cgcatcatgt atgtgccgga ggaggacctc ttcgagttcg ccttgagacta caagaacttc 3180
tccaggacgg acgcggtatta catcaagaag tggaaagtct acagctacgg caaccgcate 3240
aggatcttcc gcaaccccaa gaagaacaac gtcttcgact gggaggaggt gtgcctcacc 3300
tccgcctaca aggagctgtt caacaagtac ggcatacaact accagcaggg cgacatcagg 3360
gcgctctctg gcgagcagag cgacaaggcc ttctactcca gcttcatggc gctcatgtcc 3420
ctcatgtgct agatgcgcaa cagcatcacg ggcaggaccg acgtcgactt cctgatctcc 3480
ccggtgaaga acagcgacgg catcttctac gacagccgca actacgaggg ccaggagaac 3540
gcgatctctg caaagaacgc ggacgccaac ggcgcttaca acatcgcgag gaagggtctg 3600
tgggccaatg gccagttcaa gaaggcggag gacgagaagc tcgacaaggt caagatcgcc 3660
atctccaaca aggagtggtt ggagtagcgg cagacctcgg tgaagcaca gaggcccgct 3720
gccaccaaga aggcggggcca ggccaagaag aagaagtga 3759

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SEQ ID NO: 30      moltype = AA length = 238
FEATURE           Location/Qualifiers
REGION            1..238
                  note = tdTomato
source            1..238
                  mol_type = protein
                  organism = synthetic construct

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SEQUENCE: 30
MVSKGEEVIK EFMRFKVRME GSMNGHEFEI EGEGERPYE GTQAKLKVT KGGPLPFAWD 60
ILSPQFMYGS KAYVKHPADI PDYKKLSFPE GFKWERVMNF EDGLVTVTQ DSSLQDGTLI 120
YKVKMRGTNF PPDGPMQKK TMGWEASTER LYPRDGLVKI EHQALKLKD GGHYLVFCKT 180
IYMAKKPVQL PGYYYVDTKL DITSHNEDYT IVEQYERSEG RHHFLFLYGM D ELYKSRGT 238

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SEQ ID NO: 31      moltype = DNA length = 717
FEATURE           Location/Qualifiers
misc_feature       1..717
                  note = tdTomato
source            1..717
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 31
atggtgagca agggcgagga ggtcatcaaa gagttcatgc gttcaagggt gcgcatggag 60
ggctccatga acggccacga gttcgagatc gagggcgagg gcgagggccg cccctacgag 120
ggcaccacga ccgccaagct gaaggtgacc aaggcgggcc ccctgccctt cgctggggac 180
atcctgtccc ccagttcat gtacggctcc aaggcgtacg tgaagcacc cgcgcacatc 240
cccgattaca agaagctgtc ctcccccag ggttcaagt gggagcgct gatgaacttc 300
gaggacggcg gttcgggtgac cgtgacccag gactcctccc tgcaggacgg cacgctgac 360
tacaaggtag agatgcgcgg caccacactt cccccgcag gccccgtaac gcagaagaag 420
accatgggct gggaggcctc caccgagcgc ctgtaccccc gcgacggcgt gctgaagggc 480
gagatccacc aggcctgtaa gctgaaggac ggcggccact acctggtgga gttcaagacc 540
atctacatgg ccaagaagcc cgtgcaactg ccgcgctact actacgtgga caccaagctg 600
gacatcacct cccacaacga ggactacacc atcgtggaac agtacgagcg ctccgagggc 660
cgccaccacc tgttctctga cggcatggac gagctgtaca agtctagagg tacctga 717

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```

SEQ ID NO: 32      moltype = DNA length = 21
FEATURE           Location/Qualifiers
misc_feature       1..21
                  note = Protospacer m7GEP336
source            1..21
                  mol_type = other DNA
                  organism = synthetic construct

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```

SEQUENCE: 32
gcacgttctt gcgcatgagc a 21

```

```

SEQ ID NO: 33      moltype = DNA length = 21
FEATURE           Location/Qualifiers
misc_feature       1..21
                  note = Protospacer m7GEP337
source            1..21
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 33
gcgccaccgt cggtgacagc a 21

```

```

SEQ ID NO: 34      moltype = DNA length = 21
FEATURE           Location/Qualifiers
misc_feature       1..21
                  note = Protospacer m7GEP338
source            1..21
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 34
ctgtgcagac aacggcaacc g 21

```

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SEQ ID NO: 35      moltype = DNA length = 21

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FEATURE                               Location/Qualifiers
misc_feature                           1..21
                                         note = Protospacer m7GEP339
source                                1..21
                                         mol_type = other DNA
                                         organism = synthetic construct

SEQUENCE: 35
agctctctcca cctcccagtc g                                     21

SEQ ID NO: 36                               moltype = AA length = 1791
FEATURE                               Location/Qualifiers
REGION                                1..1791
                                         note = Zm00008a004913
source                                1..1791
                                         mol_type = protein
                                         organism = synthetic construct

SEQUENCE: 36
MLSGIIDGLT GANKHARLKG TVVLMRKINV L D L N D F G A T V V D S I S E F L G K G V T C Q L I S S T L 60
VDANNGNRGR VGAEANLEQW LTSLSPLTTG ESKFGVTFDW EVEKLGVPGA VVVKNNHAAE 120
FFLKTTITLDD VPGRGAVTFV ANSWVYPAGK YRYNRVFFSN DTYLPSQMPA ALKPYRDEL 180
RNLRGDDQQG PYQEHDRVYR YDVYNDLGEP DGGNPRPILG GSADHPYPRR CRTGRKPTKT 240
DPNSESRLSL VEQIYVPRDE RFGHLKMSDF LGYSIKAITQ GIIPAVRTYV DTPGFEFDSF 300
QDIINLYEGG IKLPKIQALE DMRKLFPLQL VKDLLPAGGD YLLKLPIQOI IQGTSRIPID 360
VRGRLLSGLH IYMCSEYEDKN AWRTEDEFAR EVLAGVNPVM ITRLTEFPPK STLDPSKYGD 420
HTSTITAETH EKNLEGLTVQ QALDGNRLYI LDHDFRMPMF LIDVNNLEGN FIYATRTLFF 480
LRGDGRLAPL AIELSEPYID GDLTVAKSKV YTPASSGVEA WVWQLAKAYV AVNDSGWHQL 540
VSHWLNTHAV MEPPVIATNR QLSVTHPVHK LLSHFRDTM TINALARQTL INGGGIFEMT 600
VPPGKYALGM SSVVYKSWNF TEQGLPADLV KRGVAVADPS SPYKVRLLIE DYPYASDGLA 660
IWHAEIQWVG EYLAIYYPDD GALRGDEELQ AWWKEVREVG HGDHKDAPWW PKMQAVSELA 720
SACTTIIWIA SALHAAVNFG QYPYAGYLPN RPTVSRRRMP EPGSKEYEEL ERDPERGFIH 780
TITSQIQTII GISLIEILSK HSSDEVYLGQ RDTPEWTSDA RALAFAKRF S DALVKIEGKV 840
VGENRDPQLR NRNGPAEFPP MLLYPNTSDH SEARERAAGQ GIHTHTHRTL PEKREARSEE 900
RPATMFHWGV ADRLTGKNKE AWWNEGKIRGT VRLVKKEVLD VGDFFNASLLD GVHRILGWDD 960
GVAFQLVSAT AADPSNRSRG KVGKAHLEE AVVSLKSTD GETVYRVSEF WDGSGQVPGA 1020
VLVRNLQHA EFLKSLTLEG VPGRGTVVVF ANSWIYPHNL YSQRVFFAN DTYLPSKMPA 1080
ALVPYRQDEL KILRGDDNPG PYKEHDRVYR YDVYNDLGEP DKGEDHARPV LGGSQEHYPY 1140
RRCRTGRRPT ETDPNSESRL FLLNLNIYVP RDERFGHLKM SDFLGYSLKA IEAVLPTLG 1200
TFVDDTPKEF DSFEDILGLY EPGPEAPNPN LVAEVRKRIP SEFLRSILPN GSHDHPLKMP 1260
LPNIIRSDVL KKAPEFKFGW RTDEEFARET LAGVNPVLK RLTFFPAKST LDPSQYGDHT 1320
SKITEAHIGH NMEGLSVQNA LKKNRLFILD HHDHFMPLYN KINELEGNFI YASRTLFLK 1380
DDGTLKPLAV ELSLPHPDGQ QHGAUSKVYT PAHSGAEGHV WQLAKAYACV NDSAWHQLIS 1440
HWNTHAVIE PFVIATNRQL SVVHPVHKLL SPHYRDTLNI NALARQTLIN ADGIFERTVF 1500
PAKYALGMS SSVVYKSWNF QALPADLVKR GVAVPDQSSP YGVRLLIKDY PYAVDGLVIV 1560
WAIERWVKEY LDVYYPNDGE LQRDVELQAW WKEVREEAHG DLKDRDWWPR MDAVQRLARA 1620
CTTVIWSAVA LHAAVNFGQY PYAGYLPNRP TVSRRRMPPE GSDDYKKLEA GQKEADAVFI 1680
RTITSQFQTI GISLIEILSK HSSDEVYLGQ QRDEPERWTS DARALDAFR FGSRLVEIEK 1740
RIRTMDSPT LKNRKGPEVM PYMLLYPNTS DVTGEKGEGL TAMGIPNSIS I 1791

SEQ ID NO: 37                               moltype = AA length = 1790
FEATURE                               Location/Qualifiers
REGION                                1..1790
                                         note = WVP18-09358-016
source                                1..1790
                                         mol_type = protein
                                         organism = synthetic construct

SEQUENCE: 37
MLSGIIDGLT GANKHARLKG TVVLMRKINV L D L N D F G A T V V D S I S E F L G K G V T C Q L I S S T L 60
VDANNGNRGR VGAEANLEQW LTSLSPLTTG ESKFGVTFDW EVEKLGVPGA VVVKNNHAAE 120
FFLKTTITLDD VPGRGAVTFV ANSWVYPAGK YRYNRVFFSN DTYLPSQMPA ALKPYRDEL 180
RNLRGDDQQG PYQEHDRVYR YDVYNDLGEP DGGNPRPILG GSADHPYPRR CRTGRKPTKT 240
DPNSESRLSL VEQIYVPRDE RFGHLKMSDF LGYSIKAITQ GIIPAVRTYV DTPGFEFDSF 300
QDIINLYEGG IKLPKIQALE DMRKLFPLQL VKDLLPAGGD YLLKLPIQOI IQGTSRIPID 360
VRGRLLSGLH IYMCSEYEDKN AWRTEDEFAR EVLAGVNPVM ITRLTEFPPK STLDPSKYGD 420
HTSTITAETH EKNLEGLTVQ QALDGNRLYI LDHDFRMPMF LIDVNNLEGN FIYATRTLFF 480
LRGDGRLAPL AIELSEPYID GDLTVAKSKV YTPASSGVEA WVWQLAKAYV AVNDSGWHQL 540
VSHWLNTHAV MEPPVIATNR QLSVTHPVHK LLSHFRDTM TINALARQTL INGGGIFEMT 600
VPPGKYALGM SSVVYKSWNF TEQGLPADLV KRGVAVADPS SPYKVRLLIE DYPYASDGLA 660
IWHAEIQWVG EYLAIYYPDD GALRGDEELQ AWWKEVREVG HGDHKDAPWW PKMQAVSELA 720
ACTTIIWIAS ALHAAVNFGQ YPYAGYLPN RPTVSRRRMP PGSKEYEEL RDPERGFHT 780
ITSQIQTII GISLIEILSK HSSDEVYLGQ RDTPEWTSDA RALAFAKRF S DALVKIEGKV 840
GENRDPQLR NRNGPAEFPP MLLYPNTSDH SEARERAAGQ GIHTHTHRTL PEKREARSEE 900
PATMFHWGVA DRLTGKNKEA WNEGKIRGT VRLVKKEVLDV GDFNASLLD VHRILGWDDG 960
VAFQLVSATA AADPSNRSRG VVSLKSTDG ETVYRVSEF WDGSGQVPGA 1020
LVRNLQHA EFLKSLTLEG VPGRGTVVVF ANSWIYPHNL YSQRVFFAN DTYLPSKMPA 1080
LVPYRQDEL KILRGDDNPG PYKEHDRVYR YDVYNDLGEP DKGEDHARPV LGGSQEHYPY 1140
RRCRTGRRPT ETDPNSESRL FLLNLNIYVP RDERFGHLKM SDFLGYSLKA IEAVLPTLG 1200
TFVDDTPKEF DSFEDILGLY EPGPEAPNPN LVAEVRKRIP SEFLRSILPN GSHDHPLKMP 1260
PNIIRSDVL KKAPEFKFGW RTDEEFARET LAGVNPVLK RLTFFPAKST LDPSQYGDHT 1320
KITEAHIGH NMEGLSVQNA LKKNRLFILD HHDHFMPLYN KINELEGNFI YASRTLFLK 1380

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DGTLKPLAVE	LSLPHPDGQQ	HGAVSKVYTP	AHSGAEGHVW	QLAKAYACVN	DSAWHQLISH	1440
WLNTHAVIEP	FVIATNRQLS	VVHPVHKLLS	PHYRDTLNI	ALARQTLINA	DGIFERTVFP	1500
AKYALGMSSD	VYKSWNFNEQ	ALPADLVKRG	VAVPDQSSPY	GVRLLIKDYD	YAVDGLVIWW	1560
AIERWVKEYL	DVYYPNDGEL	QRDELQAWW	KEVREEAHGD	LKDRDWWPRM	DAVQRLARAC	1620
TTVIWVASAL	HAAVNFGQYP	YAGYLPNRP	VSRRPMPEPG	SDDYKKLEAG	QKEADAVFIR	1680
TITSQFQITL	GISLIEILSK	HSSDEVYLGQ	RDEPERWTS	ARALDAFRFR	GSRLVEIEKR	1740
IRTMNDSPTL	KNRKGVPVEM	YMLLYPNTSD	VTGEKGEGLT	AMGIPNSISI		1790

SEQ ID NO: 38 moltype = AA length = 1791
 FEATURE Location/Qualifiers
 REGION 1..1791
 note = WVP18-09344-014
 source 1..1791
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 38

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VDANNNGRR	VGAEANLEQW	LTSLSPLTTG	ESKFGVTFDW	EVEKLGVPGA	VVVKNNHAAE	120
FFLKTTITLDD	VPGRGAVTFV	ANSWVYPAGK	YRYNRVFFSN	DTYLP SQMPA	ALPKYRDEL	180
RNLRGDDQQG	PYQEHDRVYR	YDVYNDLGEP	DGGNRPILG	GSADHPYPRR	CRTGRKPTKT	240
DPNSESRLSL	VEQIYVPRDE	RFGHLKMSDF	LGYSIKAITQ	GIIPAVRTYV	DTTPGEFDSF	300
QDIINLYEGG	IKLPKIQALE	DMRKLFPQLQ	VKDLLPAGGD	YLLKLPIQOI	IQGTSRIPID	360
VRGRLLSGLH	IYMCSEYEDN	AWRTDEEFAR	EVLAVGNPMV	ITRLTEFPPK	STLDPSKYGD	420
HTSTITAETH	EKNLEGLTVQ	QALDGNRLYI	LDHHRFMPF	LIDVNNLEGN	FIYATRTLFF	480
LRGDGRLAPL	AIELSEPYID	GDLTVAKSKV	YTPASSGVEA	WVWQLAKAYV	AVNDSGWHQL	540
VSHWLNTHAV	MEPFVIATNR	QLSVTHPVHK	LLSSHFRDTM	TINALARQTL	INGGGIFEMT	600
VFPQYALGMS	SSVYKSWNF	TEQGLSADLV	KRGVAVADPS	SPYKVRLLIE	DYPYASDGLA	660
IWHAIEQVVG	EYLAIIYPDD	GALRGDEELQ	AWWKEVREVG	HGDHKDAPWW	PKMQAVSELA	720
SACTTIWIA	SALHAAVNFG	QYPYAGYLPN	RPTVSRRRMP	EPGSKEYEEL	ERDPERGFH	780
TITSQIQTII	GISLIEILSK	HSSDEVYLGQ	RDTPEWTS	DALAAFKRFS	DALVKIEGKV	840
VGENRDPQLR	NRNGPAEFPP	MLLYPNTSDH	SEAREERAAG	GIHTHTHRTL	PEKREARSEE	900
RPATMFHWGV	ADRLTGKNKE	AWNKGKIRGT	VRLVKKEVLD	VGDFNASLLD	GVHRILGWDD	960
GVAFQLVSAT	AADPSNGSRG	KVGKAAHLEE	AVVSLKSTTD	GETVYRVSEF	WDGSGQVPGA	1020
VLVRNLQHA	FLKSLTLEGG	VPGRGTVVVF	ANSWIYPHNL	YSQERVFAN	DTYLP SKMPA	1080
ALVPYRQDEL	KILRGDDNPG	PYKEHDRVYR	YDYNDLGEP	DKGEDHARPV	LGGSQEHYPY	1140
RCRGTGRRPT	ETDPNSESRL	FLNLNIYVPR	RDERFGHLKM	SDFLGYSILKA	IEAVLPTLGT	1200
TFVDDTPKEF	DSFEDILGLY	EPGPEAPNPN	LVAEVRKRIP	SEFLRSILPN	GSHDHPKMLP	1260
LPNIIRSDVL	KAPEFKFGWR	RTDEEFARET	LAGVNPVLIK	RLTEFPAKST	LDPSQYGDHT	1320
SKITEAHIQH	NMEGLSVQNA	LKKNRLFILD	HHDFMPYLNK	KINELEGNFI	YASRTLLFLK	1380
DGTLKPLAVE	LSLPHPDGQQ	HGAVSKVYTP	PAHSGAEGHV	WQLAKAYACV	NDSAWHQLIS	1440
HWLNTHAVIE	PFVIATNRQL	SVVHPVHKLL	SPHYRDTLNI	NALARQTLIN	ADGIFERTVFP	1500
PAKYALGMS	DVYKSWNFNE	QALPADLVKR	GAVVPDQSSP	YGVRLLIKDY	PYAVDGLVIW	1560
WAIERWVKEY	LDVYYPNDG	LQRDELQAWW	WKEVREEAHG	DLKDRDWWPR	MDAVQRLARA	1620
CTTVIWVASA	LHAAVNFGQY	PYAGYLPNRP	TVSRRPMPEP	GSDDYKKLEA	GQKEADAVFI	1680
RTITSQFQIT	LGISLIEILS	KHSDEVYLGQ	QRDEPERWTS	DARALDAFRF	FGSRLVEIEK	1740
RIRTMNDSPT	LKNRKGVPVEM	PYMLLYPNTS	DVTGEKGEGL	TAMGIPNSIS	I	1791

SEQ ID NO: 39 moltype = AA length = 1790
 FEATURE Location/Qualifiers
 REGION 1..1790
 note = WVP18-09309-014_339-009
 source 1..1790
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 39

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FFLKTTITLDD	VPGRGAVTFV	ANSWVYPAGK	YRYNRVFFSN	DTYLP SQMPA	ALPKYRDEL	180
RNLRGDDQQG	PYQEHDRVYR	YDVYNDLGEP	DGGNRPILG	GSADHPYPRR	CRTGRKPTKT	240
DPNSESRLSL	VEQIYVPRDE	RFGHLKMSDF	LGYSIKAITQ	GIIPAVRTYV	DTTPGEFDSF	300
QDIINLYEGG	IKLPKIQALE	DMRKLFPQLQ	VKDLLPAGGD	YLLKLPIQOI	IQGTSRIPID	360
VRGRLLSGLH	IYMCSEYEDN	AWRTDEEFAR	EVLAVGNPMV	ITRLTEFPPK	STLDPSKYGD	420
HTSTITAETH	EKNLEGLTVQ	QALDGNRLYI	LDHHRFMPF	LIDVNNLEGN	FIYATRTLFF	480
LRGDGRLAPL	AIELSEPYID	GDLTVAKSKV	YTPASSGVEA	WVWQLAKAYV	AVNDSGWHQL	540
VSHWLNTHAV	MEPFVIATNR	QLSVTHPVHK	LLSSHFRDTM	TINALARQTL	INGGGIFEMT	600
VFPQYALGMS	SSVYKSWNF	TEQGLPADLV	KRGVAVADPS	SPYKVRLLIE	DYPYASDGLA	660
IWHAIEQVGE	YLAIIYPDDG	ALRGDEELQA	WWKEVREVG	GDHDKDAPWW	KMQAVSELAS	720
ACTTIWIAS	ALHAAVNFGQ	PYPYAGYLPN	PTVSRRRMPE	PGSKEYEELE	RDPERGFH	780
ITSQIQTII	ISLIEILSKH	SSDEVYLGQR	DTPEWTS	DALAAFKRFS	ALVKIEGKV	840
GENRDPQLRN	RNGPAEFPPY	MLLYPNTSDH	SEAREERAAG	GIHTHTHRTL	EKREARSEE	900
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VAFQLVSATA	ADPSNGSRGK	VGKAAHLEE	VVSLKSTTDG	ETVYRVSEF	DGSGQVPGAV	1020
LVNRNLQHA	FLKSLTLEGG	VPGRGTVVFA	NSWIYPHNLY	YSQERVFAND	TYLP SKMPA	1080
LVVPYRQDEL	KILRGDDNPG	PYKEHDRVYR	YDYNDLGEP	DKGEDHARPV	LGGSQEHYPY	1140
RCRTGRRPTE	ETDPNSESRL	FLNLNIYVPR	RDERFGHLKMS	DFLGYSLKAI	IEAVLPTLGT	1200
FVDDTPKEFD	SFEDILGLYE	PGPEAPNPNL	VAEVRKRIPS	SEFLRSILPN	GSHDHPKMLP	1260
LPNIIRSDVL	KAPEFKFGWR	RTDEEFARET	LAGVNPVLIK	RLTEFPAKST	DPSQYGDHT	1320
KITEAHIQH	NMEGLSVQNA	LKKNRLFILD	HHDFMPYLNK	KINELEGNFI	YASRTLLFLK	1380
DGTLKPLAVE	LSLPHPDGQQ	HGAVSKVYTP	AHSGAEGHVW	QLAKAYACVN	DSAWHQLISH	1440

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WLNTHAVIEP	FVIATNRQLS	VVHPVHKLLS	PHYRDTLNIN	ALARQTLINA	DGIFERTVFP	1500
AKYALGMSSD	VYKSWNFNEQ	ALPADLVKRG	VAVPDQSSPY	GVRLLIKDYP	YAVDGLVIWW	1560
AIERWVKEYL	DVYYPNDGEL	QRDVELQAWW	KEVREEAHGD	LKDRDWWPRM	DAVQRLARAC	1620
TTVIWVASAL	HAAVNFGQYP	YAGYLPNRPT	VSRPRMPEPG	SDDYKKLEAG	QKEADAVFIR	1680
TITSQFQITL	GISLIEILSK	HSSDEVYLGQ	RDEPERWTS	ARALDAFRFR	GSRLVEIEKR	1740
IRTMNDSPTL	KNRKGPVEMP	YMLLYPNTSD	VTGEKGEGLT	AMGIPNSISI		1790

SEQ ID NO: 40 moltype = AA length = 1791
 FEATURE Location/Qualifiers
 REGION 1..1791
 note = WVP18-09307-014
 source 1..1791
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 40

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FFLKTITLDD	VPGRGAVTFV	ANSWVYPAGK	YRYNRVFFSN	DTYLP SQMPA	ALPKYRDDDEL	180
RNLRGDDQGG	PYQRHDRVYR	YDVYNDLGEP	DGGRPRPILG	GSADHPYPRR	CRTGRKPTKT	240
DPNSESRLSL	VEQIYVPRDE	RFGHLKMSDF	LGYSIKAITQ	GIIPAVRTYV	DTTPGEFDSF	300
QDIINLYEGG	IKLPKIQALE	DMRKLFPQL	VKDLPAGGD	YLLKLPIQI	IQGTSRIPID	360
VRGRLLSGLH	IYMCSEYEDN	AWRTDEEFAR	EVLAGVNPV	ITRLTEFPK	STLDPSKYGD	420
HTSTITAETH	EKNLEGLTVQ	QALDGNRLYI	LDHHRFMPF	LIDVNNLEGN	FIYATRTLFF	480
LRGDGRLAPL	AIELSEPYID	GLTVAKSKV	YTPASSGVEA	WVWQLAKAYV	AVNDSGWHQL	540
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VFPKYALGMM	SSVYKSWNF	TEQGLPADLV	KRGVAVADPS	SPYKVRLLIE	DYPYASDGLA	660
IWHAEIQWVG	EYLAIYYPDD	GALRGDEELQ	AWWKEVREVG	HGDHKDAPFW	PKMQAVSELA	720
SACTTIIWIA	SALHAAVNFG	QYPYAGYLPN	RPTVSRRRMP	EPSSKEYEEL	ERDPERGFIH	780
TITSQIQITII	GISLIEILSK	HSSDEVYLGQ	RDTPEWTSDA	RALAAFKRFS	DALVKIEGKV	840
VGENRDPQLR	NRNGPAEFYP	MLLYPNTSDH	SEARERAAGQ	GIHTHTHRTL	PEKREARSEE	900
RPATMFHWGV	ADRLTGKNKE	AWNKGKIRGT	VRLVKKEVLD	VGDPNASLLD	GVHRLGWDD	960
GVAFLQVSAT	AADPSNGSRG	KVGKAAHLEE	AVVSLKSTTD	GETVYRVSE	WDGSQGVPGA	1020
VLVRNLQHA	EFLKSLTLEG	VPGRGTVVFV	ANSWIYPHNL	YSQERVFFAN	DTYLP SKMPPA	1080
ALVPRYQDEL	KILRGDDVYR	PYKEHDRVYR	YDYNDLGEP	DGEDHARPV	LGSQEHYPY	1140
RRCRTGRRPT	ETDPNSESRL	FLLNLNIYVP	RDERFGHLKM	SDFLGYSLKA	IEAVLPTLG	1200
TFVDDTPKEF	DSFEDIIGLY	EPGPEAPNPN	LVAEVRKRIP	SEFLRSILPN	GSHDHPKMP	1260
LPNIIRSDVL	KKAPEFKFGW	RTDEEFARET	LAGVNPVLIK	RLTEFPKST	LDPSQYGDHT	1320
SKITEAHIQH	NMEGLSVQNA	LKKNRLFILD	HHDFMPYLN	KINELEGNFI	YASRTLLFLK	1380
DDGTLKPLAV	ELSLPHPDGQ	QHGAUSKYYT	PAHSGAEGHV	WQLAKAYACV	NDSAWHQLIS	1440
HWNTHAVIE	FVVIATNRQL	SVVHPVHKLL	SPHYRDTLNI	NALARQTLIN	ADGIFERTVF	1500
PAKYALGMSS	DVYKSWNFNE	QALPADLVKR	GVAVPDQSSP	YGVRLLIKDY	PYAVDGLVIW	1560
WAIERWVKEY	LDVYYPNDGE	LQRDVELQAW	WKEVREEAHG	DLKDRDWWPR	MDAVQRLARA	1620
CTTVIWVASA	LHAAVNFGQY	PYAGYLPNRPT	TVSRPRMPEP	GSDDYKKLEA	QKEADAVFI	1680
RTITSQFQITI	LGISLIEILS	KHSSDEVYLG	QRDEPERWTS	DARALDAFRF	FGSRLVEIEK	1740
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SEQ ID NO: 41 moltype = DNA length = 5376
 FEATURE Location/Qualifiers
 misc_feature 1..5376
 note = Zm00008a004913
 source 1..5376
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 41

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SEQ ID NO: 42      moltype = DNA length = 5376
FEATURE            Location/Qualifiers
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                    note = WVP18-09358-016
source             1..5376
                    mol_type = other DNA
                    organism = synthetic construct

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SEQ ID NO: 47          moltype = DNA length = 24
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SEQ ID NO: 51      moltype = DNA length = 18
FEATURE           Location/Qualifiers
misc_feature       1..18
                   note = cruaxxxxxxf02x
source             1..18
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SEQUENCE: 51
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SEQ ID NO: 52      moltype = DNA length = 21
FEATURE           Location/Qualifiers
misc_feature       1..21
                   note = cruaxxxxxxr01x
source             1..21
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                   organism = synthetic construct
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SEQUENCE: 52
ccgtcgttgt agaaccattg g 21
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SEQ ID NO: 53      moltype = DNA length = 19
FEATURE           Location/Qualifiers
misc_feature       1..19
                   note = nptIIxxxf01
source             1..19
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SEQUENCE: 53
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SEQ ID NO: 54      moltype = DNA length = 18
FEATURE           Location/Qualifiers
misc_feature       1..18
                   note = nptIIxxxr01
source             1..18
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                   organism = synthetic construct
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SEQUENCE: 54
cggcacttcg cccaatag 18
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SEQ ID NO: 55      moltype = DNA length = 20
FEATURE           Location/Qualifiers
misc_feature       1..20
                   note = tDTxxxf04
source             1..20
                   mol_type = other DNA
                   organism = synthetic construct
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SEQUENCE: 55
tcctcgttgt gggaggtgat 20
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```
SEQ ID NO: 56      moltype = DNA length = 19
FEATURE           Location/Qualifiers
misc_feature       1..19
                   note = tDTxxxr01
source             1..19
                   mol_type = other DNA
                   organism = synthetic construct
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SEQUENCE: 56
gtgcaactgc ccggctact 19
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SEQ ID NO: 57      moltype = DNA length = 29
FEATURE           Location/Qualifiers
misc_feature       1..29
                   note = CruaxxxMGB
source             1..29
                   mol_type = other DNA
                   organism = synthetic construct
```

```
SEQUENCE: 57
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SEQ ID NO: 58      moltype = DNA length = 15
FEATURE           Location/Qualifiers
misc_feature       1..15
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source	note = nptIIxxxMGB 1..15 mol_type = other DNA organism = synthetic construct	
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source	1..18 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 59 agcttggtgt ccacgtag		18
SEQ ID NO: 60 FEATURE misc_feature	moltype = DNA length = 19 Location/Qualifiers 1..19 note = LbCpf1- Fw	
source	1..19 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 60 tgccggaaga ggatctttt		19
SEQ ID NO: 61 FEATURE misc_feature	moltype = DNA length = 23 Location/Qualifiers 1..23 note = LbCpf1- Rv	
source	1..23 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 61 agtcggcgtc ggtcctgctg aag		23
SEQ ID NO: 62 FEATURE misc_feature	moltype = DNA length = 18 Location/Qualifiers 1..18 note = Probe for nuclease test	
source	1..18 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 62 agttcgctct cgactata		18
SEQ ID NO: 63 FEATURE misc_feature	moltype = DNA length = 24 Location/Qualifiers 1..24 note = IR106_lox3_F1	
source	1..24 mol_type = other DNA organism = synthetic construct	
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SEQ ID NO: 64 FEATURE misc_feature	moltype = DNA length = 22 Location/Qualifiers 1..22 note = IR106_lox3_F2	
source	1..22 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 64 ccattacttg ttggtcggc gt		22
SEQ ID NO: 65 FEATURE misc_feature	moltype = DNA length = 21 Location/Qualifiers 1..21 note = IR106_lox3_F3	
source	1..21 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 65 gccatcactt gttttctcgg c		21
SEQ ID NO: 66 FEATURE	moltype = DNA length = 20 Location/Qualifiers	

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	note = IR106_lox3_F4	
source	1..20	
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SEQ ID NO: 67	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
misc_feature	1..25	
	note = IR106_lox3_R1	
source	1..25	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 67		
tcaaaggcta atataactga cacgt		25
SEQ ID NO: 68	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
misc_feature	1..23	
	note = IR106_lox3_R2	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 68		
ggcaaacgcg tttccttaat tca		23
SEQ ID NO: 69	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = IR106_lox3_R3	
source	1..20	
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ataacgctga tgtgctaagc		20
SEQ ID NO: 70	moltype = DNA length = 24	
FEATURE	Location/Qualifiers	
misc_feature	1..24	
	note = IR106_lox3_R4	
source	1..24	
	mol_type = other DNA	
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gttaataacg ctgatgtact aagc		24
SEQ ID NO: 71	moltype = DNA length = 22	
FEATURE	Location/Qualifiers	
misc_feature	1..22	
	note = 946R	
source	1..22	
	mol_type = other DNA	
	organism = synthetic construct	
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SEQ ID NO: 72	moltype = DNA length = 20	
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misc_feature	1..20	
	note = 1145F	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 72		
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SEQ ID NO: 73	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = 346F	
source	1..20	
	mol_type = other DNA	
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SEQ ID NO: 74	moltype = DNA length = 20	

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FEATURE	Location/Qualifiers	
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	note = 1948R	
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SEQ ID NO: 75	moltype = DNA length = 5145	
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	mol_type = genomic DNA	
	organism = Brassica napus	
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SEQ ID NO: 76 moltype = DNA length = 5366
 FEATURE Location/Qualifiers
 source 1..5366
 mol_type = genomic DNA
 organism = Brassica napus

SEQUENCE: 76

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SEQ ID NO: 79 moltype = AA length = 919
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 source 1..919
 mol_type = protein
 organism = Brassica napus

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HQKEFFLESI TIEGFALGPV HFPNCNSWVQS QKDHPGKRIF FTNKPLLPSE TPDGLKKLRE 240
RELKNLRGDG SGARKLSMRI YDFDVYNDLG NPDKSSELSR PKLGGKEIPY PRCRTGRHP 300
TDTDKAESR VEKPLPMYVP RDEQFEETKQ KTFAGRLKA VLHHLVPSLK ASILAEDFAD 360
FGEIDGLYKE GLLKLKGFQD EIFDKFPLPK AIVNTLQESS KGLLKYDTPK ILSKDKNAWL 420
RDDEFARQAI AGINPVNIER VRTFPPVSNL DPEIYGPQHS ALTSDHII GH LDGLSVQQAL 480
EENRLYMILNY HDIFLPLFLDR INALDGRKAY ATRTIFFLTR LGTLKPVAIE LSLPPHGNH 540
RSKRVTTPPV DATSNWVWQL AKAHVSSNDA GVHQLVNHWWL RTHACLEPFI LAAHRQLSAM 600
HPIFKLLDPH MRYTLEINAL ARQSLISADG VIEEGFTAGS YGEMMSAAAY KSSWRFDMEG 660
LPADLIRRG I AVDPDTQPHG VKLLIEDYPY ANDGLLLWSA IQTWVRTYVE RYYPNPNIK 720
TDSELQSWYS ESINVGHADL RDAEWWPENL TVDDLVSILT TLIWLASAQH AALNFGQYPY 780
GGYVPMRPL MRRLIPDES PLYASFISDP EKIFYFSAMPS LLQTSKFMAV VDTLSTHSPD 840
EEYIGERQQP SIWTGDGEIV DAFYGFAAEI GRIEKEIEKR NSDPDRNRNC GAGVLPYELL 900
VPSSEPGVTC RGVPSVSI 919

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SEQ ID NO: 80      moltype = AA  length = 919
FEATURE            Location/Qualifiers
source             1..919
                   mol_type = protein
                   organism = Brassica napus

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SEQUENCE: 80
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LELISTELDP KTNLPKKSNA AVLKDWSEKS KTKAERVHYT AEFTVDAAFG TPGAITIMNK 180
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KELKNLRGDG TGVKRLSDRI YDFDVYNDLG NPDKSSELSR PKLGGGERPY PRCRTGRQP 300
TDTDNEAESR VEKPLPMYVP RDEQFEETKQ DTFAGRLKA VLHHLVPSLK ASILADDFSD 360
FGEIDDLKYK GLLKLKGFQD EIFNKFPLPK GIVNTLQESS KGLLKYDTPK ILSKDKNAWL 420
RDDEFARQAI AGINPVNIER VKTFFPVSNL DPEIYGPQHS ALTSDHII GH LDGLSVQQAL 480
EENRLYMILNY HDIFLPLDQ INALDGRKAY ATRTIFFLTR LGTLKPVAIE LSLPSHGNH 540
RSKRVTTPPV DATSNWVWQL AKAHVSSNDA GVHQLVNHWWL RTHACLEPFI LAAHRQLSAM 600
HPIFKLLDPH MRYTLEINAL ARQSLISADG VIEEGFTAGQ YGLEMSAAAY KSSWRFDMEG 660
LPADLIRRG I AVDPDTQPHG VKLLIEDYPY ANDGLLIWSA IQTWVRTYVE RYYPNSNSIQ 720
TDSELQSWYS ESINVGHADL REAEWWPKLD TVDDLVSILT TLVWLASAQH AALNFGQYPY 780
GGYVPMRPL MRRLIPDES PEYASFISDP EKIFYFSMPS LLQTSKFMAV VDTLSTHSPD 840
EEYIGERQQP STWTGDAIIV DAFYGFAAEI GRIEKEIEKR NSDPDRNRNC GAGVLPYELL 900
VPSSEPGVTC RGVPSVSI 919

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SEQ ID NO: 81      moltype = AA  length = 919
FEATURE            Location/Qualifiers
source             1..919
                   mol_type = protein
                   organism = Brassica napus

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SEQUENCE: 81
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TDTDKAESR VEKPLPMYVP RDEQFEETKQ KTFAGRLKA VLHHLVPSLK ASILAEDFAD 360
FGEIDGLYKE GLLKLKGFQD EIFNKFPLPK AIVNTLQESS KGLLKYDTPK ILSKDKNAWL 420
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EENRLYKLDY HDIFLPLFLDR INALDGRKAY ATRTIFFLTR LGTLKPVAIE LSLPPHGNH 540
RSKRVTTPPV DATSNWVWQL AKAHVSSNDA GVHQLVNHWWL RTHACLEPFI LAAHRQMSAM 600
HPIFKLLDPH MRYTLEINAL ARQSLISADG VIEEGFTAGS YGEMMSAAAY KSSWRFDMEG 660
LPADLIRRG I AVDPDTQPHG VKLLIEDYPY ANDGLLLWSA IQTWVRTYVE RYYPNPNIK 720
TDSELQSWYS ESINVGHADL RDAEWWPENL TVDDLVSILT TLIWLASAQH AALNFGQYPY 780
GGYVPMRPL MRRLIPDES PEYASFISDP EKIFYFSAMPS LLQTSKFMAV VDTLSTHSPD 840
EEYIGERQQP SIWTGDAIIV DAFYGFAAEI GRIEKEIEER NSDPDRNRNC GAGVLPYELL 900
VPSSEPGVTC RGVPSVSI 919

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SEQ ID NO: 82      moltype = AA  length = 919
FEATURE            Location/Qualifiers
source             1..919
                   mol_type = protein
                   organism = Brassica napus

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SEQUENCE: 82
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LELISTELDP KTNLPKKSNA AVLKDWSEKS KTKAERVHYT AEFTVDAAFG TPGAITIMNK 180
HQKEFFLECI TIEGFALGPV HFPNCNSWVQS QNDHPEKRIF FTNQPFLLPSE TPEGLRKLRLG 240
KELKNLRGDG TGVKRLSDRI YDFDVYNDLG NPDKSYELSR PKLGGGERPY PRCRTGRQP 300
TDTDNEAESR VEKPLPMYVP RDEQFEETKQ DTFAGRLKA VLHHLVPSLK ASILADDFSD 360
FGEIDDLKYK GLLKLKGFQD EIFNKFPLPK GIVNTLQESS KGLLKYDTPK ILSKDKNAWL 420
RDDEFARQAI AGINPVNIER VKTFFPVSNL DPEIYGPQHS ALTSDHII GH LDGLSVQQAL 480
EENRLFMILNY HDIFLPLFLDR INALDGRKAY ATRTIFFLTR LGTLKPVAIE LSLPSHGNH 540
RSKRVTTPPV DATSNWVWQL AKAHVSSNDA GVHQLVNHWWL RTHACLEPFI LAAHRQLSAM 600
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SEQ ID NO: 83 moltype = DNA length = 2760
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 mol_type = genomic DNA
 organism = Brassica napus

SEQUENCE: 83

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SEQ ID NO: 84 moltype = DNA length = 2760
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 mol_type = genomic DNA
 organism = Brassica napus

SEQUENCE: 84

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 organism = Brassica napus

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 organism = Brassica napus

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 mol_type = genomic DNA
 organism = Zea mays

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 Location/Qualifiers

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The invention claimed is:

1. A method for conferring or increasing resistance or tolerance to an insect and comprising the steps of:

- (i) providing at least one *Zea mays* plant cell;
- (ii) introducing into the at least one *Zea mays* plant cell at least one gene silencing construct, at least one genome editing system or a genome modification, which leads to a targeted knock-down of a Lox3 gene endogenous to the *Zea mays* plant;
- (iii) obtaining at least one modified *Zea mays* plant cell having reduced expression of the Lox3 gene;
- (iv) obtaining at least one *Zea mays* plant cell, tissue, organ, plant, or seed having reduced expression of the Lox3 gene after an additional step of regenerating the *Zea mays* plant tissue, organ, plant, or seed from the at least one modified *Zea mays* plant cell; and
- (v) testing the at least one *Zea mays* plant cell, tissue, organ, plant, or seed having reduced expression of the Lox3 gene for increased resistance or tolerance to the insect,

wherein the Lox3 gene is represented by the nucleic acid sequence of SEQ ID NO: 6, 7, 9, 10, 12, 13, 15, 16, 87, or 88 or a nucleic acid sequence having a sequence identity of at least 95% to the sequence of SEQ ID NO: 6, 7, 9, 10, 12, 13, 15, 16, 87, or 88, and

wherein the Lox3 gene encodes the amino acid sequence of SEQ ID NO: 8, 11, 14, 17 or 89 or an amino acid sequence having a sequence identity of at least 95% to the sequence of SEQ ID NO: 8, 11, 14, 17 or 89.

2. The method according to claim 1, wherein the method is for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*).

3. The method according to claim 1, wherein the method is for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of green peach aphid (*Myzus persicae*), diamondback moth (*Plutella xylostella*), cabbage stem flea beetle (*Psylliodes chrysocephala*), crucifer flea beetle (*Phyllotreta cruciferae*), striped flea beetle (*Phyllotreta striolata*), hop flea beetle (*Psylliodes punctulata*), rape stem weevil (*Ceutorhynchus picipitarsis*) and cabbage stem weevil (*Ceutorhynchus quadridens*).

4. The method according to claim 1, wherein in step (ii) a construct is introduced into the at least one *Zea mays* plant cell, which targets the Lox3 gene for gene silencing.

5. The method according to claim 4, wherein the construct is or wherein the construct encodes an RNAi construct comprising a sense and an antisense sequence targeting the Lox3 gene, the RNAi construct forming an RNA hairpin upon transcription.

6. The method according to claim 5, wherein the method is for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*), and wherein the sense sequence is encoded by the nucleic acid sequence of SEQ ID NO: 1, or a nucleic acid sequence having a sequence identity of at least 95% to the sequence of SEQ ID NO: 1.

7. The method according to claim 6, wherein the method is for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*), and wherein the antisense sequence is encoded by the nucleic acid sequence of SEQ ID NO: 2, or a nucleic acid sequence having a sequence identity of at least 95% to the sequence of SEQ ID NO: 2, or

wherein the method is for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*), and wherein the RNA hairpin has an intervening intron loop sequence comprising the nucleic acid sequence of SEQ ID NO: 3, or a nucleic acid sequence having a sequence identity of at least 95% to the sequence of SEQ ID NO: 3.

8. The method according to claim 4, wherein the method is for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*), and wherein the construct comprises the nucleic acid sequence of SEQ ID NO: 4, or a nucleic acid sequence having a sequence identity of at least 95% to the sequence of SEQ ID NO: 4, or

wherein the method is for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of fall army worm (*Spodoptera frugiperda*), corn leafhopper (*Dalbulus maidis*) and green belly stink bug (*Dichelops melacanthus*), and wherein a vector is introduced into the plant cell, which vector comprises or consists of the nucleic acid

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sequence of SEQ ID NO: 5, or a nucleic acid sequence having a sequence identity of at least 95% to the sequence of SEQ ID NO: 5.

9. The method according to claim 4, wherein the construct is introduced into the at least one *Zea mays* plant cell by transformation or transfection mediated by biolistic bombardment, *Agrobacterium*-mediated transformation, micro- or nanoparticle delivery, chemical transfection, or a combination thereof.

10. The method according to claim 1, wherein in step (ii) at least one genome editing system is introduced into the at least one *Zea mays* plant cell, which targets the Lox3 gene, wherein the at least one genome editing system comprises

- (a) at least one site-specific nuclease or site-specific nickase, and, in case a CRISPR system is used, at least one guide molecule or a sequence encoding the same, and/or
- (b) at least one repair template, or a sequence encoding the same.

11. The method according to claim 10, wherein the at least one genome editing system is selected from a CRISPR/MAD7 system, a CRISPR/Cpf1 (CRISPR/Cas12a) system, a CRISPR/MAD2 system, a CRISPR/Cas9 system, a CRISPR/CasX system, a CRISPR/CasY system, a CRISPR/Cas13 system, and a CRISPR/Csm system, or wherein the at least one genome editing system is selected from a zinc finger nuclease system, or a transcription activator-like nuclease system, or a meganuclease system, or any combination, variant, or an active fragment thereof.

12. The method of claim 10, wherein the at least one genome editing system is introduced into the at least one *Zea mays* cell by transformation or transfection mediated by biolistic bombardment, *Agrobacterium*-mediated transfor-

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mation, micro- or nanoparticle delivery, chemical transfection, or a combination thereof.

13. The method of claim 10, wherein the method is for conferring or increasing resistance or tolerance to one or more insect(s) selected from the group consisting of green peach aphid (*Myzus persicae*), diamondback moth (*Plutella xylostella*), cabbage stem flea beetle (*Psylliodes chrysocephala*), crucifer flea beetle (*Phyllotreta cruciferae*), striped flea beetle (*Phyllotreta striolata*), hop flea beetle (*Psylliodes punctulata*), rape stem weevil (*Ceutorhynchus piciparsis*) and cabbage stem weevil (*Ceutorhynchus quadridens*), wherein the at least one genome editing system comprises a crRNA encoded by the nucleic acid sequence of any of SEQ ID NOs: 46 to 49, or a nucleic acid sequence having a sequence identity of at least 95% to the sequence of any of SEQ ID NOs: 46 to 49.

14. The method of claim 13, wherein the genome editing system is encoded by a plasmid of the nucleic acid sequence of SEQ ID NO: 50 or a nucleic acid sequence having a sequence identity of at least 95% to the sequence of SEQ ID NO: 50.

15. The method according to claim 1, wherein in step (ii) a mutagenesis is performed on a single *Zea mays* plant cell or on a plurality of *Zea mays* plant cells by applying chemicals or radiation, and/or

wherein an alkylating agent is applied to the single *Zea mays* plant cell or the plurality of *Zea mays* plant cells to induce mutagenesis and/or

wherein one or more mutations in the Lox3 gene are inserted and identified by TILLING in step (ii).

16. The method according to claim 15, wherein one or more *Zea mays* cell(s) with knock-down mutations in the Lox3 gene are selected in step (ii).

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